



**Department of Computer Science**  
**COMP4300-Graduation Project**  
**Project Report-2024/2025**

Title of Project:  
**User Risk Assessment and Payment Security Enhancement in Online Auctions**



**MAZADCO**

**Auction**

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## **Abstract**

This research proposes the development of an auction website tailored for users in the Middle East, a region where online payment methods are underutilized due to cultural barriers and unfamiliarity with digital transactions. The study aims to design an algorithm that categorizes users based on various risk factors, ultimately encouraging participation in online auctions. By leveraging insights from existing literature on online learning in auctions, consumer behavior, and security concerns, this research seeks to facilitate a more inclusive online auction experience. The findings will contribute to enhancing user confidence and engagement in digital marketplaces, fostering economic growth in the region.

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## Chapter 1: Introduction

The rapid growth of e-commerce has revolutionized the way consumers engage in buying and selling goods. However, in the Middle East, online auction platforms face significant challenges due to cultural barriers and the limited adoption of online payment methods. Many potential users remain hesitant to participate in digital transactions, primarily due to concerns about security, trust, and unfamiliarity with technology.

To address these challenges, this research focuses on creating a user-friendly auction website that not only allows for seamless transactions but also builds user confidence through tailored algorithms. By categorizing users based on identified risk factors, the proposed algorithm aims to provide personalized experiences and recommendations that encourage participation.

### Aim and Objectives

The primary aim of this research is to develop a comprehensive workflow for risk categorization and payment security enhancement in the context of online auction platforms. The specific objectives are as follows:

- Establish a robust data collection mechanism to gather comprehensive information on user behavior, bidding history, payment methods and performance on the online auction platform.
- Design an algorithm for risk categorization that accurately classifies users into high-risk, medium-risk, and low-risk categories based on defined criteria derived from the collected user data.
- Provide personalized payment method recommendations for each risk category to enhance security and build trust among platform users.
- Evaluate the effectiveness of the proposed risk categorization workflow and payment security enhancement strategies in improving user trust, reducing fraud, and increasing overall participation and engagement within the online auction platform.

## 1.3 Overview

Online auction platforms must ensure secure transactions and foster consumer trust in the digital marketplace. This research presents a comprehensive workflow that leverages user behavior data to categorize participants into distinct risk levels - high, medium, and low - and provides tailored payment method recommendations and security enhancement strategies.

The workflow begins with the collection of user data through cookies, capturing key information such as browsing behavior, bidding history, and payment method usage. This data is then aggregated over a 30-day period to generate a user profile that reflects their overall patterns and tendencies on the platform.

Using predefined risk criteria, the algorithm categorizes users based on factors such as the frequency of failed payments, high-value failed deliveries, win/loss ratios, and the consistency of payment method usage. Users classified as high-risk exhibit characteristics like frequent failed payments, low win rates, and inconsistent payment methods. Medium-risk users display occasional payment issues and moderate win rates, while low-risk users demonstrate a successful payment history, high win rates, and consistent use of reliable payment options.

To enhance security and build trust, the workflow provides personalized recommendations for each risk category. High-risk users are encouraged to utilize secure payment options with buyer protection, and are prompted to enable two-factor authentication and adopt strong password practices. Medium-risk users are advised to use credit cards with fraud protection features and monitor their account statements. Low-risk users are offered a broader range of flexible payment options, including digital wallets and direct bank transfers, while still receiving guidance on maintaining robust security measures.

By implementing this risk categorization workflow, online auction platforms can effectively assess user risk, recommend appropriate payment methods, and enhance security, ultimately contributing to a more trustworthy and engaging digital marketplace for buyers and sellers



## Chapter 2: Background and Literature Review

Auction websites function as dynamic digital marketplaces where buyers and sellers engage in a competitive bidding process for various goods and services. The process begins with user registration, requiring participants to create accounts that often necessitate verification to enhance security. Sellers can list items by providing detailed descriptions, images, and setting auction parameters such as starting prices and duration. Once the auction is live, registered users can place bids, with many platforms offering proxy bidding, which allows users to set maximum bid limits that the system will automatically adjust to maintain their position as the highest bidder. As the auction progresses, participants receive real-time notifications about bidding activity and can monitor items through personalized watch lists. At the conclusion of the auction, the highest bidder wins the item and is prompted to complete payment, after which the seller is responsible for shipping.

In many Middle Eastern countries, cultural norms and traditions significantly influence consumer behavior. There is often a preference for face-to-face transactions and a reluctance to engage in online commerce. This contrasts with regions like North America and Europe, where online shopping and auctions have become mainstream. Cultural practices surrounding negotiation and the value placed on personal relationships can hinder the acceptance of impersonal online auction formats [1]

The proposed research on the Risk Categorization Workflow for an Online Auction Platform builds upon previous studies that have examined the key drivers of consumers' intention to purchase in online auctions. Specifically, the work by [2] on the most important factors influencing consumer decision-making in online auctions in Thailand provides a valuable foundation for this current research.

The three key factors identified by [2] - the quality of the online information system, the price competitiveness of the online auction, and the quality of the online service - are highly relevant to the development of the risk categorization workflow. These factors directly impact the user experience and trust in the online auction platform, which are crucial considerations for the proposed workflow.

For instance, the quality of the online information system, as described [2] is reflected in the data collection stage of the risk categorization workflow, where user behavior, bidding history, and payment methods are captured. Ensuring the clarity, accuracy, and ease of access to this information is essential for accurately assessing user risk profiles.

Similarly, the price competitiveness of the online auction, as highlighted by [2], is a crucial factor in the risk categorization process. Users with a history of successful bids at competitive prices are more likely to be classified as low-risk, while those with a pattern of failed payments or high-value failed deliveries may be categorized as high-risk.

Furthermore, the quality of the online service, as discussed in [3], aligns with the trust-building features incorporated into the proposed workflow, such as the display of seller ratings, clear return policies, and timely notifications. These elements aim to enhance the overall user experience and foster confidence in the online auction platform, which is a key objective of the risk categorization workflow.

By building upon the insights from [3] and incorporating the identified key drivers of consumer intention to purchase, the current research on the Risk Categorization Workflow for an Online Auction Platform can provide a more comprehensive and user-centric approach to addressing

security and trust concerns in the digital auction environment.

Research has shown that cultural attitudes significantly influence online shopping behaviors in the Middle East. A study by [2] highlighted how cultural factors, including trust and personal relationships, affect e-commerce participation in the region.

Here are some key studies and findings that align with the challenges faced by users in the Middle East when engaging with online auction platforms:

### **1. Trust and Security Concerns**

study by: [3] examined the factors affecting user acceptance of e-commerce in arab world. The findings indicated that perceived security and trust were critical factors influencing online shopping intentions. In many countries within the region, consumer protection laws and regulations governing online transactions may be underdeveloped, leading to increased vulnerability for buyers.

### **2. Limited Familiarity with Online Payment Methods**

In a Research by [4] Primary data were collected online from nine countries in the Middle East indicated that many Arab consumers exhibit hesitance towards mobile payments due to lack of awareness and familiarity with online payment systems hampers the adoption of e-commerce in the Middle East. This study found that many consumers preferred cash transactions due to concerns over security.

### **3. Regulatory and Legal Frameworks**

The World Bank's "Doing Business" reports [5] have highlighted the varying regulatory environments across countries, including the challenges faced by e-commerce in several Middle Eastern nations. These reports often point to inadequate consumer protection laws as a barrier to online commerce.

Many Middle Eastern countries have not yet established comprehensive consumer protection laws specific to e-commerce. the regulatory environment significantly influences consumer trust.

Without clear regulations and protections, users may be wary of participating in online auctions due to fears of fraud and lack of recourse.

### **4. Payment Fraud and Charge backs**

A report by [1] focused on the risks associated with online transactions in the arab countries, and general payment risks in the Middle East noting that concerns over fraud and chargeback issues are prevalent among consumers. Many online payment systems may not have advanced security measures in place, making them susceptible to hacking and data breaches.

This research builds upon the foundations of previous studies aiming to create an algorithm that not only addresses the unique needs of Middle Eastern users but also aligns with best practices in online auction design. By fostering a secure and supportive online environment, this study aspires to enhance the overall user experience and promote the adoption of auction platforms in the region.

## 2.2 Review of past work

### 2.1 Details of relevant theory

**Digital Market Theory:** This theory is the core objective of our project, and aims to simplify the traditional process of communicating with auctioneers by making it electronic. Sellers can easily register and list their items on our platform, while suppliers can browse auctions and choose what suits them.

**User Interaction and Experience Theory:** We focus on understanding how users interact with technology and enhancing their experiences. By applying this theory, we designed our platform to be intuitive and easy to use, ensuring a responsive and enjoyable experience for all users.

**Data Privacy and Security Theory:** Protecting user data is of utmost importance. We adhere to privacy regulations and implement strong security measures to prevent unauthorized access or breaches. Prioritizing data privacy and security builds trust among users, ensuring that their information is handled with care and integrity.

#### 2.2.1 Catawiki

It is a safe selling application where more than 600 auctions are held every week across 80 diverse categories of items, including luxury goods, classic cars, collectibles, art and antiques. Auctions start every Friday and are supervised by specialized experts. [6]

#### 2.2.2 Copart

It is an online auction application that specializes in selling motor vehicles and contains a wide range including cars, trucks, SUVs, and bikes. It operates with a network of more than 200 locations, which allows access to millions of vehicles in more than 190 countries through its easy-to-use website and mobile applications.

Table 1 :Details of relevant theory

| Application                                                         | Catawiki                                                         | Copart                                                       | Mazadco (Our App)                                                                                                                               |
|---------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| categorized user Risk                                               | No                                                               | No                                                           | Yes, it depends on the user's control and behavior in transactions, and is classified as a risk and granted powers based on its classification. |
| categories type                                                     | Arts and antiques                                                | Cars                                                         | 10 different categories                                                                                                                         |
| Geographic spread                                                   | America, Europe, and Asia                                        | Europe and Asia                                              | The West Eastern with scalability                                                                                                               |
| Unified Interface (Buyer + Seller)                                  | Separate the buyer and seller                                    | Only buyers                                                  | Yes – one user manages everything                                                                                                               |
| Bidding on specific types<br>Payment methods by user classification | NO                                                               | No                                                           | Yes – depending on user level                                                                                                                   |
| Chatbot as a user guide                                             | Just a traditional help center                                   | NO                                                           | Yes – Chatbot guides the user step by step and answers frequently asked questions                                                               |
| Payment methods by user classification                              | No - there is no classification, all users have the same methods | No - payment is unified (pre-deposit and bank transfer only) | Yes – it depends on the user's classification and gradual payment options are available to him                                                  |

## 2.3 Tools and Technology

### 2.3.1 Flutter :

It is a useful tool kit that enables easy ways to create new applications. It is the best choice for creating applications with an amazing user interface and high performance. It is an open source program developed by Google to build user interfaces for Android, iOS, Windows and web applications. It also works with existing free software. Its ability to target multiple platforms with a single database simplifies the development process and ensures a faster time to market. \* It is characterized by the fast loading feature, which saves time, as the developer sees the changes he applies in real time and can stop the execution of the code, make changes to the code and continue the code from the same place, making it more productive. It is easy to learn and use, very flexible and scalable. It is also suitable for building applications of all sizes, from small prototypes to large and complex enterprise applications for its rapid maintenance. [7]

### 2.3.2 Database using PHP and MySQL via XAMPP

In this project, we used XAMPP as a local development environment, along with PHP and phpMyAdmin for database management. We chose these tools because they are easy to use, suitable for our project requirements, and provide us with complete control over data in a clear and simple manner.

We wrote PHP files to process data sent and received from the application, such as logins and account creation. We also used phpMyAdmin because it allows us to manage the database through

a graphical interface within the browser, making it easier to create tables, modify data, and execute queries without having to write complex commands.

Using these tools helped us work efficiently and organized, and enabled us to test the application's functionality during the development phase, without the need for external server configuration.

### 2.3.3 Python

Python was the primary programming language used in our project due to its simplicity, flexibility, and vast ecosystem of libraries. It served as the backbone of both the machine learning and API components, enabling smooth integration between data handling, prediction logic, and web services.

Key Python Technologies and Libraries Used:

#### 1. Flask

A lightweight Python web framework used to build the RESTful API endpoints (`/predict-risk` and `/retrain`). Flask enabled us to:

- \* Accept POST requests with JSON data.
- \* Handle responses in a structured and secure manner.
- \* Manage errors and logging effectively.

#### 2. Scikit-learn

This library was used to train and manage the Random Forest Classifier, which is responsible for risk-level prediction. Scikit-learn made it easy to:

- \* Preprocess data.
- \* Train models using existing labeled data.
- \* Export and reuse trained models.

#### 3. Joblib

Used for model serialization and persistence. Once trained, the model is saved (`.pkl` file) and loaded dynamically during predictions and retraining.

#### 4. Pandas

The Pandas library played a critical role in handling structured data, especially when working with data from the MySQL database. It allowed for easy transformation of rows into a `DataFrame` for analysis and model training.

#### 5. MySQL Connector for Python

Used to connect the Python backend to a MySQL database. It enabled secure data queries, insertions, and updates for both prediction logging and retraining purposes.

#### 6. Threading and Locking

The `threading` module and a `Lock` object were used to ensure thread safety during model

prediction and retraining processes, preventing race conditions in concurrent requests.

#### 7. Logging and Traceback

Logging was implemented for monitoring API activity and debugging. The `'traceback'` module helped capture detailed error reports.

## Chapter 3: Approach

The rise of online auction platforms has introduced new challenges in ensuring secure transactions and fostering consumer trust. This algorithm addresses these challenges by analyzing user behavior and historical transaction data to assess risk levels and propose tailored security enhancements.

This section presents an algorithm designed to categorize users of online auction platforms into risk levels—high, medium, and low—based on their behavior. **By retrieving data directly from the database for collection**, the algorithm combines real-time user behavior with historical data on bidding, payment methods. The ultimate goal is to enhance user trust and security in online transactions by recommending personalized payment options and security measures.

The Random Forest algorithm is an ensemble learning model that randomly constructs a large number of decision trees and then combines their results to produce a more accurate and stable prediction or classification than a single decision tree [8]

In the context of online auction platforms, the algorithm is used to analyze user data and in-app behavior to classify users into different risk levels (high, medium, low).

### 1. Data Collection

It starts with a large dataset containing features and labels that describe users and their behavior. The data is pulled directly from the auction platform's database to collect current and historical data.

### 2. Select the number of trees.

We determined 100 trees as an optimal balance between accuracy and computational efficiency. This number is sufficient to effectively reduce variance and produce stable, reliable results without causing a significant increase in computational resources. Moreover, practical research and extensive experience in the field of machine learning have demonstrated that, in the vast majority of cases, random forest models achieve stable and accurate performance when using between 100 and 200 trees. Therefore, choosing 100 trees provides a practical compromise between classification quality and processing speed, making it well-suited for real-time applications

### 3. Create random data samples (Bootstrap Sampling)

Each decision tree is built on its own data sample, called a bootstrap sample. This sample is randomly selected from the original dataset, with the sample size equal to the original data size, but allowing for redundancy. This means that some samples (for example, a particular user's data) may appear more than once in the same sample, while others may not. This technique is known as bootstrap sampling with replacement.

The goal of this process is to independently diversify the training data for each tree, allowing each tree to learn a slightly different model from the others. This diversification reduces interdependence between trees and increases the robustness

of the model as a whole, as combining the results of multiple disparate models reduces generalization error and improves classification accuracy.

Thus, using bootstrap samples helps the Random Forest become more flexible and capable of handling diverse and complex data, and also reduces the risk of overfitting the original training data.

#### 4. Build Decision Tree for each sample

##### 1. Selecting a random set of features at each split point:

At each step in building the tree (at each "node" or split point), not all available features are used to analyze the data; rather, only a small set of features are selected. For example, if we have 10 features (cancellation rate, number of bids, number of reports, etc.), the algorithm randomly selects only 3 of them to consider at that step. This helps diversify the trees and prevents oversimilarity.

##### 2. Selecting the best feature and the best split point from the selected features:

From the randomly selected features, each feature is evaluated to determine the best value that divides the data into two groups so that these groups are more taxonomically pure (i.e., the data in each group more likely belongs to the same class).

##### 3. Repeating the steps until a stopping condition is reached:

This process is repeated repeatedly on each branch of the tree, with the data at each new node being split in the same way. The tree continues to grow until a stopping condition is reached, such as:

A specific tree depth is reached or there is too little data at a node.

##### 4. The outcome of each tree:

Each tree becomes an independent model that can classify users based on the analyzed characteristics.

#### 5. Repeat steps 3 and 4 for each tree.

The process of selecting the sample and building the tree is repeated for each pre-specified number of trees.

#### 6. Voting/Aggregation and take final decision :

After building all the trees, when using the algorithm for prediction:

Each decision tree is assigned a classification (e.g., low/medium/high risk).

The results of all the trees are summed.

In classification: We take the majority vote (i.e., the most frequent classification).

In regression: We calculate the average of the results of all the trees.



## Model Evaluation:

To build and evaluate the Random Forest classification model, the dataset was first split into a training set (70%) and a test set (30%).

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y_num, test_size=0.3, stratify=y_num, random_state=42
)
```

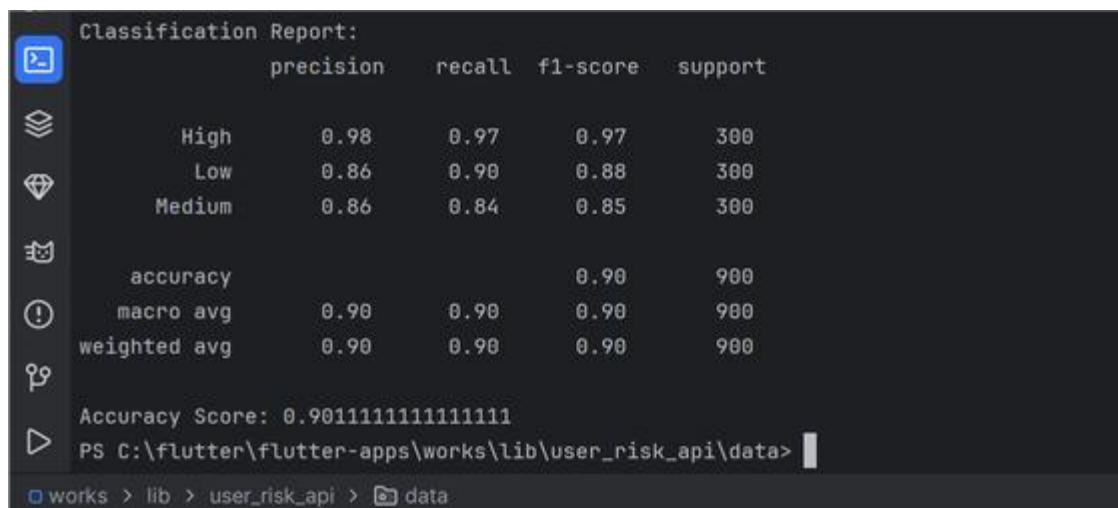
Figure 1 : testing and training set

The training set was used to train the model, allowing it to learn patterns and relationships between user behavior features and their associated risk levels. A Random Forest classifier with 100 decision trees was trained using this data. After training, the model was applied to the test set to evaluate its generalization ability and classification performance.

```
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train_scaled, y_train_resampled)
```

Figure 2 : random forest depth

We evaluate the model's performance using a test dataset , We measure classification accuracy, or any other appropriate metric.



|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| High         | 0.98      | 0.97   | 0.97     | 300     |
| Low          | 0.86      | 0.90   | 0.88     | 300     |
| Medium       | 0.86      | 0.84   | 0.85     | 300     |
| accuracy     |           |        | 0.90     | 900     |
| macro avg    | 0.90      | 0.90   | 0.90     | 900     |
| weighted avg | 0.90      | 0.90   | 0.90     | 900     |

Accuracy Score: 0.9011111111111111

PS C:\flutter\flutter-apps\works\lib\user\_risk\_api\data>

Figure 3: data set report

The evaluation results indicate that the user risk classification model performs exceptionally well across all risk levels—High, Medium, and Low. making the model reliable for real-world deployment in risk-sensitive applications

Scientific Justification and Detailed Analysis:

#### **Overall Accuracy (90.1%):**

This is a strong performance for a multi-class classification problem. It means that the model correctly predicts 9 out of every 10 cases

#### **High-Risk Class:**

The model excels in identifying high-risk users, with **precision = 0.98**, **recall = 0.97**, and **F1-score = 0.97**. This is a critical strength, as correctly detecting high-risk users is essential for ensuring security and minimizing fraudulent behavior.

#### **Medium-Risk Class**

The model achieves solid results in identifying medium-risk users, with a precision of 0.86, recall of 0.84, and an F1-score of 0.85. These values indicate that the model is generally effective at detecting users who show moderate signs of risk.

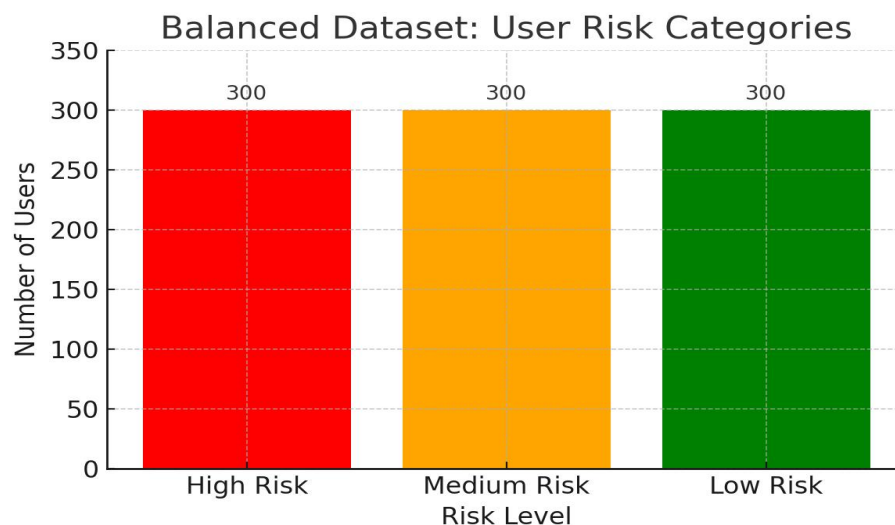
#### **Low Risk Class**

The low-risk class also demonstrates strong performance, with a precision of 0.86, a recall of 0.90, and an F1-score of 0.88. The high recall indicates that the model is highly effective at correctly identifying most low-risk users, which is valuable in maintaining a positive user experience

#### **Macro and Weighted Averages:**

Both macro and weighted averages for precision, recall, and F1-score are 0.90, which reflects balanced and fair performance across all classes, without bias toward any specific risk level.

**Support Size:** Equal support (300 users per class) ensures fair and balanced training and evaluation.



*Figure 4: user risk categorization*

## 3.2 Why Choose This Algorithm

### 1- High Accuracy:

Because it combines the results of multiple decision trees, it reduces errors and provides more accurate predictions than a single decision tree.

### 2-Reduces Over-fitting:

Using multiple trees based on random samples and features reduces the likelihood that the algorithm will learn very fine details in the training data alone.

### 3- Works with Large and Complex Data:

It can handle a large number of features and large data sets without requiring significant reduction or modification.

### 4- Good Handling of Missing Data:

Random Forest can handle some missing data or missing values better than many other algorithms.

### 5- Feature Importance Evaluation:

Provides a measure to determine which features have the greatest impact on decision-making, which is useful for analyzing and understanding data.

### 6- Robustness:

Because it relies on the aggregation of multiple decision trees, it is less sensitive to noise or random errors in the data.

### 7- Relatively easy to use:

It does not require complex parameter tuning compared to other algorithms such as neural networks.

## 3.1 Data Collection

The algorithm employs a database to gather comprehensive data on user interactions with the auction platform. Key data points include:

### **User Behavior:**

User interactions within the system, such as pressing buttons, performing various operations, entering data, or making decisions (such as placing a bid or confirming a purchase and report seller ), are recorded and analyzed in order to update their risk assessment.

### **Bidding History:**

The table contains detailed information about each bid, including Bid ID, Item ID, Bid Time, and User ID, as well as the Bid Amount, Title, Description, and Starting Price. It also displays the Image of the item, the Status of the bid, the Start Time and End Time of the auction, the Location,

and finally the user Name. These columns help provide a comprehensive and clear overview of all auction details.

**Payment Methods:**

Types of payment methods used (credit card -visa, paypal , cash on delivery ).

**fraud report :**

The number of times this account is reported is fetched from the database to be used in assessing the severity score.

### 3.2 Data Aggregation

User risk rating data is collected from a localhost environment using phpMyAdmin and a MySQL database. The system continuously collects and updates user data based on key evaluation criteria such as account authentication, bidding history, transaction history, and the number of fraud reports. Each time a user logs into the system, their risk rating is dynamically recalculated, ensuring that the rating remains accurate and up-to-date.

### 3.3 Risk Categorization

The algorithm categorizes users into risk levels based on defined criteria derived from their aggregated data. The risk assessment process involves:

**High Risk:**

Users with frequent failed payments, a low win rate, and inconsistent use of payment methods.

**Medium Risk and new user :**

Users with occasional payment , moderate win rates, and a tendency to use a single payment method have shown some issues.

Classifying new users as **medium risk** is considered a fair and balanced strategy in risk management. This approach ensures that unknown users are not overly trusted by avoiding their immediate placement in the low-risk category, which could pose potential security threats due to lack of behavioral data. At the same time, it prevents unfair judgment by not assigning them to the high-risk category without any evidence of suspicious activity. By placing new users in the medium risk level, the system maintains a cautious yet flexible stance, allowing for a dynamic reassessment of their risk classification as more behavioral data becomes available over time. This decision reflects a practical trade-off between trust and caution when dealing with limited information.

**Low Risk:**

Users demonstrate a high win rate with low-value bids, a successful payment history, and consistent use of reliable payment methods.

### 3.4 Payment Method Recommendation

Based on the user's risk category, the algorithm suggests appropriate payment methods to enhance security:

High Risk: Recommend secure payment options with user protection, such as visa .

Medium Risk: Encourage the use of credit cards with fraud protection features, emphasizing security measures provided by financial institutions like visa .

Low Risk: Offer a broader range of payment options, including direct bank transfers and digital wallets, to provide flexibility like visa ,palpay and cash on delivery (all options are allowed for this category ).

we use **Luhn algorithm** that check if the user enter validate visa card details

It is one of the most widely used and reliable algorithms for verifying bank card numbers, such as Visa cards. It is characterized by its simplicity and high efficiency in detecting errors resulting from incorrect input, making it an effective tool for ensuring the validity of the number before proceeding with any financial transaction. The algorithm relies on performing simple mathematical operations on numbers: the number is passed from right to left, and every second digit is doubled. If the result exceeds 9, 9 is subtracted from it. All values are then added together. If the final result is a multiple of 10, the number is considered valid. This method significantly helps improve the reliability of input data and enhances the quality of verification in payment applications and digital transactions [9]

### 3.5 Categorization

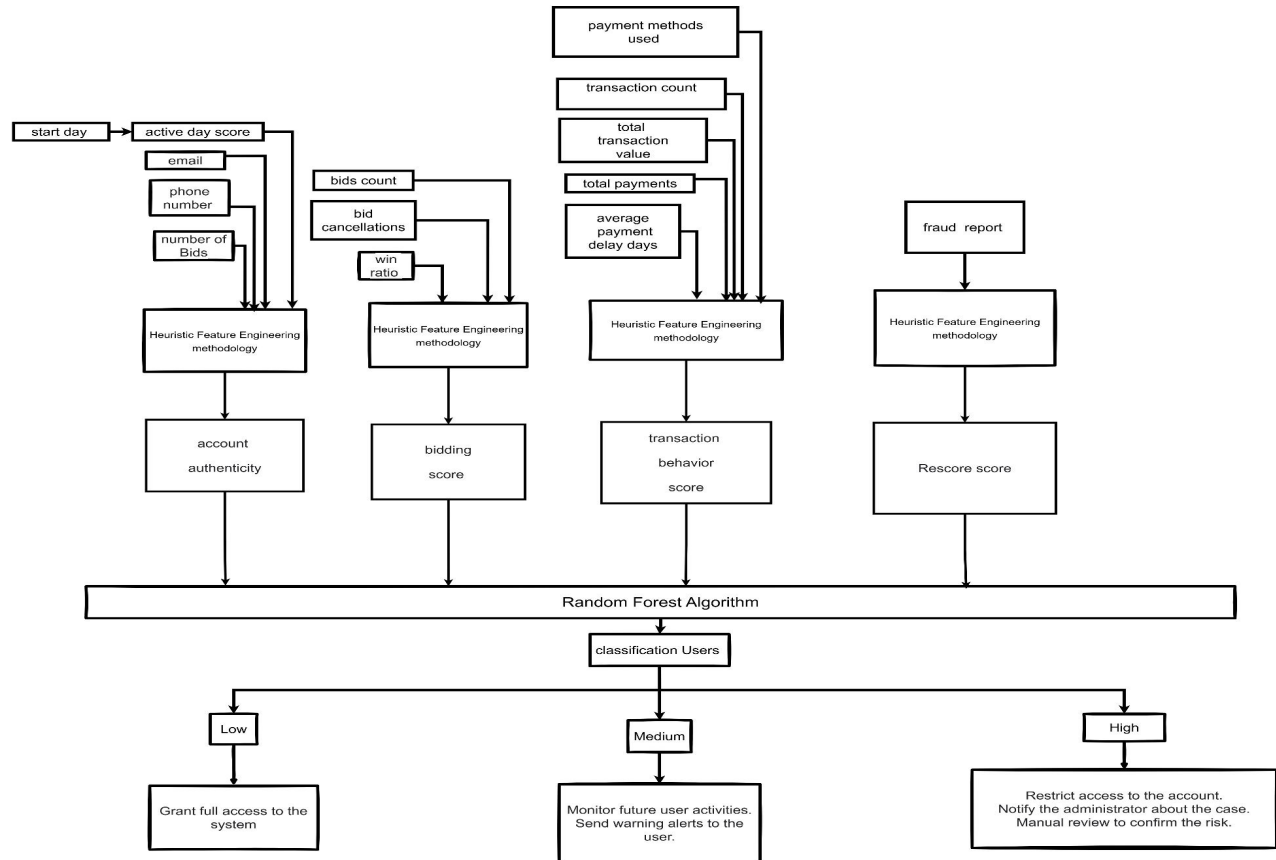


Figure 5: Categorization Process Flow

The partial weights for each feature within the composite indicators (such as `account_authenticity` or `bidding_score`) were chosen based on the Heuristic Feature Engineering methodology, which is very common in cases where we have a clear understanding of user behavior within the system. The weights were distributed in a way that reflects the impact of each behavior on the overall reliability, while ensuring that the combined values do not exceed a threshold of 1.0 to maintain consistency in training.

In this project, the Heuristic Feature Engineering methodology was used to improve data representation and reduce the number of features entered into the prediction model. This methodology relies on integrating field experience and an understanding of user behavior within the system to design composite indicators with statistical and behavioral meaning. Instead of directly using the primary features (such as email address, phone number, duration of activity, and number of bids), we reformulated them into a single indicator representing "account authenticity." Similarly, features related to bidding activity were combined into the "Bidding Score" indicator,

and transaction characteristics into the "Financial Worth" indicator. This approach aims to simplify the model structure and reduce noise resulting from duplicate features or their weak correlation with the outcome, enhancing the model's generalization ability and prediction accuracy. This methodology is particularly suitable in situations with limited data or where a deep understanding of the domain is present, such as in electronic auction systems [10]

## 1. Account\_Authenticity

The account\_authenticity score represents how authentic and trustworthy a user's account appears. It is calculated by combining four sub-scores using  $\diamond$  Heuristic Feature Engineering, where values are manually assigned based on logical rules and domain knowledge, rather than automated algorithms.

If the user has an email address, they receive 0.2 (email\_score = 0.2), since providing an email is a basic but essential verification step.

If the user has a registered mobile number, they also receive 0.2 (mobile\_score = 0.2).

The number of days since account creation (days\_active) contributes as follows:

- $\geq 365$  days  $\rightarrow 0.4$
- $\geq 180$  days  $\rightarrow 0.3$
- $\geq 90$  days  $\rightarrow 0.2$
- $\geq 30$  days  $\rightarrow 0.1$

Additionally, if the user has placed at least 5 bids, they receive another 0.2 (bidding\_score\_contribution = 0.2) to indicate early engagement.

All these scores are summed and capped at 1.0 using min() to ensure normalization. This engineered feature simplifies multiple behavioral signals into one consistent metric representing account legitimacy.

## 2. Bidding\_Score

The bidding\_score reflects the user's behavior and success in auctions, also crafted through  $\diamond$  Heuristic Feature Engineering.

First, the number of bids (bids\_count) is translated into a score:

$\geq 20$  bids  $\rightarrow 0.4$

- 10–19 bids  $\rightarrow 0.3$
- 5–9 bids  $\rightarrow 0.2$
- 1–4 bids  $\rightarrow 0.1$

Next, the user's win ratio (win\_ratio) is considered if they have placed any bids:

- $\geq 0.7 \rightarrow 0.4$
- $0.4-0.69 \rightarrow 0.3$
- $0 \rightarrow 0.2$

To penalize risky or unreliable behavior, the number of bid cancellations (bid\_cancellations) is deducted:

- $\geq 5$  cancellations  $\rightarrow$  subtract 0.2
- $2-4$  cancellations  $\rightarrow$  subtract 0.1

Finally, the total bidding\_score is clamped between 0.0 and 1.0 to avoid overrepresentation. This feature balances user activity, success, and discipline in bidding behavior.

### 3. Transaction\_Score

The transaction\_score captures the user's financial reliability and diversity of payment behavior, computed using  $\diamond$  Heuristic Feature Engineering by converting raw metrics into fractional scores. Transaction count (transaction\_count) contributes:

- $\geq 20 \rightarrow 0.4$
- $10-19 \rightarrow 0.3$
- $5-9 \rightarrow 0.2$
- $1-4 \rightarrow 0.1$

Total transaction value (total\_transaction\_value) adds:

- $1000 \rightarrow 0.3$
- $500-1000 \rightarrow 0.2$
- $100-499 \rightarrow 0.1$

Payment success ratio (calculated as successful\_payments / total\_payments) affects the score:

- $\geq 0.9 \rightarrow 0.3$



- $0.7-0.89 \rightarrow 0.2$
- $0.5-0.69 \rightarrow 0.1$
- $< 0.5 \rightarrow -0.1$  (penalty for poor reliability)

Payment method diversity (payment\_methods\_used) also adds:

- $\geq 3$  methods  $\rightarrow 0.2$
- 2 methods  $\rightarrow 0.1$

All components are added and normalized between 0.0 and 1.0. This composite score helps model financial trustworthiness in a simplified and scalable way.

## 4. Fraud\_Reports

The fraud\_reports feature indicates how many times the user has been reported for fraud. While this value is passed directly to the model without conversion, its internal interpretation is guided by a scoring reference built using Heuristic Feature Engineering.

Internally, the severity of fraud reports is mapped as:

- 0 reports  $\rightarrow 0.4$
- 1 report  $\rightarrow 0.3$
- 2 reports  $\rightarrow 0.2$
- 3 reports  $\rightarrow 0.1$
- $\geq 4$  reports  $\rightarrow 0.0$

This mapping reflects a logical degradation of trust as fraud complaints increase. Even though the raw number is sent to the model, it is conceptually treated as a trust penalty based on real-world intuition.

## 3.6 Algorithms used in our App:

### 3.6.1 : Percentile Algorithm

First Validity: Classifying Auction Prices Based on User Rating Using the Percentile Algorithm:

The percentile algorithm is a statistical method used to divide data into equal parts based on percentages. In this project, I divided the data into three parts using percentiles as follows:

The first part (Low): From the beginning of the data to the 33rd percentile (P33)  $\rightarrow$  represents low

values.

The second part (Medium): From P33 to P66 → represents medium values.

The third part (High): From P66 to the end of the data → represents high values.

To implement this, the data was sorted in ascending order, and then the desired percentile locations were calculated using the equation:

$$[k \times (n+1)] / 100 = r \quad \text{where} \quad P_k = X_r$$

- $P_k$  : The value of the desired percentile (e.g., 33rd percentile or 66th percentile).
- $k$ : The percentile number (for example, 33 or 66).
- $n$ : The total number of data points.
- $X_r$  : The data value at position  $r$  after sorting the data in ascending order.

In this application, we used this classification to control the display of items based on the user's risk level:

A high-risk user: Only items in the first part (up to the 33rd percentile) are shown, meaning the lowest prices or risk.

A medium-risk user: Items in both the first and second parts (up to the 66th percentile) are shown, meaning the low and medium prices.

Low-risk user: All items are shown to them without exception, from lowest to highest [11]

(note that this algorithm will add in landing page , search page and category page )

### 3.6.2 Second Authority:

Classifying payment methods based on the user's classification based on reference:

Based on the user's classification, the permitted payment methods are determined through the Place Bid interface in the program, as follows:

High Risk: Recommend secure payment options with user protection, such as visa due to this [12] that Explains the layers of security Visa provides, including fraud protection and secure authorization processes.

Medium Risk: Encourage the use of credit cards with fraud protection features, emphasizing security measures provided by financial institutions like visa and apple pay .

due to this [13] its content : Details how Apple Pay uses tokenization and device-based

authentication to protect user data.

Low Risk: Offer a broader range of payment options, including direct bank transfers and digital wallets, to provide flexibility like visa ,PayPal , apple pay and cash on delivery (all option is allowed for this category ).

This category has been allowed to have a wider range of payment methods based on the fact that the remaining payment methods are less secure than those mentioned previously, because the user is classified as more secure than others.

that also depend on some reference [14] that Describes PayPal's dispute system, highlighting the risk of chargebacks and potential abuse by users [15] that Discusses the operational risks of cash-on-delivery, including non-payment, fraud, and logistics delays.

### **3.7 Permissions of each user based on his classification**

#### **3.7.1 Bidding Restriction Based on Item Price:**

After research and analysis, the bidding system was modified to be based on the item's base price, and two bidding methods were adopted based on the price category:

First: If the item's price is less than 100:

Bidding is permitted through two options:

Fixed Bidding with Three Specific Values:

The user can choose one of the following values to add to the original price: 5, 10, or 15.

These smaller values were chosen to accommodate the item's low price and to keep bidding accessible and logical for the user.

Final Bid (Buyout) to Close the Auction:

The user is allowed to set the highest possible price to close the auction directly, provided that this value is higher than 40% of the original price, and that the increase is reasonable and not excessive.

Second: If the item's price is equal to or greater than 100:

Bidding is permitted in the same two ways, with the increment values adjusted:

Fixed Bidding with Three Higher Values:

The user can choose an increment of 20, 30, or 40 over the original price.

Higher values were adopted to avoid boredom and speed up the auction process, as reported in some studies on user behavior.

Final Bidding (Buyout):

Under the same conditions, the auction may be closed at a price greater than 40% of the original price, provided that the increment is reasonable and consistent with market value

Although there is no established academic standard setting a maximum closing bid of 40%, the

literature suggests that the Buy-It-Now (BIT) price should be set as a positive threshold that users are likely to choose, and that it depends primarily on the nature of the market and the costs of participation. In general, studies prefer that the price be reasonable compared to the expected auction price, and strike a balance between encouraging the user to make an immediate decision and not raising the price excessively [16]

### **3.7.2 Explanation of Bidding Limits Based on Risk Levels:**

eBay puts bidding restrictions on new, or unethical accounts to control fairness and auction integrity. High Risk accounts are normally limited to one bid per day, Medium Risk accounts may place up to three bids daily, while Low Risk accounts normally have no such restrictions. These limits are any of internal platform policies designed to prevent fraudulent bidding behavior. However, these restrictions are implemented practically within the platform to ensure auction fairness.

This definition is supported by user discussions and experiences shared on the official eBay Community forums, which give into how these bidding restrictions practice [17].

### **3.7.3 Category Access Control Based on User Risk Classification :**

Based on a user's classification, access to the various categories of products offered in auctions is determined according to the risk level of each category. User access is restricted as follows:

Users "High" Risk can access and browse specific categories, namely "Books," "Fashion," and "Toys." Other categories are not available.

Users "Medium" Risk can only access categories rated "Medium" and "High", including "Books," "Fashion," "Toys," "Musical Instruments," "Home Appliances," and "Furniture." They cannot interact with categories rated "Low." This is to ensure compliance with security standards in higher-value auctions.

Users "Low" Risk can access all categories without restrictions, including "Low," "Medium," and "High" risk categories, to encourage interaction and participation.

### **3.7.4 Determining the number of auction entries concurrent based on the user's Risk Level :**

To regulate activity on the platform and ensure a safe and fair bidding environment, the maximum number of auctions a user can participate in simultaneously is determined based on their risk level. The mechanism is as follows:

- Users classified as "Low Risk" are allowed to participate in up to 10 auctions at the same time.

-Users classified as "Medium Risk" are allowed to participate in up to 3 auctions.

-Users classified as "High Risk" are allowed to participate in only one auction at a time.

This policy aims to reduce potential risks, balance opportunities, and encourage users to improve their rankings [17].

### **3.7.5 Restricting Auction Creation Based on Risk Level**

A new feature has been added to the system to enhance security and reduce fraud. Only users categorized as "low" and "medium" risk are allowed to create new auctions.

## Chapter 4: System Analysis and Design

### 4.1 Product Description

The Auction that we will design is an interactive and innovative platform that aims to provide a safe, easy and local auction experience for users. The most important feature of this system is user classification, where participants are classified based on their behavior, such as payment history, level of interaction in previous auctions and presence of any suspicious activity . This classification will enhance security within the system and ensure that participants are real and trustworthy individuals and helps monitor activities and prevent fraud. Users can view a wide range of products including household appliances, electronics, clothing, in addition to luxury goods such as cars and jewelry. This diversity allows users from different economic classes to participate in auctions, which enhances interaction within the platform and provides various opportunities for users to stay informed about the offers that suit them. This will make the system combine security, convenience, diversity, and transparency, making this platform a unique experience for users in the world of online auctions.

#### 4.1.1 System Objectives

The Objectives for our project are:

- Offering different products from many Users.
- The user chooses one of the auctions as he wants.
- When the auction starts the page will show product description, photos for a product, and payment , time left, current bid, bid amount, place bid, product specifics .
- There is a privacy of the customer, the other will not see you, the name is hidden.
- Auction depends on competition between bids, and the one that gives a high price wins the auction.
- The order will be delivered to the customer by the shipping company.

#### 4.1.2 System Main Features

##### Main features of the system

In this project, an android application will be developed and designed, and here are the main features:

1. Registration (sign up), Login and Logout.
2. Home page, about us and contact us.
3. Bot User.

Main features of users

○ Admin main features

Auction executive: View and supervise all auctions within the system.

Category executive: Add, edit or delete categories in the system.

Product executive: Access a complete list of products, including products in current and completed auctions.

Edit App Data: Update details about the app, such as descriptions or other related information.

delete and show risk level user in the system

○ user main features :

- 1- the potential to add more than one product using his home page
- 2- the potential to establish the auction begin and end
- 3- the potential to make conversation with users and chat bot
- 4- the potential to forage all categories in main page
- 5- the potential to add item to Favorite page
- 6- the potential to add new offer and price

### 4.1.3 Operating Environments

A smartphone with a recent version of the operating system, such as iOS 11 or later, or Android 5.0 or later.

### 4.1.4 Constraints

❖ Internet restrictions:

Users must be connected to the Internet to access and use the Site appropriately.

❖ Device compatibility:

The system must be available and work on various mobile devices (Android phones, iPhones, and tablets).

❖ Limits on user acceptance:

The system's success depends largely on its acceptance by users.

❖ Data Entry restrictions:

The user must enter the required data correctly (such as the identity of the User, Phone number, private email, etc.) into the system.

❖ Privacy restrictions:

The system must maintain the confidentiality of user data and adhere to privacy laws.

### 4.1.5 Functional Requirements

#### **1-Users must create an account and log in**

1.1. The system must display a registration form that contains the following fields: name, password, email, and phone number. The login must contain username and password

1.2. The authentication process must take place in a short time, estimated at a fraction of a second

1.3 The system must verify the validity of the entered information and store it securely.

#### **2-The user should be able to easily buy, sell and bid on products and services.**

2.1. The system should enable medium or low risk users and Admin to create new auctions by entering the product name, description, images, and starting price, and also allow specifying the start and end time of the auction.

#### **3- Users should be able to browse open auctions.**

3.1. The system should display a list of all open auctions dynamically in the user interface.

3.2. The system must clearly display the auction details.

3.3. The page should display a clear description and image for each product entered by system administrators or vendors.

**4- Users should be able to see a clear description and images of products and services before making a purchase decision.**

4.1. The system should display a clear description and images of each product or service entered by vendors or system administrators.

**5- Users must have instant chat to ask questions and receive responses in real-time.**

5.1. The system should support an instant chat feature that enables users to ask questions and inquire about products and auctions.

**6- Users should receive notifications when auctions start or end, and when bids are received or declined.**

6.1. The system provides multiple notifications including push notifications, email, and in-app notifications for auction-related events.

**7-The user should be able to choose their preferred and convenient payment method from a variety of local and global options, such as credit cards, e-wallets, or bank transfers.**

7.1. The system supports integration with global and local payment services to facilitate payment processes.

#### **4.1.6 Non-Functional Requirements**

##### **1. Usability**

The system should support a user-friendly environment, characterized by a simple and intuitive user interface, easily accessible information

- The user interface (UI) across all platforms maintains consistency in design elements, navigation patterns, and terminology.
- Navigation within the application is intuitive and straightforward, allowing users to easily access different features and sections. Clear and descriptive labels should be used for navigation menus and buttons.
- The application's user interface is responsive and adapts to different screen sizes and orientations.

##### **2. Reliability**

In the event of a major system failure, full functionality should be restored within one to two hours.

- Automatic recovery for simple errors takes place within approximately 30 seconds.



- The system must copy important data across multiple servers to ensure its availability.
- The system contains backup servers ready to take over operations if the primary server fails.
- System health is constantly monitored with alerts sent if problems occur.
- Notify users in the event of scheduled maintenance or unexpected downtime.
- Verify the validity of user data and ensure its safety.
- Maintain backup copies of data off-site.

### **3. Security**

- Control access to documents based on user roles and categorization .
- Establish transparent privacy policies and implement approval mechanisms for data use.
- Integration with payment gateways and application of PCI DSS standards to handle payment information.
- Record and monitor user activities, system events and security incidents.
- Store records securely in accordance with regulatory requirements.
- Restricting physical access to data centers, servers and networking equipment.

### **4. Performance**

- Speed of searches: Search results are displayed quickly.
- Payment processing speed
- Deal with a large number of concurrent users: The system deals with concurrent users efficiently.
- Stable performance during peak times: The system maintains its performance even during heavy use.
- API response speed: access time does not exceed the required limit.

### **5. Maintainability**

- Organizing the software base into standard components for ease of maintenance and updates.
- Place documentary comments on the code to make it easy to understand and modify.
- Follow industry coding standards to ensure consistency and readability.
- Maintain a version control system to track changes and manage branches.

- Regularly update dependencies to collect bug fixes, security and performance improvements.
- Providing comprehensive technical documentation including system architecture, APIs, data models, and deployment procedures.
- Maintain change logs to document updates, fixes, and improvements.
- Use continuous integration tools to accelerate build, testing and deployment processes.
- Regularly update components, libraries and dependencies to patch security vulnerabilities.
- Conduct periodic security audits to proactively identify and address vulnerabilities.
- Implement regular backups of data to protect it from loss or damage.
- Document institutional knowledge and lessons learned to facilitate knowledge transfer within the team. Infrastructure planning to ensure scalability and meet increased traffic and new features.

## 4.2 Functional Decomposition

### 4.2.1 Actors

The table provided offers a cogent delineation of the roles and responsibilities of each actor within application, elucidating their interactions and relationships with other actors in the system:

*Table 1 :Actors in system*

| Actors                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>User</b>                 | A user on the platform can perform both buyer and seller roles. They can browse available products and services, submit bids on open auctions, and complete purchases. Additionally, they can create and upload their own auctions by providing product details (title, description, images, pricing, and auction timing). The user is responsible for managing their own listings and interacting with other users through bidding or messaging. |
| <b>Admin</b>                | The person responsible for managing the entire system, which includes supervising auctions, managing users, ensuring that policies are implemented correctly, and ensuring smooth operations. He also supervises solving technical problems and communicating with uses                                                                                                                                                                           |
| <b>Payment Gateway</b>      | An external system used to securely process financial payments, such as credit cards, e-wallets, or bank transfers. It helps complete financial transactions between the buyer and the seller quickly and securely.                                                                                                                                                                                                                               |
| <b>Shipping Provider</b>    | A company concerned with conducting shipping calculations, managing logistics, and tracking sold products, cooperating with the platform to ensure the provision of real shipping rates and tracking of packages to users in various regions.                                                                                                                                                                                                     |
| <b>Notification Service</b> | A service that sends notifications to users (such as payment notifications, bids, auction changes) , in-app push notifications                                                                                                                                                                                                                                                                                                                    |
| <b>chat bot</b>             | The chat bot handles user inquiries and solves problems It works to provide effective and quick solutions to ensure customer satisfaction.                                                                                                                                                                                                                                                                                                        |

## 4.2.2 Use Cases (for All System):

### **Use Case: Registration**

Users register on the platform by providing personal details such as name, phone number, email, and password. The system verifies this information and sends a verification code generated by the API to ensure its accuracy. Upon successful registration, they are granted access to explore and participate in auctions.

### **Use Case: log in**

Pres-registered users can access their accounts by logging in with their registered phone number and password. The system verifies the provided credentials and grants them access to the platform. If their password is forgotten, users can retrieve it through a forgotten password verification process that includes the phone number. Once logged in, users can browse auctions, place bids, and use other platform features.

### **Use Case: Creating Auctions**

Users can list products by creating new auctions and providing details such as item description, starting price, duration, and images. The system verifies the completeness of the listing and ensures compliance with the auction policies before publishing it.

### **Use Case: Product Search**

users search for products to find auctions that interest them, and the platform retrieves and displays results based on relevance, including auction details, bid status, and product images

### **Use Case: Bidding**

Registered users participate in live auctions by bidding on items. The system ensures real-time updates, notifying users if their bid is exceeded or if they are the highest bidder. users can set automatic bid increments within a specified range to stay competitive.

### **Use Case: Payment Processing**

Upon winning an auction, users are prompted to complete their payment. The platform supports multiple payment options, including credit/debit cards, bank transfers, and digital wallets. The system securely processes transactions and verifies successful payment

### **Use Case: Send Notifications**

The platform sends real-time notifications to users such as bid status updates, auction end, payment confirmations, shipment tracking, and status changes.

### **Use case: edit User Profiles**

Users can edit their profiles to update personal details such as their name, contact information, and payment preferences. The platform ensures that all updates are validated and securely stored.

### **Use case: Generate Reports**

Admin can generate reports on platform usage and auction outcomes.

### **Use case: Browse Auction :**

The users can browse through various products and services listed in the auction, and view product details such as description, bidding status, and remaining time. Users can also track the price trends and place bids on their chosen items.

#### **Use Instant Chat :**

The instant chat feature enables communication between the users in real-time.

### **Use Case Description:**

#### **Use Case Login:**

The login process for an Mazadco application remains active from the moment the application is opened until the login process is completed, as shown in Table 3

*Table 2 : Use Case: login*

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case Name     | Login to Auction                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Actors            | users, Admin, Notification Service                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description       | To describe the process by which users (securely log in to the auction system)                                                                                                                                                                                                                                                                                                                                                                                             |
| Preconditions     | The user must have an active account.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Postconditions    | - Successful login grants the user access to their respective dashboard.                                                                                                                                                                                                                                                                                                                                                                                                   |
| Main Flow         | <p>The user provides login credentials (username and password)</p> <p>The system validates the credentials against the database.</p> <p>3. If valid, the user is authenticated and granted access to their respective dashboard.</p> <p>4. The system updates the login status and records the login attempt and Update to user risk level</p> <p>5. Notifications are sent to inform the user about the successful login.</p>                                             |
| Alternative Flows | <p>Invalid Credentials:</p> <ul style="list-style-type: none"> <li>- If the credentials are incorrect, the system displays an error message.</li> <li>- The user can attempt to log in again or reset their password.</li> </ul> <p>Account Locked:</p> <ul style="list-style-type: none"> <li>- After multiple failed login attempts, the account may be temporarily locked.</li> <li>- The system notifies the user via email and the support team is alerted</li> </ul> |

## Use Case: Placing a Bid

The process of placing a bid in the Mazadco application allows users to participate in auctions by submitting their bid amounts, which remain valid until the auction ends, as shown in Table 4.

Table 3 : Placing a Bid

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case Name     | Placing a Bid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Actors            | users, Admin, Payment Gateway, Notification Service, Support Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Description       | A user places a bid on a product in an auction. Admin verifies the bid and updates the auction. Notifications are requested to the interested party, and if the user wins, payment and shipping processes are brought about                                                                                                                                                                                                                                                                                                                                                                                  |
| preconditions     | To place a bid, the user should first be logged into the system. The auctions must be open and active, establishing that they are presently accepting bids. Also, the bid placed by the user must be more than the current highest bid, if one exists. Last, the user should have sufficient funds available in their account to place the bid. If any of these conditions are not accepted, the bid cannot be successfully placed.                                                                                                                                                                          |
| postconditions    | The bid is recorded, and the auction is updated with the new highest bid. Notifications are requested to the user about the update.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Main Flow         | user selects an open auction category.<br>The system displays auction details (product, current bid, and auction end time).<br>The user enters a bid amount and clicks "Place Bid" and choose the amount of increasing<br>System checks if the bid is valid (more than the current highest bid and sufficient funds are available).<br>user confirms the bid.<br>The system records the bid and updates the auction information.<br>Notification Service sends confirmation<br>Payment Gateway prepares for activity processing if the bid wins.<br>The system updates the auction with the new highest bid. |
| Alternative Flows | If invalid:<br>System asks user to add fit price                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Exceptions        | Network Failure:<br>- If the system encounters a network issue, the bid is not placed, and the user is asked to try again later.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Use Case : Account Authenticity

The account authenticity process ensures that all accounts on the Mazadco platform are verified and secure to maintain trust and minimize fraudulent as shown in Table 5

*Table 4 : Account Authenticity*

|                    |                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase Name       | Account Authenticity                                                                                                                                                                                                                                                                                                                                |
| Actors             | user, Admin, Notification Service                                                                                                                                                                                                                                                                                                                   |
| Trigger            | A user builds or updates their account, beginning the verification step.                                                                                                                                                                                                                                                                            |
|                    | The system verifies the trustworthiness of accounts through email, phone, or ID verification, account age, and profile integrity. This ensures validity and reduces fake activities.                                                                                                                                                                |
| Pre-Conditions     | <ul style="list-style-type: none"><li>-user must have an account in the system.</li><li>- Verification methods(email, phone, ID) must be possible.</li><li>- Profile information must be usable.</li></ul>                                                                                                                                          |
| Flow of Events     | <ol style="list-style-type: none"><li>1. user annals or updates their account.</li><li>2. System prompts the user to replete profile information</li><li>3. user provides verification</li><li>4. Notification Service tells the user about their verification status.</li><li>5. chatbot helps users with account verification problems.</li></ol> |
| Data               | <ul style="list-style-type: none"><li>-User minutiae (name, email, phone,).</li><li>- Account establishment date.</li><li>- Profile minutiae</li><li>- Verification status (email verified, phone verified, ID verified).</li></ul>                                                                                                                 |
| Post-Conditions    | - user accounts are flagged as verified .                                                                                                                                                                                                                                                                                                           |
| Alternative Flows: | If verification is unsuccessful via email, the system helps the user to retry or choose a backup verification method(phone or ID).                                                                                                                                                                                                                  |
| Exceptions:        | 1-Email, phone, or ID verification fails because of technical problems or incorrect inputs.                                                                                                                                                                                                                                                         |

## Use Case: Transaction behavioral and User Classification

The transaction behavior use case monitors and records the behavior of users during the buying and selling process within the Mazadco application, tracking the status of each transaction as shown in

*Table 5 : Transaction behavioral*

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case Name     | Transaction behavioral and User Classification                                                                                                                                                                                                                                                                                                                                                                                 |
| Actors            | Admin, Notification System, user                                                                                                                                                                                                                                                                                                                                                                                               |
| Description       | Evaluate a transaction behavior and classify them into risk levels (Low, Medium, High) according to algorithm                                                                                                                                                                                                                                                                                                                  |
| preconditions     | The user account exists, and all data can be retrieved.                                                                                                                                                                                                                                                                                                                                                                        |
| postconditions    | The user is classified into a risk category<br>- user profile is updated with the risk level.<br>- Admin is notified for high-risk users.                                                                                                                                                                                                                                                                                      |
| Main Flow         | 1.retrieve account authenticity data to check if the account is verified<br>2. calculates and assigns a score for account authenticity based on the verification status<br>3.retrieve bidding, transaction behavior, payment, and fraud indicator data<br>4.assigns scores to each factor based on predefined conditions<br>5. all data enter to algorithm and categorized user and according to his risk level add permission |
| Alternative Flows | if user don't have scores to make classification its take a new users status and take medium risk permission                                                                                                                                                                                                                                                                                                                   |
| Exceptions        | If the Risk Score is calculated but fails to meet the predefined thresholds due to system error                                                                                                                                                                                                                                                                                                                                |



## Use case : Cancellation Rate

The cancellation rate use case tracks the number of canceled transactions or bids in the Mazadco application, as shown in Table 7

*Table 6 : Cancellation Rate*

| Test case Name    | Cancellation Rate                                                                                                                                                                                                                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors            | Notification System, user                                                                                                                                                                                                                                                                                 |
| Description       | This use case evaluates the cancellation rate of auctions. The system tracks cancellations, calculates the cancellation rate,.                                                                                                                                                                            |
| Preconditions     | Cancellation reasons are recorded in the system                                                                                                                                                                                                                                                           |
| Post conditions   | A cancellation report is generated and sent to the admin.                                                                                                                                                                                                                                                 |
| Main Flow         | <ol style="list-style-type: none"><li>1.Retrieve Auction Data</li><li>2.Identify Cancellations</li><li>3.Generate Cancellation Report</li><li>4. sends the cancellation report to the admin for review</li><li>5.Notification System sends them a notification with the reason for cancellation</li></ol> |
| Alternative Flows | If the cancellation report fails to generate logs an error and notifies the admin for resolution                                                                                                                                                                                                          |

### 3.2.3 Use Cases Diagram:

The Table 10 that is shown below is about the use cases diagram of our system:

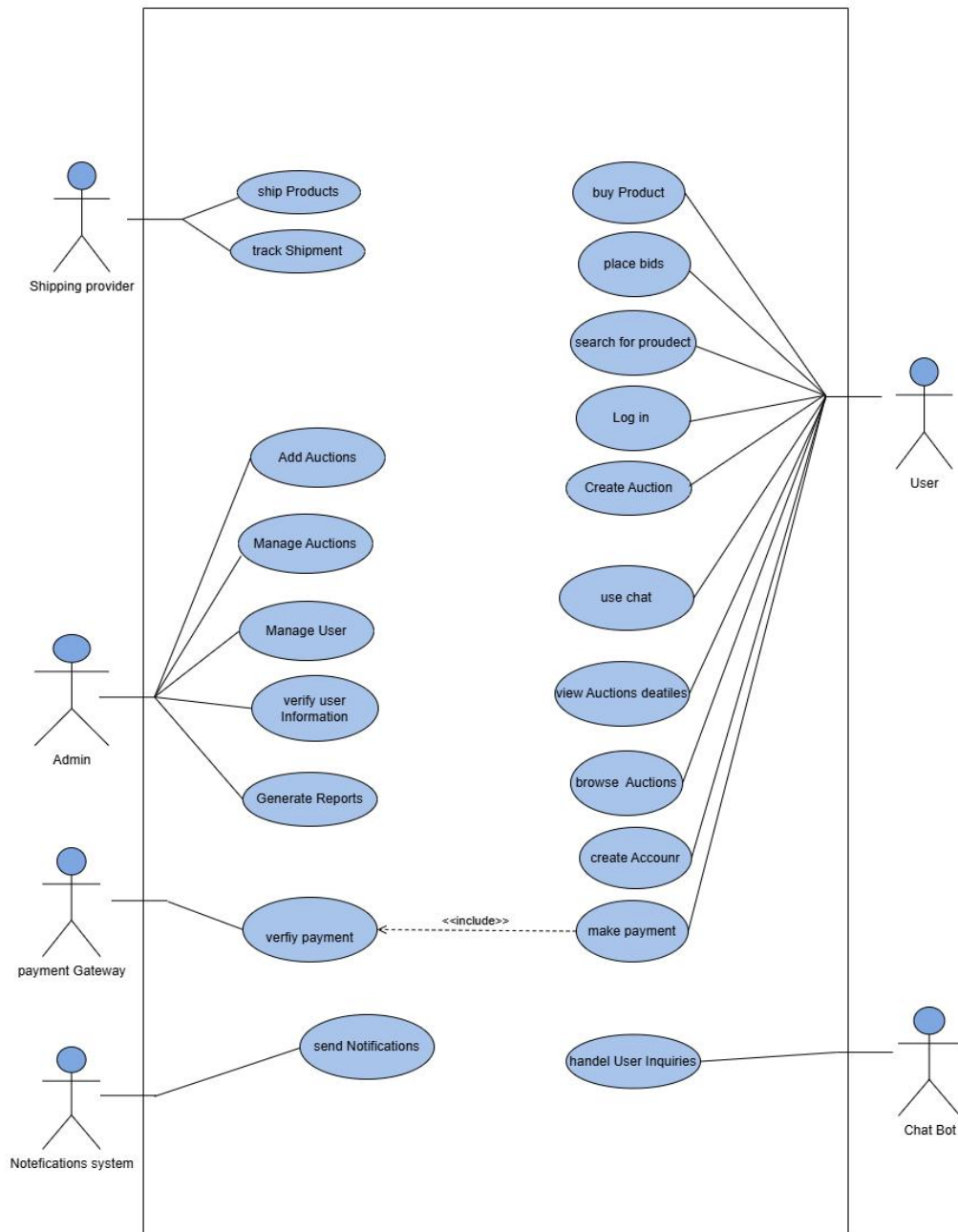


Figure 6 :Use Case Diagram

## 4.3 System Models

### 4.3.1 Class Diagram

The class diagram provides a static representation of an application, illustrating the various types of objects within the system and their interrelationships. Each class is composed of its respective objects and may inherit properties and behaviors from other classes. This diagram serves multiple purposes: facilitating the visualization, description, and documentation of different facets of the system, as well as aiding in the construction of executable software code shown at the Table 11

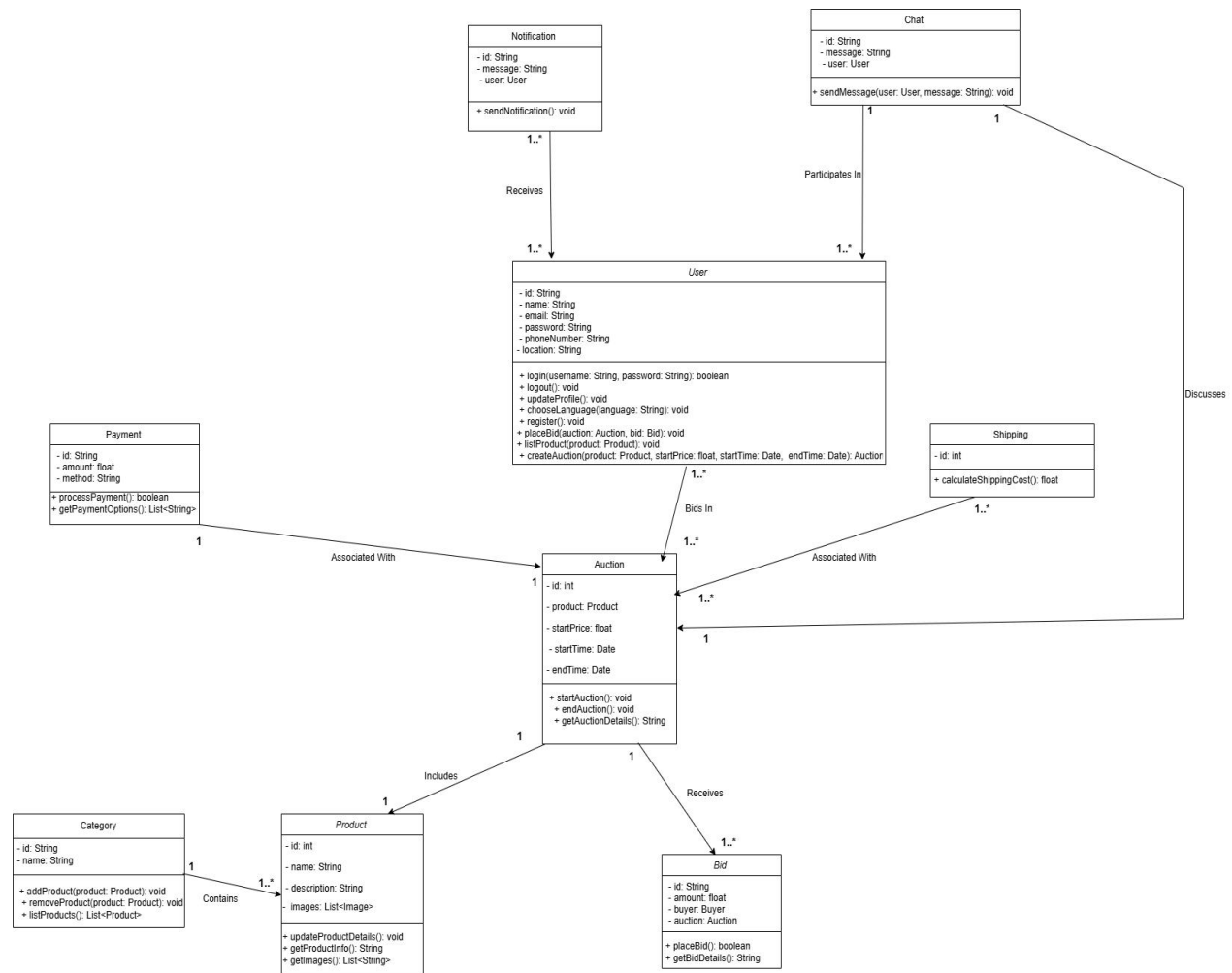


Figure 7 : Class Diagram

### 4.3.2 Sequence Diagram

A sequence diagram is a type of UML (Unified Modeling Language) diagram that depicts the interactions and flow of messages between objects or components in a software system. It shows the order in which these interactions occur, and helps to visualize the flow of control and data between the objects. Sequence diagrams are useful for modeling and analyzing complex systems and processes, and can help to identify potential problems or inefficiencies in the system design. They are also used to communicate design information between developers, stakeholders, and user

#### Sequence Diagram: Account Authenticity

A Sequence Diagram for Account Authenticity illustrates the flow of interactions between users, systems, and services involved in verifying the authenticity of an account

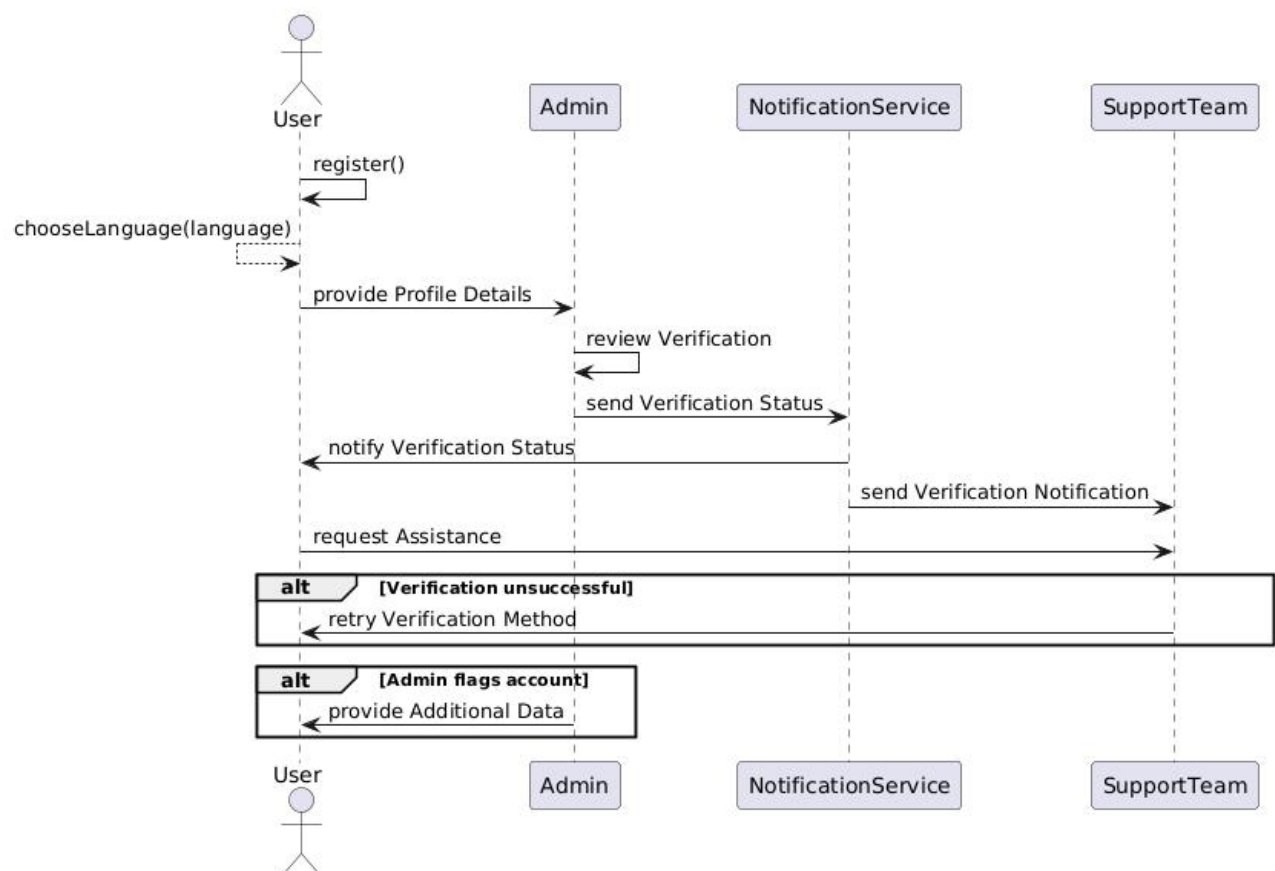


Figure 8 : Account Authenticity

## Sequence Diagram: Transaction behavioral

to illustrates how classification users on system according to many factors retrieve from transaction and behavioral

**Use Case: Transaction Behavioral and User Classification**

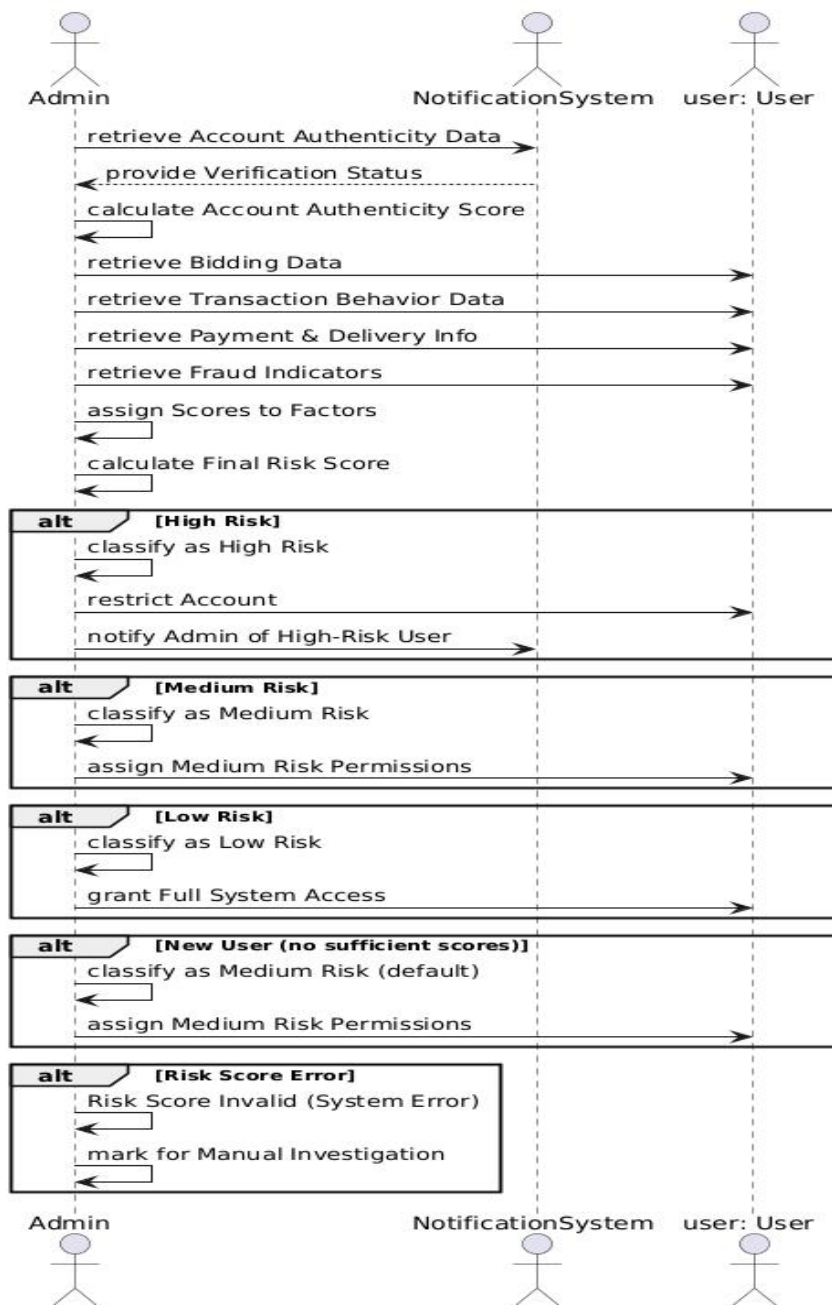


Figure 9 :Transaction Behavioral

## Sequence Diagram: Cancellation Rate

represents the interaction between different components or actors involved in the process of cancellation

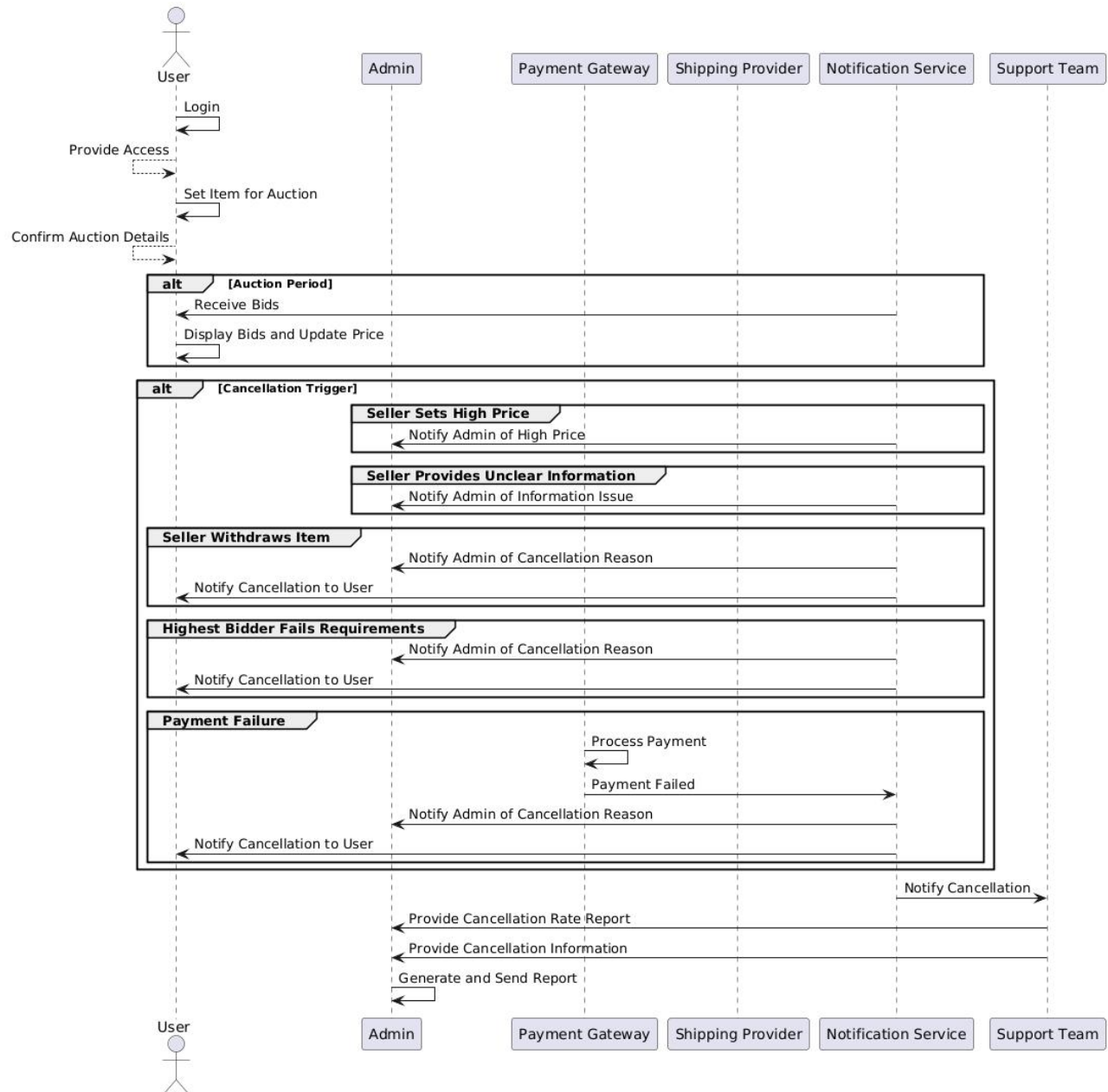


Figure 10 :cancellation Rate

### 4.3.3 Activity Diagram

An activity diagram is a type of UML (Unified Modeling Language) diagram that visually represents the workflow or activities within a system or process. It shows the sequence of actions, decision points, and concurrent activities, illustrating how different activities are carried out from start to finish

#### Activity Diagram :Cancellation Rate:

illustrates the process flow of calculation in a system. It shows the key steps

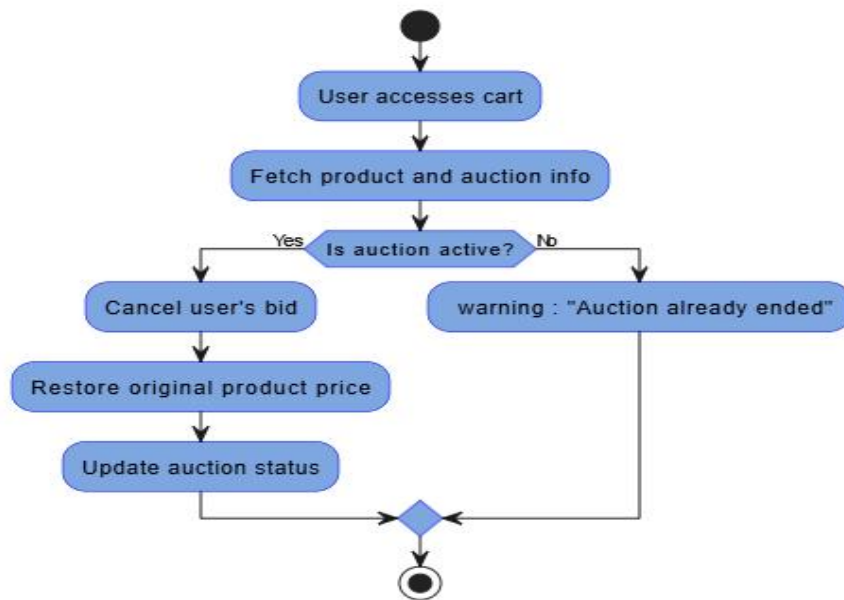


Figure 11: Cancellation Rate

## Activity Diagram :Account Authenticity

illustrates the flow of actions taken to verify the legitimacy of a user's account during the login process.

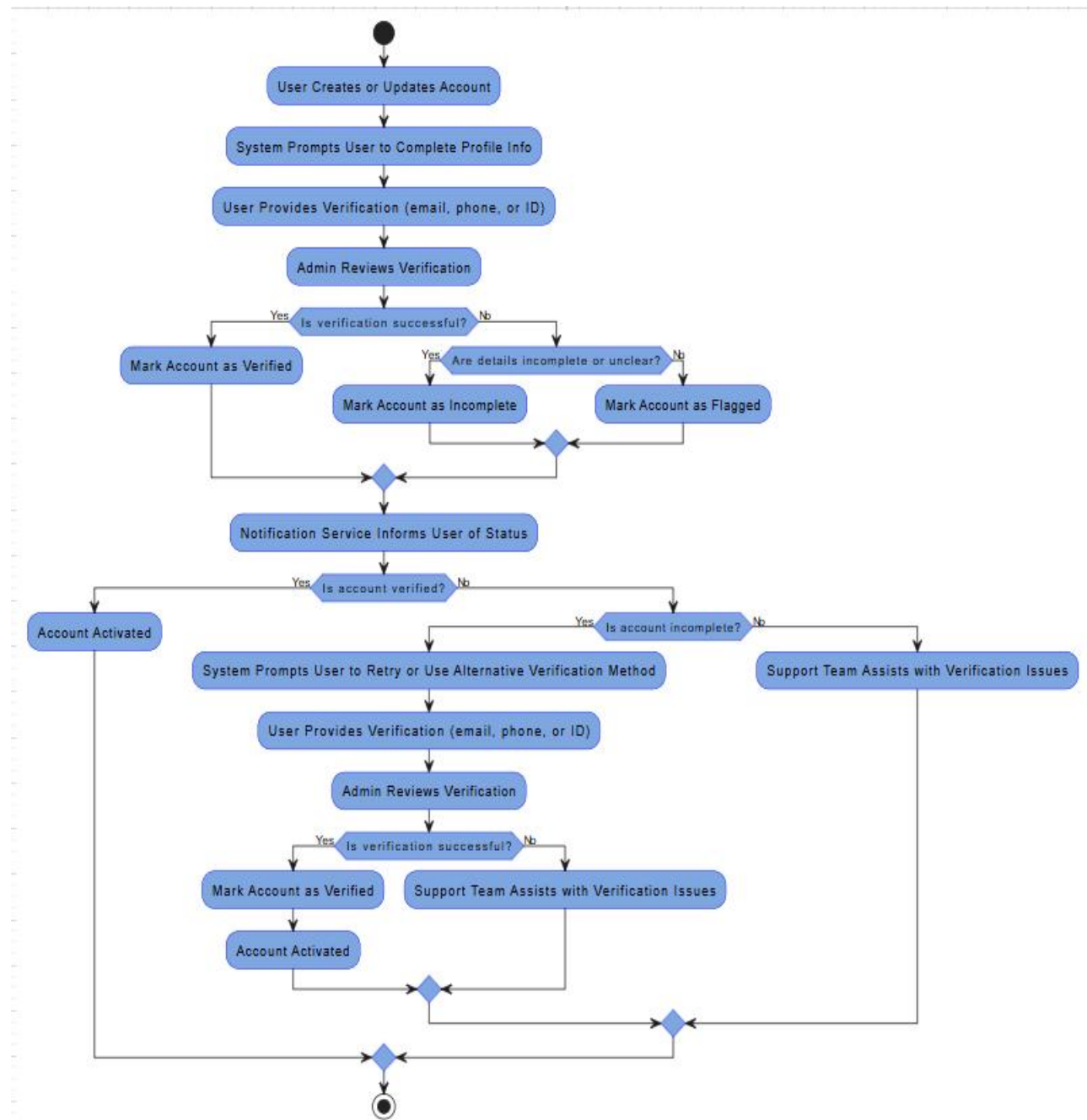


Figure 12 :Account Authenticity



## Activity Diagram :Transaction Behavioral and user classification

It shows the flow of data retrieved in the system and the classification of users to give them permissions according to their classification.

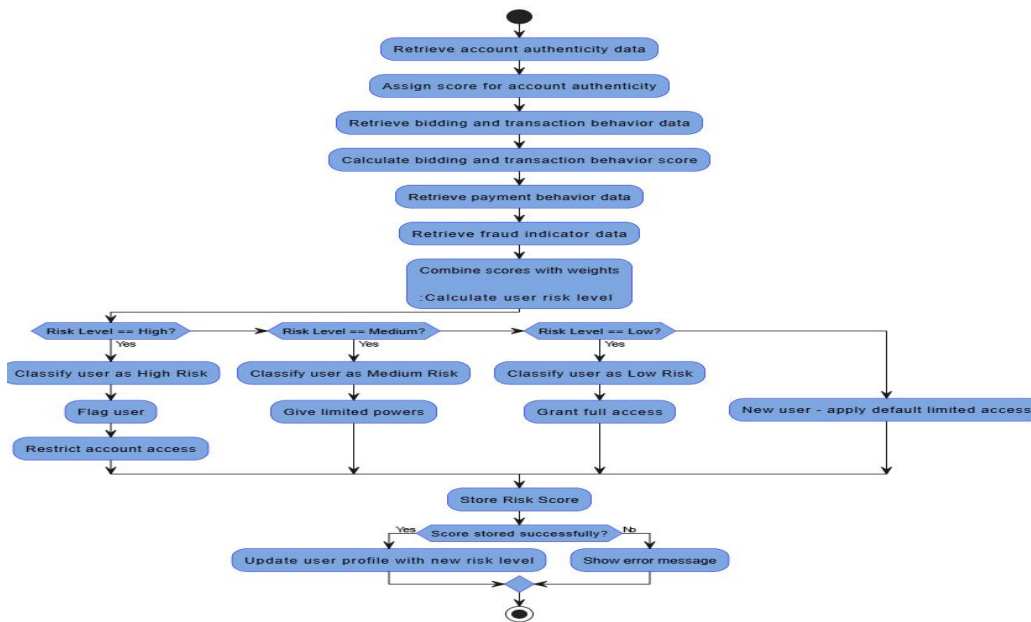


Figure 13 :Transaction Behavioral and user classification

### 3.3.4 State Chart Diagram :

is an illustration of all the possible behavioral states a our system

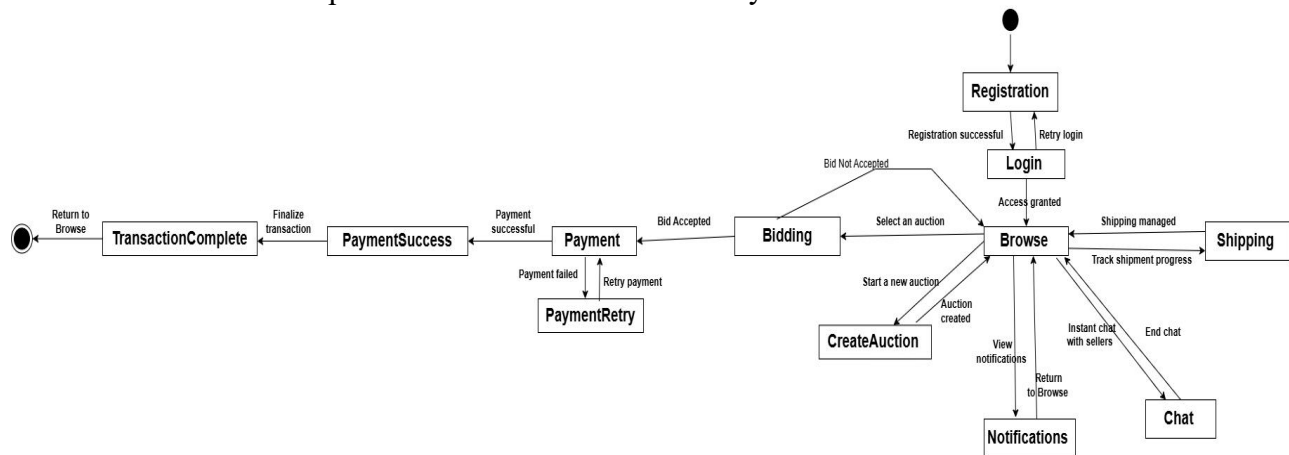


Figure 14 : Chart Diagram

## 4.4 System Architecture

### 4.4.1 Sub-System

1-Payment Gateway: Handles all financial transactions of users, ensuring secure payments for purchases in the auction.

- Payment Processing: Facilitates successful bidding through various methods such as credit cards, bank transfers, and e-wallets.
- Transaction Verification: Verifies the validity and security of each transaction.
- Refunds: Manages refund requests.

2-Notification System: Handles communications between the platform and users through bid notifications, auction status updates, and important alerts.

- Real-Time Notifications: Sends alerts to users about new bids, auction closing times, winning status, and payment reminders.

### 4.4.2 Software Architecture

graphical representation of a system that illustrates the relationships and interactions between its components in our system

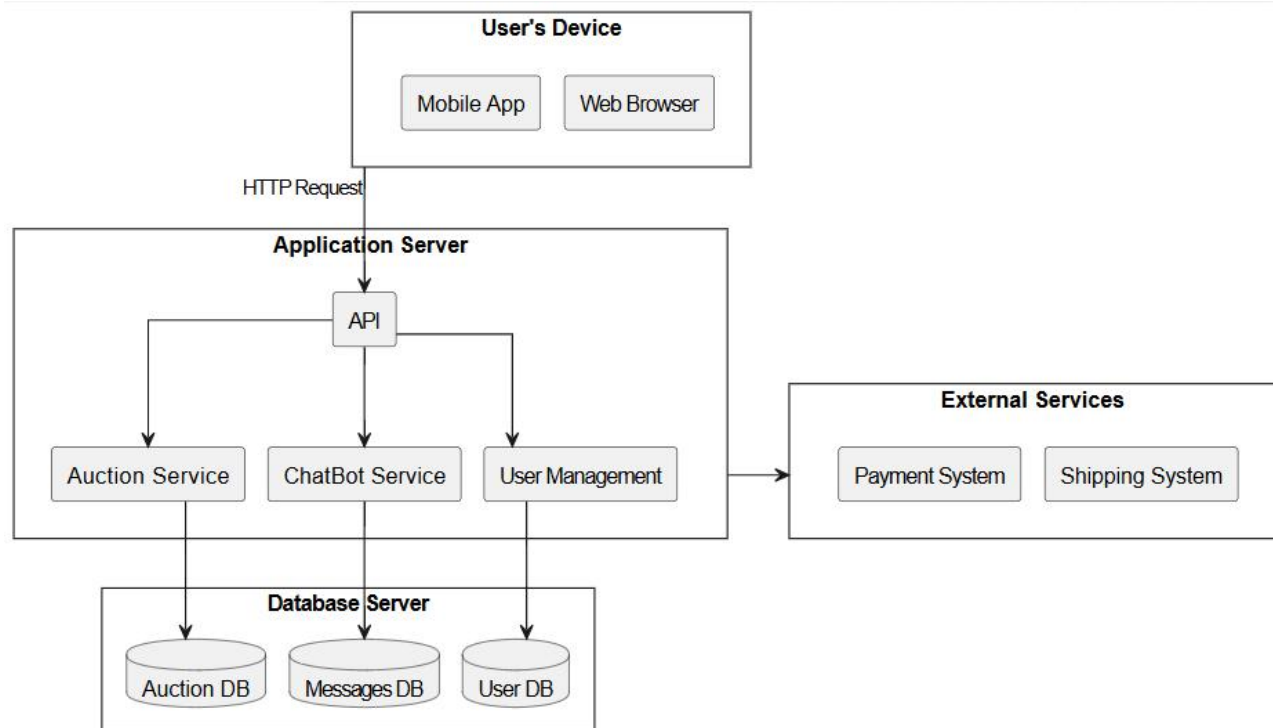


Figure 15 : Software Architecture

### 4.4.3 Deployment diagram

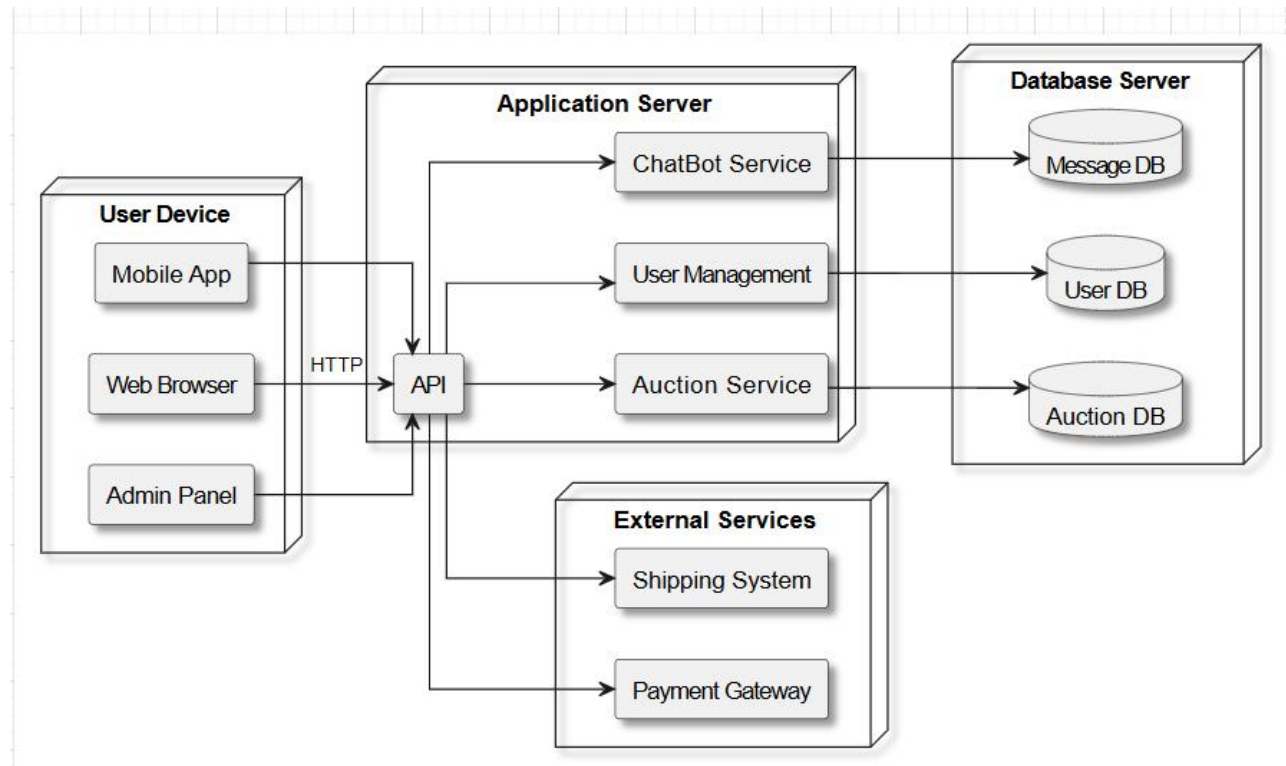


Figure 16 :Deployment Diagram

### 4.4.4 Components' Diagram

illustrates the key modules of the system and highlights how these components interact with each other to deliver the functionalities.

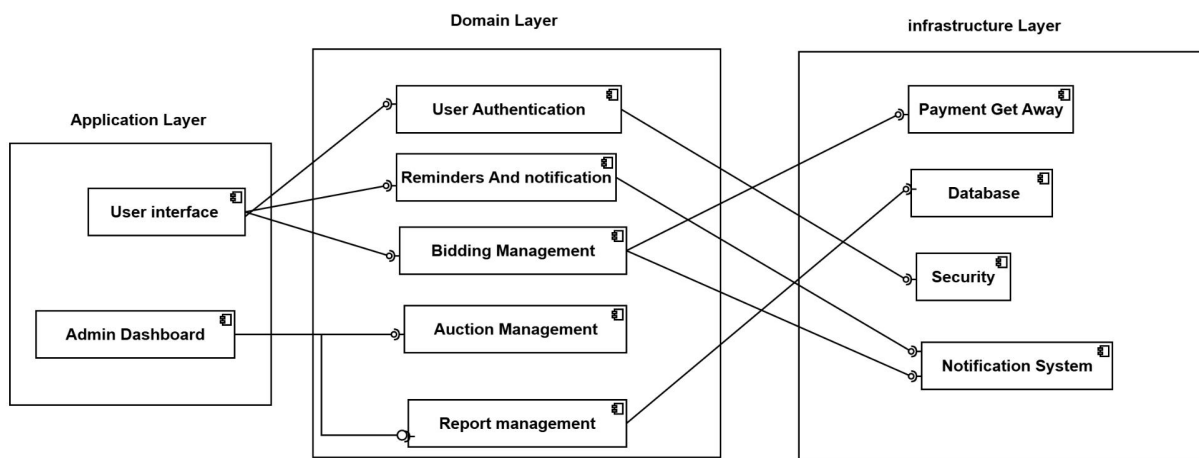


Figure 17 :Components' Diagram

## 4.6 Data Management and ERD

An Entity-Relationship Diagram (ERD) serves as an invaluable instrument for articulating the interconnections between various entities within a database. By visually representing entities as boxes and their relationships as lines

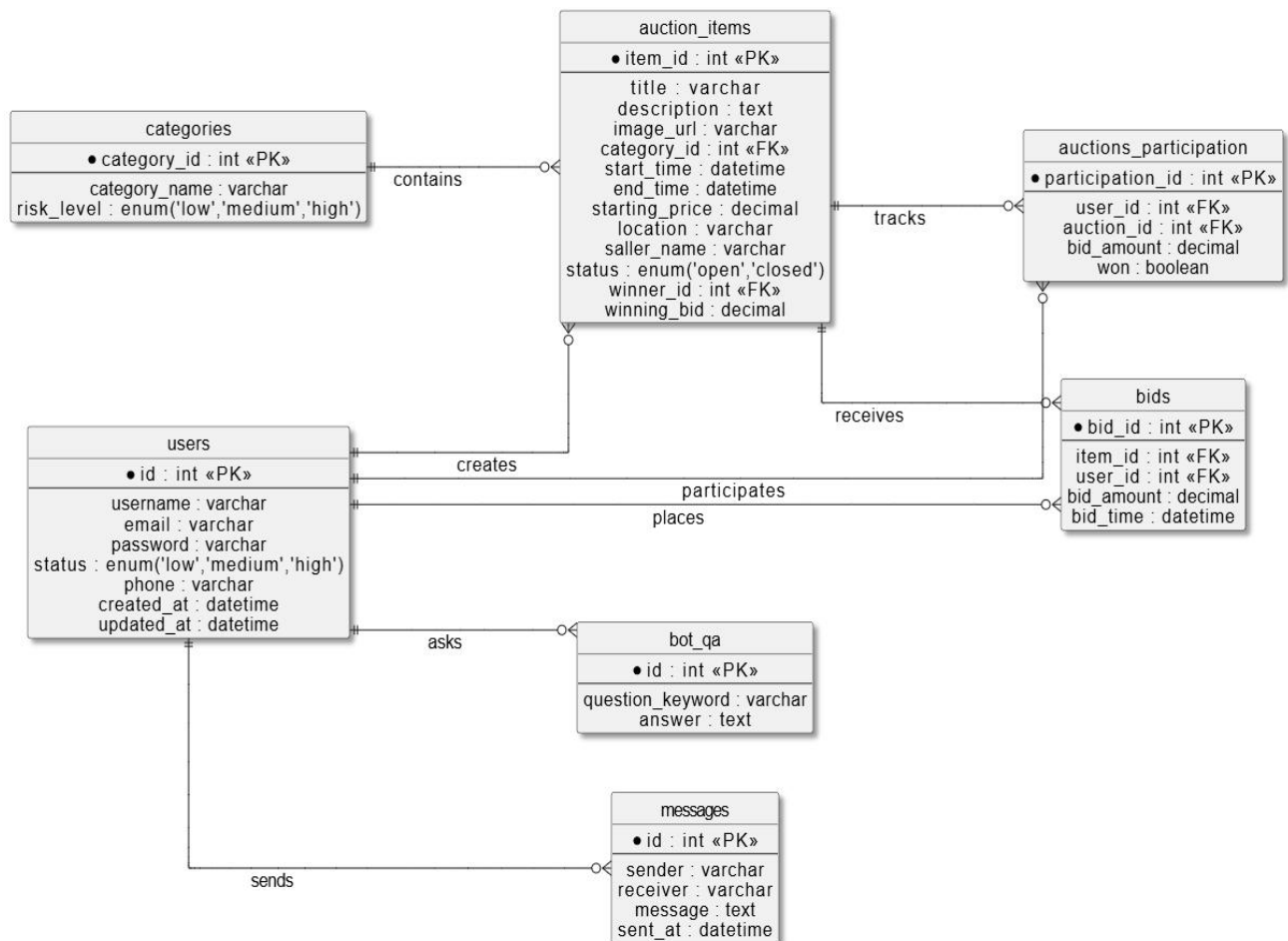


Figure 18 : ERD Diagram

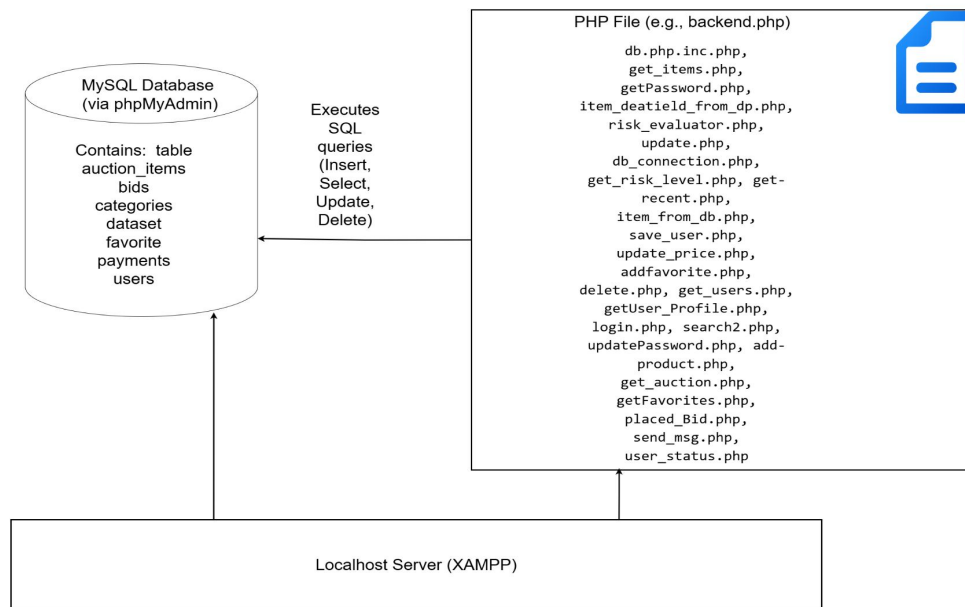


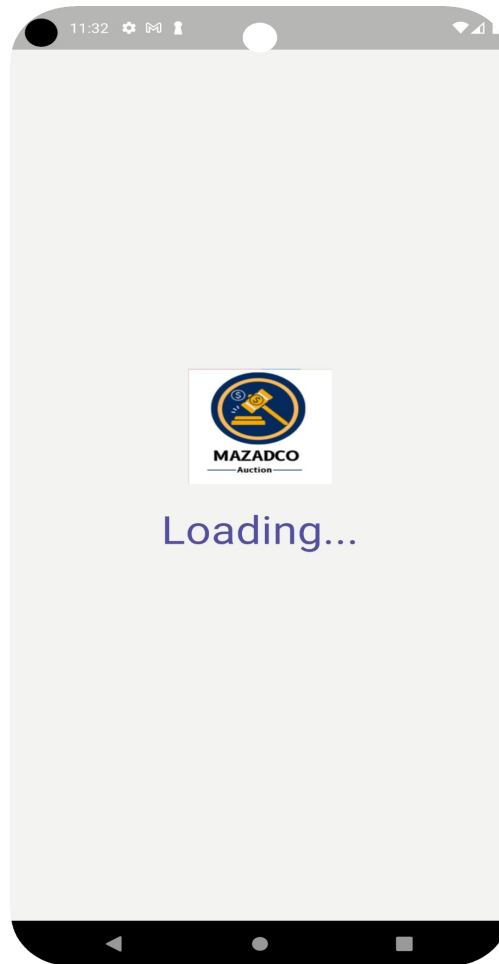
Figure 19 : API and database interaction

This diagram illustrates the basic architecture of an auction application, showing the relationship between the API and the database. The API services process data, such as entering a new auction or placing a bid, and then communicate with the database to store or retrieve the necessary information.

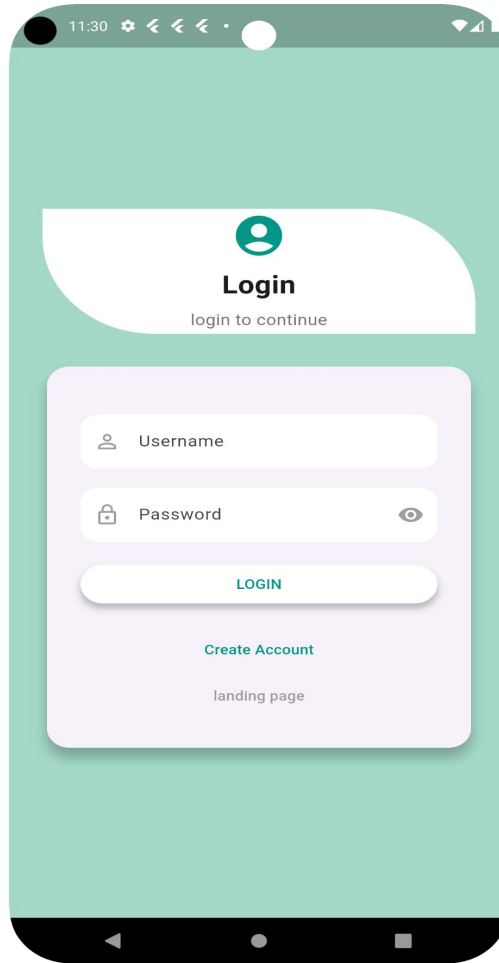
## Chapter 5 : Implementation

### Loading page :

This page displays when you first launch the application, loading the basic resources and proceeding to login.



*Figure 20 : introduction*



*Figure 21 : login page*

here is the first interface for our project the user have many option to start ( first option that the the user need to create account , second option ( landing page ) the user want to Browsing the site without login, and the last one user already have account with valid data .

second interface : create account step one ( first option ):

10:17 76% 5G

## Create New Account

Email Address

Phone Number

Created At  
2025-06-18

Payment Method

Address

Next

\* All fields are required

*Figure 22 : : create account step1*

the user must enter all valid data to start registration , and cannot move to the second step unless all the data is completed according to the required conditions.



The image displays two side-by-side mobile app screenshots of a "Create New Account" form. Both screens have a teal header and a white title "Create New Account".

**Left Screenshot (Negative Test Case):**

- Full Name:** Input field contains "safaalawi".
- Email Address:** Input field contains "safa". Below the field is a red error message: "Invalid email format".
- Phone Number:** Input field is empty. Below the field is a red error message: "Enter your phone number".
- Created At:** Date picker shows "2025-06-18".
- Payment Method:** Dropdown menu is open, showing "Payment Method" with a downward arrow. Below the field is a red error message: "Please select a payment method".

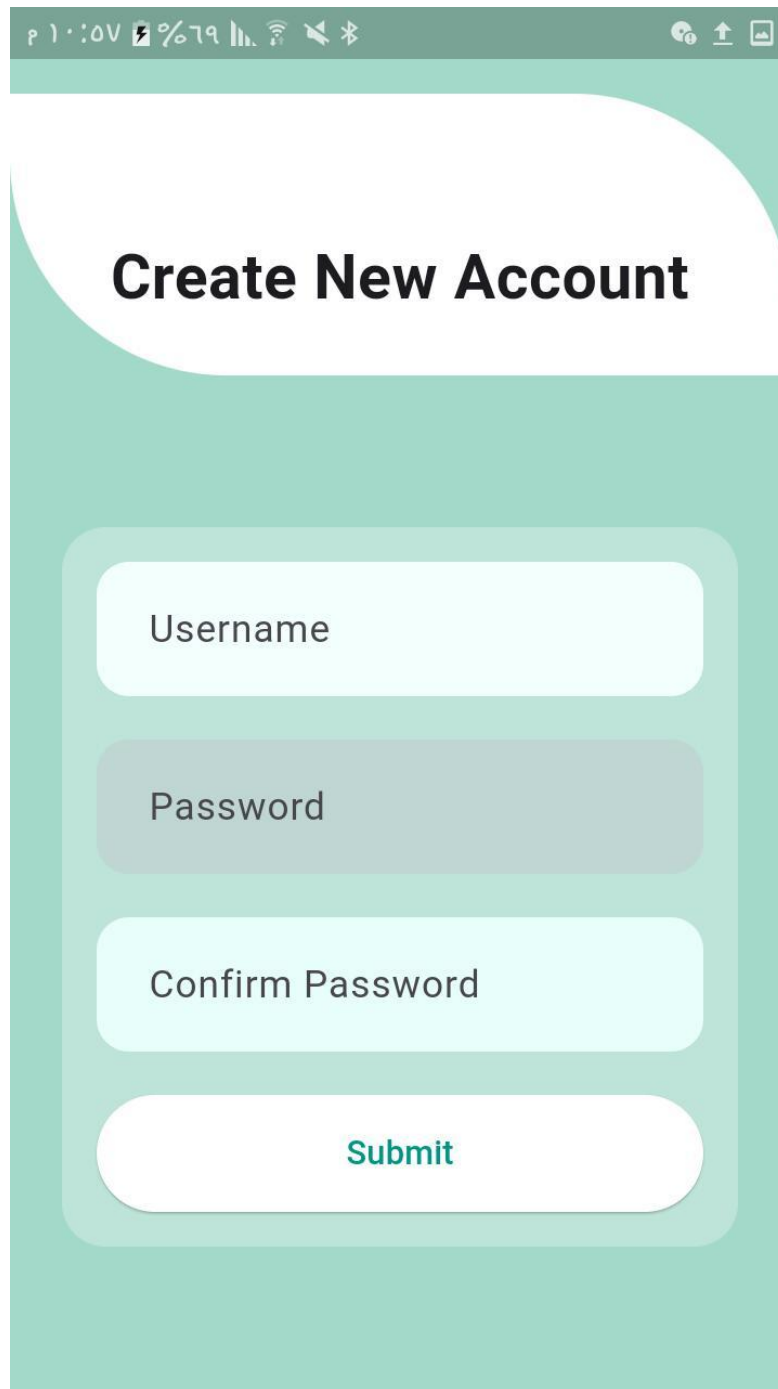
**Right Screenshot (Positive Test Case):**

- Full Name:** Input field contains "safaalawi".
- Email Address:** Input field contains "safaalawi19@gmail.com".
- Phone Number:** Input field contains "0852369852".
- Created At:** Date picker shows "2025-06-18".
- Payment Method:** Dropdown menu is open, showing "Visa" with a downward arrow.
- Address:** Input field contains "rammallah".

unaccepted data (negative test case )

accepted data(positive test case )

third interface : create account Step Two :



**Create New Account**

Username

Password

Confirm Password

**Submit**

*Figure 23 : : create account step 2*

the user must enter accepted data for username , password and confirm password to move to the last step .

**Create New Account**

Username

Please enter a username

Password

Please enter a password

Confirm Password

Please confirm your password

Submit

**Create New Account**

slaawi123

.....

must include at least one uppercas...

....

Passwords do not match

Submit

missing data in each field  
(negative test case )

the password must be match  
(negative test case )

10:08 79% [icons]

## Create New Account

safaalawi12

.....

.....|

Submit

(positive test case )

here is valid data in second step registration .

- username at least 4 character
- password at least 6 character and must have at least one upper case and one number

fourth interface :

**Create New Account**

|                    |                       |
|--------------------|-----------------------|
| <b>Username</b>    | safaalawi12           |
| <b>Full Name</b>   | safaalawi             |
| <b>Email</b>       | safaalawi19@gmail.com |
| <b>Phone</b>       | 0852369852            |
| <b>Address</b>     | rammallah             |
| <b>Created At</b>  | 2025-06-18            |
| <b>Payment Way</b> | Visa                  |

**Create New Account**

|                    |                       |
|--------------------|-----------------------|
| <b>Username</b>    | safaalawi12           |
| <b>Full Name</b>   | safaalawi             |
| <b>Email</b>       | safaalawi19@gmail.com |
| <b>Phone</b>       | 0852369852            |
| <b>Address</b>     | rammallah             |
| <b>Created At</b>  | 2025-06-18            |
| <b>Payment Way</b> | Visa                  |

**Complete Registration**

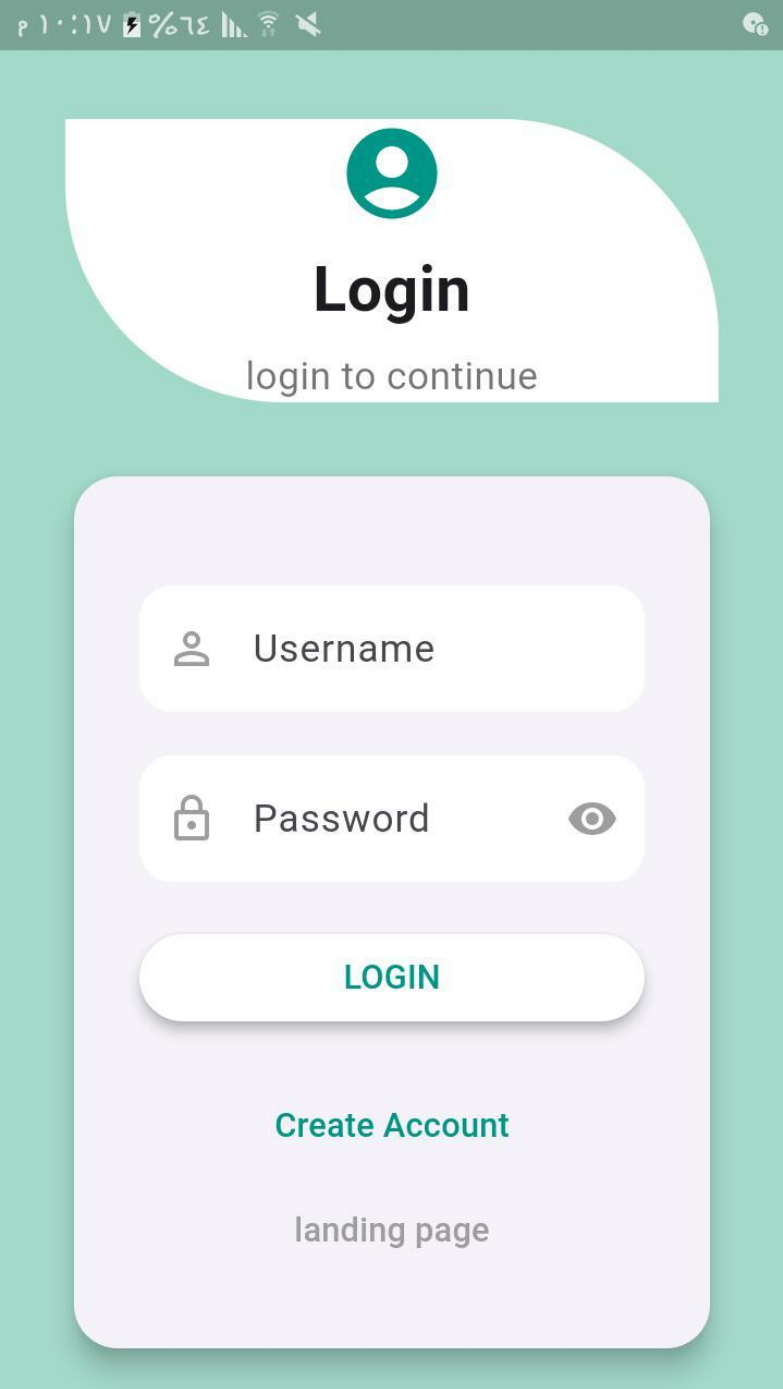
User registered successfully!

(positive test case )

Figure 24 : : create account step 3

in the last step of registration this table contain all important data for user to complete and confirm his data , when click on the button all data will save in database also after 3 second the login page will shown again .

Back to second option ( landing page ) the user want to Browsing the site without login ) :

A mobile application login screen with a teal background. At the top, there's a status bar with various icons. Below it, a white rounded rectangle contains a teal user icon, the word "Login" in bold, and the text "login to continue". Below this, a light purple rounded rectangle contains a login form. The form has two input fields: "Username" with a user icon and "Password" with a lock icon and a toggle eye icon. Below the fields is a white "LOGIN" button. At the bottom of the purple rectangle is a teal "Create Account" link and a grey "landing page" link.

10:17 76% 5G



# Login

login to continue

 Username

 Password 

LOGIN

Create Account

landing page

Figure 25 : login for user

when the user click on ( landing page button ) this page will appear :

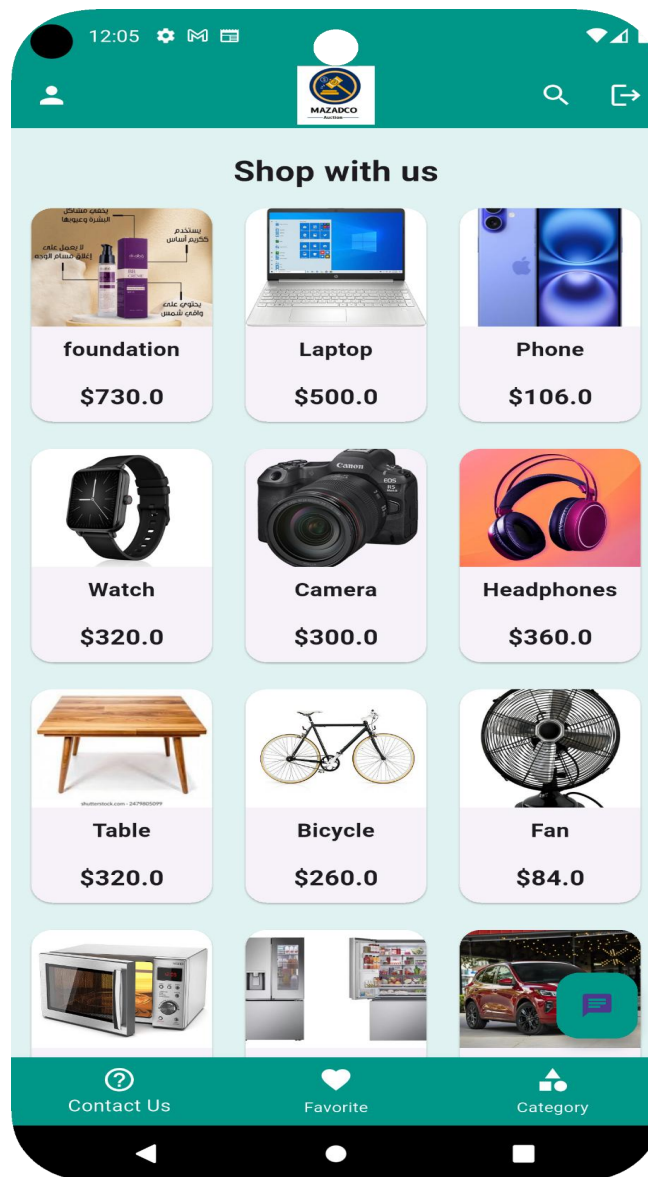
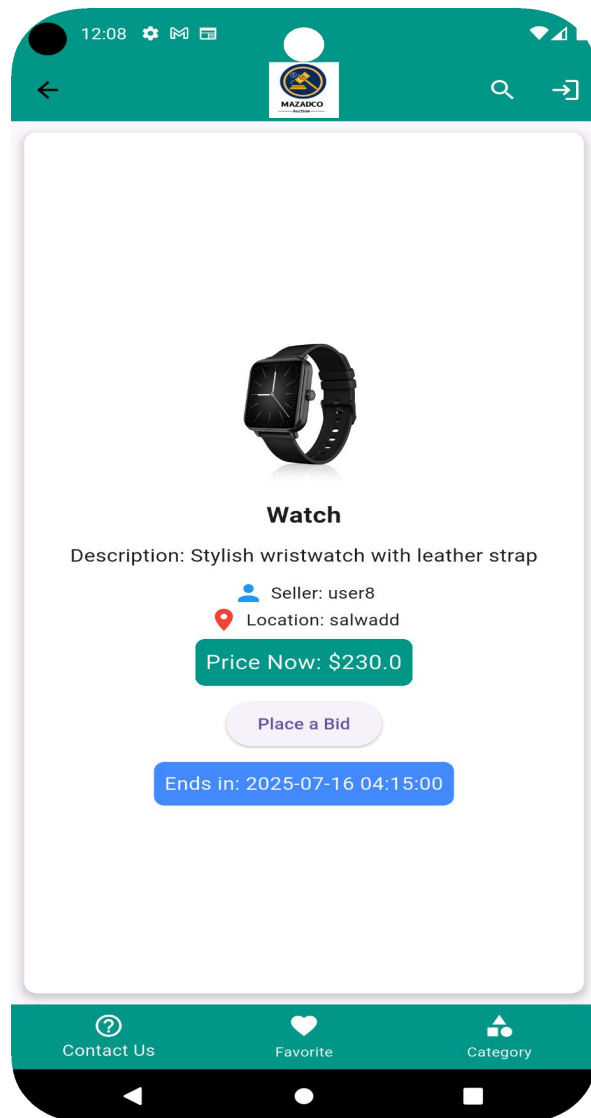


Figure 26 : landing page without login

this is landing page where contains Most of the main interfaces in our auction start with (user profile icon page ,search , category , etc. ..) , but in this case the user can only search item , Browsing item , user cant do anything else then Browsing .

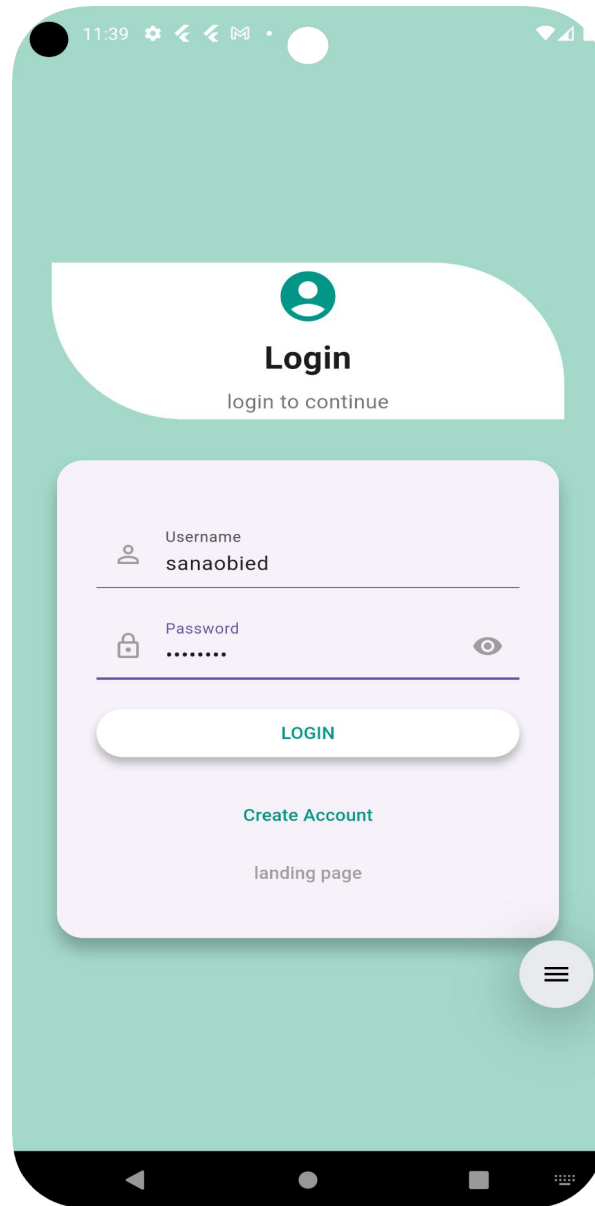
when the user click any of the above icon he will be returned to the login page.  
other case the user also can see the details of item when click it like this :



*Figure 27 : item details page without login*

but when click on place a bid It will be returned to the login page also .  
the last option in login page ( user already have account with valid data ) :





*Figure 28 : back to login page*

when the user enter valid data and click login button it will be redirect to Landing page

The fifth interface : Landing page

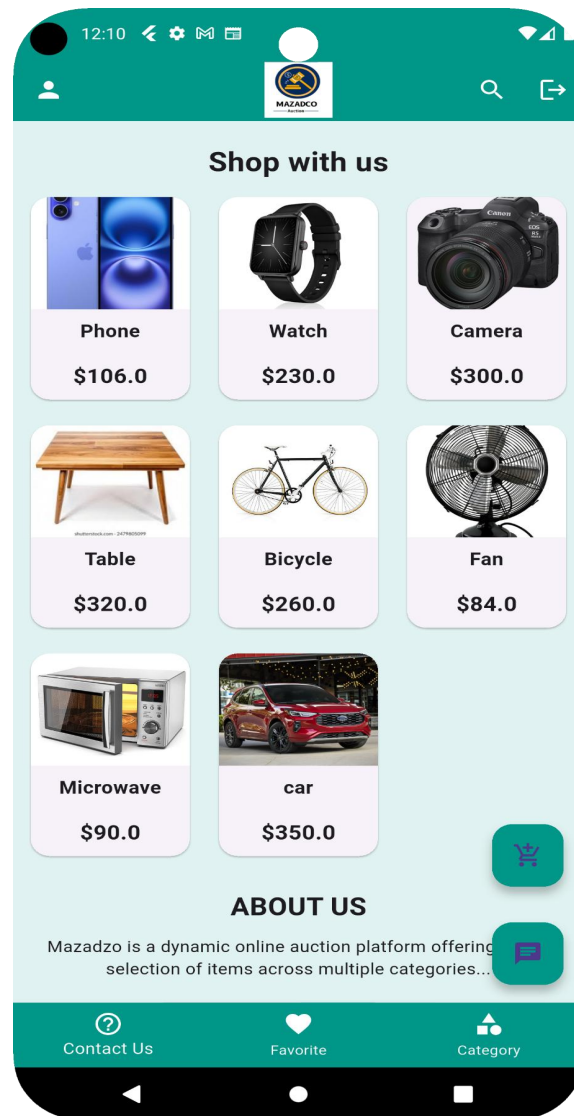


Figure 29 : landing page after login

in this page while the user was entered his data all icon will be valid



this one contains the user profile page



this for search item



this for logout



this for the bidding for the user account



this is for chat pot

and the other icon are already clear

In this interface, we have achieved the first authority to classify users according to their classification by classifying items based on price based on the percentile algorithm, where items are divided into three parts based on what was mentioned previously.

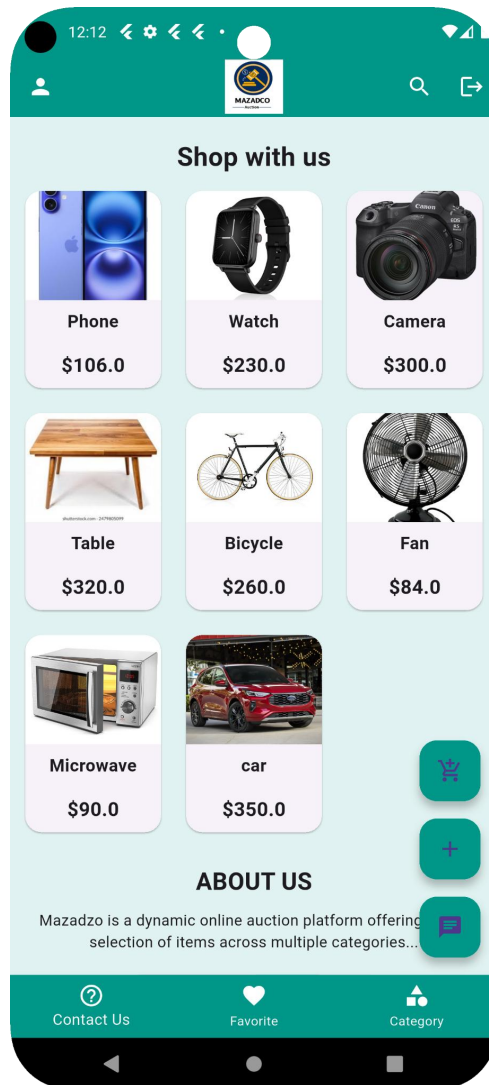


Figure 30 : medium risk - for item

this one based on ( medium risk )

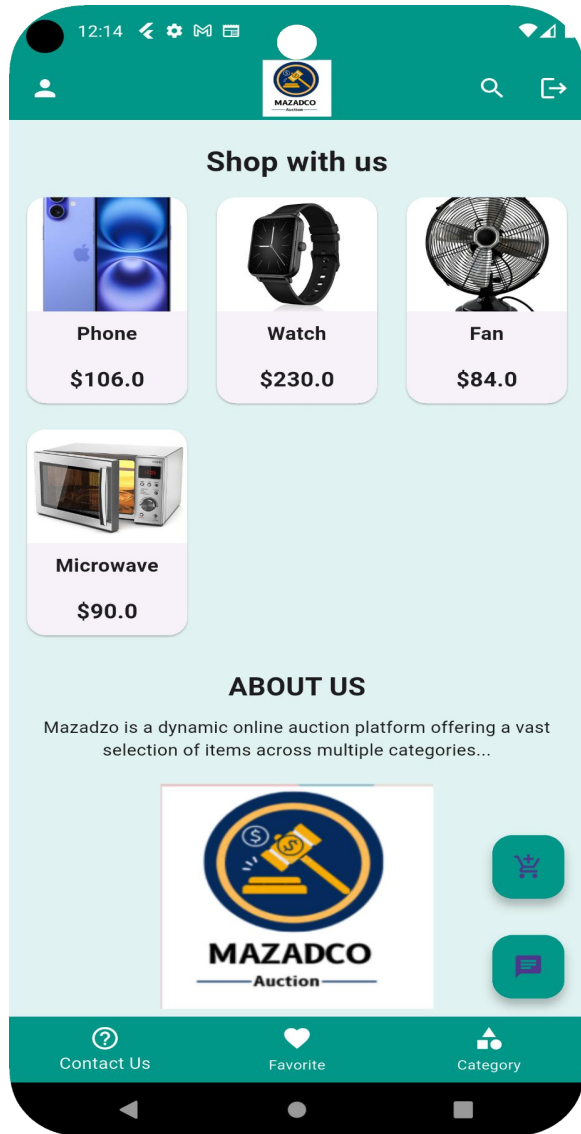


Figure 31 : high risk categorization item

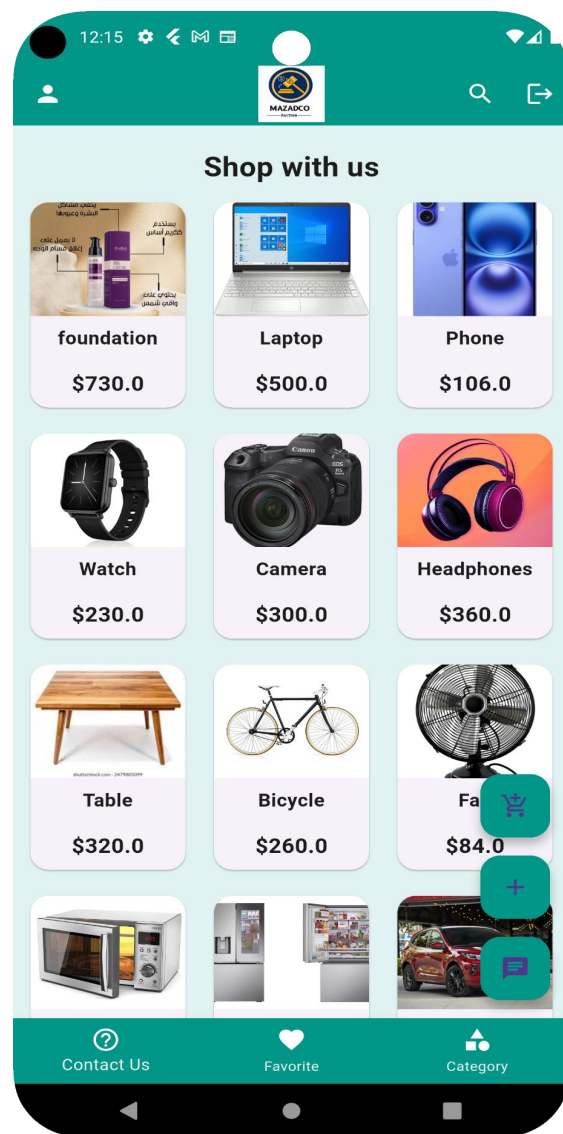
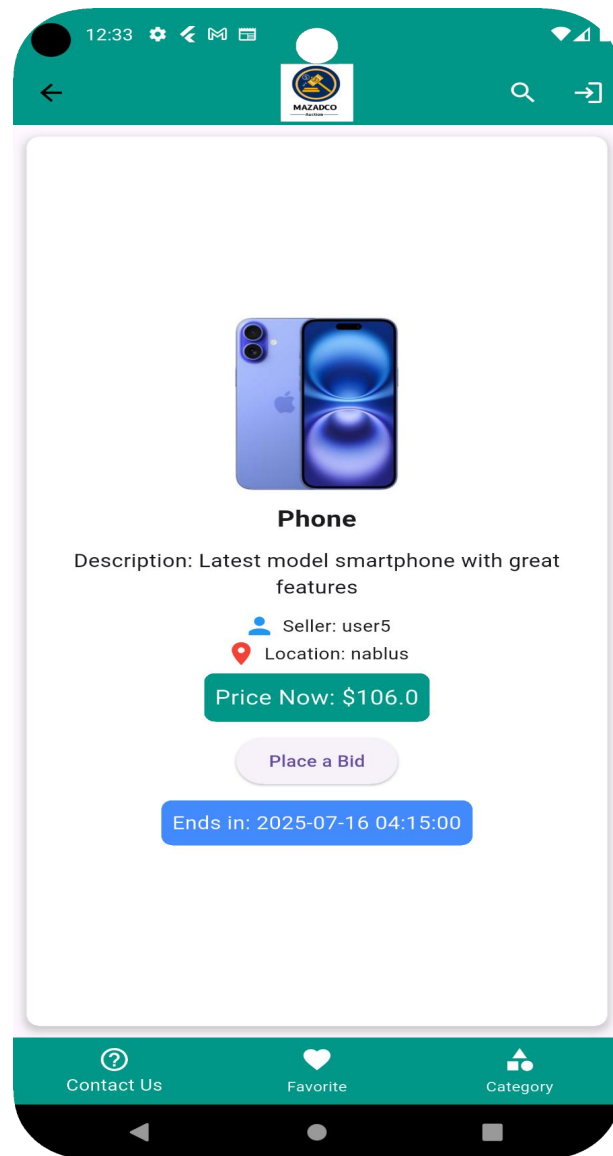


Figure 32 : low risk categorization item

these image from different accounts in the same category for all (one table for all users )

**The sixth interface : ( item details ) :**

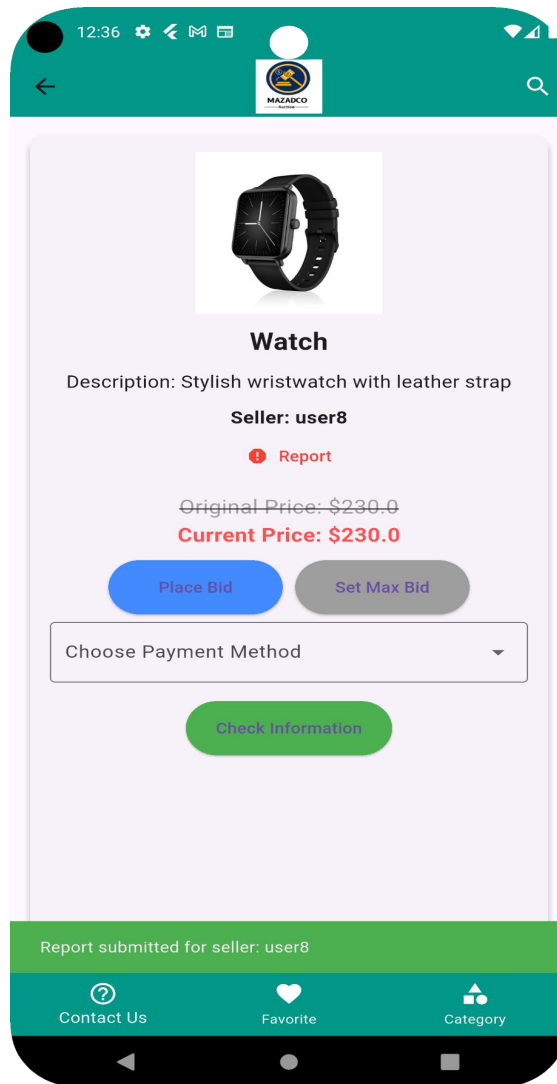


*Figure 33 : item details after login*

This interface enables you to view the item details in terms of a clear image of the item, a description of the product, the seller's name and location, the current item price, and the auction's end date. In addition, there is a button that takes you to the auction page. Again, the user must be logged in to go to the auction page.

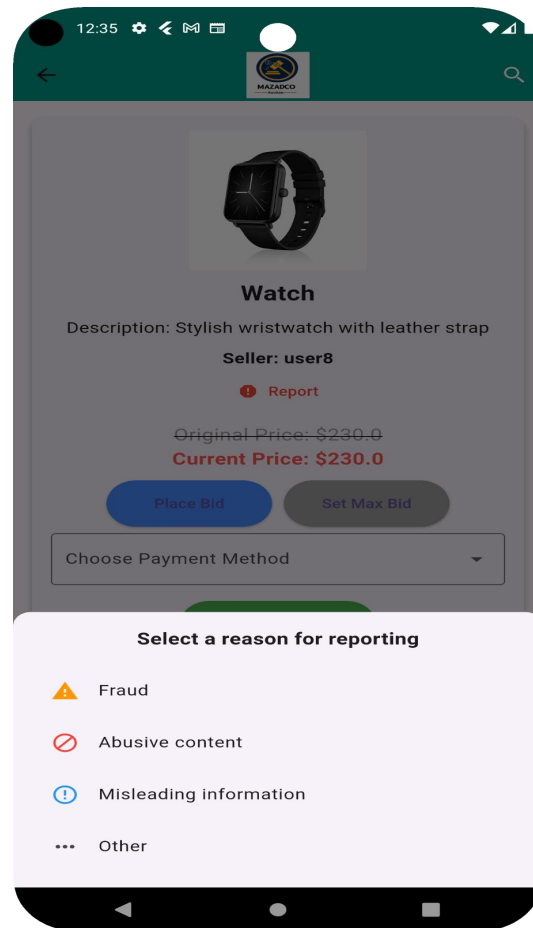
when the user click in place bid button its will re direct to the place bid page below

### The seven interface (place bid page )



*Figure 34 : place bid page*

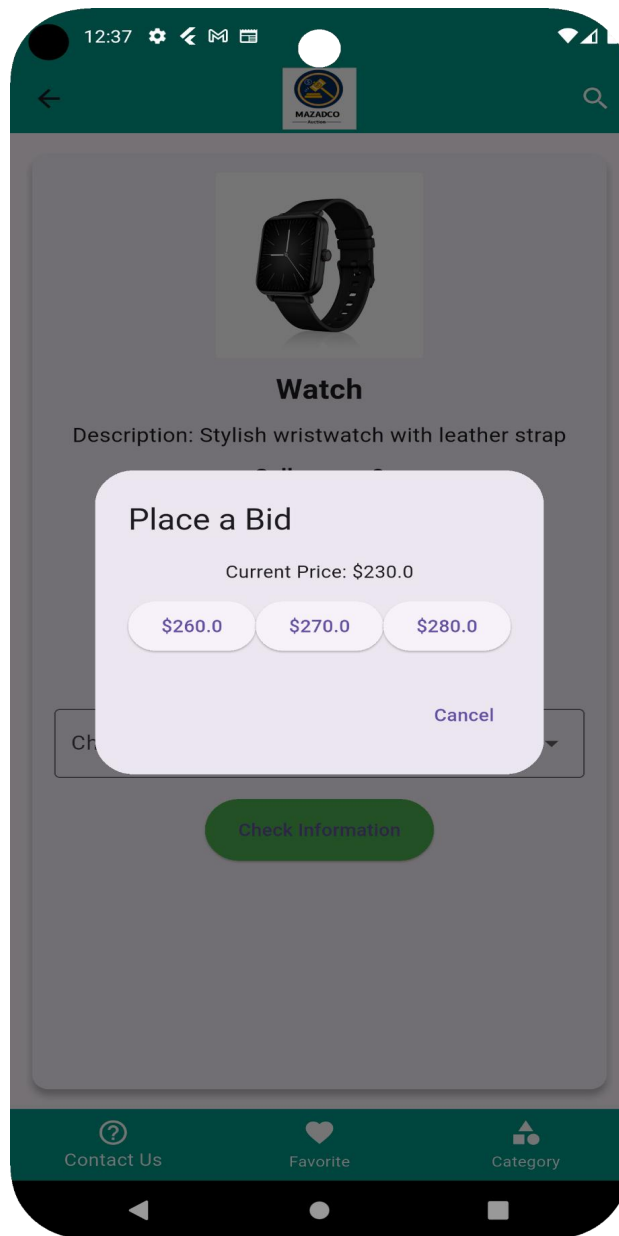
in this interface the bidding for item will be in many step :  
at first We have a button to report the seller



*Figure 35: report for seller*

for many reason the user will report the seller that will review and update in data base to place a bid process the user have 2 option to make the bid the first one :

### Click Place Bid:

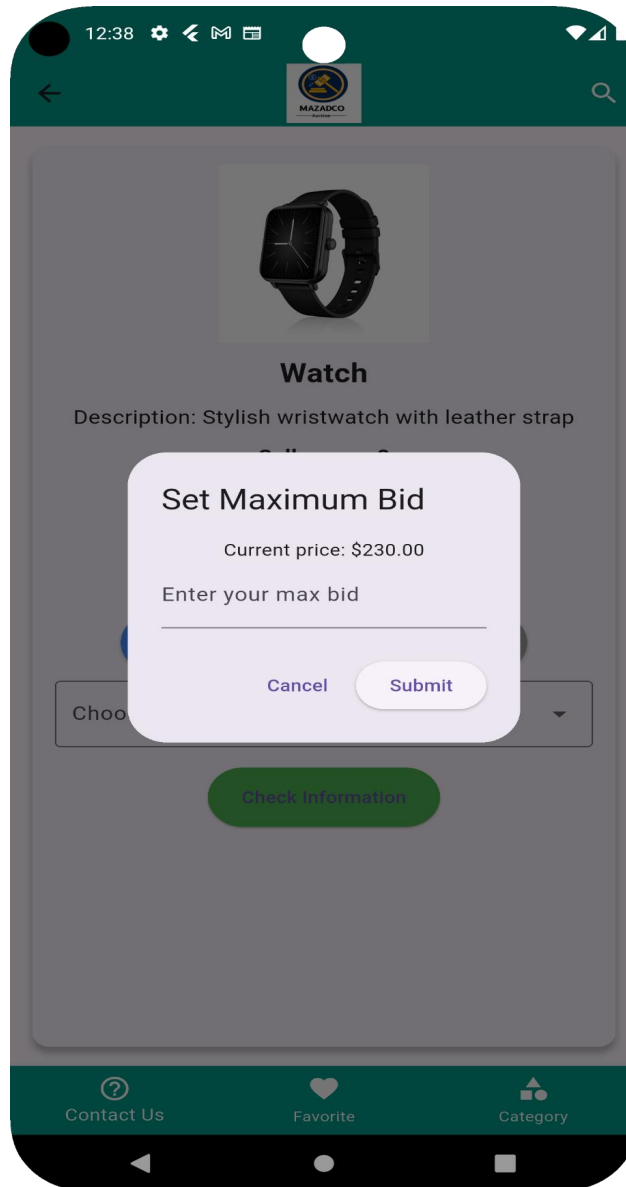


*Figure 36 : place bid with option*

If the item's price is over 100, the price option will increase by 30, 40, or 50. This is because, after reading and researching scientific sources, users tend to prefer a competitive spirit at reasonable prices. If the bidding is low on an expensive product, this may bore the bidder. If the item's price is less than 100, the bidding will increase by 5, 10, or 15 points for low prices, to avoid exaggeration and making the user feel defrauded.



**when click set max bid:**



*Figure 37: place bid with max bid option*

if the user select number more then 40 % he can current buy it and the status for this bidding will be close then the item is sold

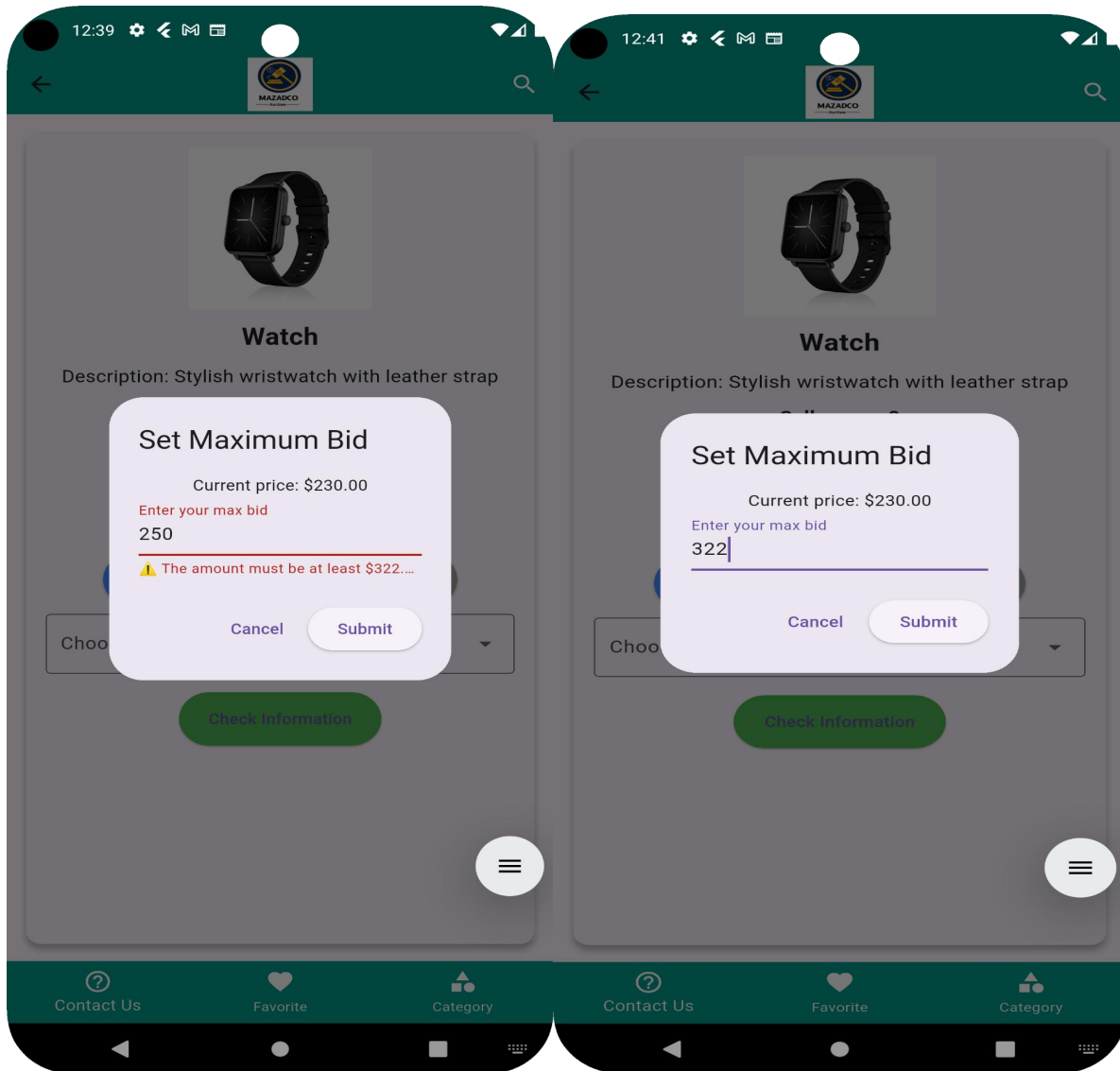


Figure 38 : set max bid

There an option with price below 100 \$ that increase the amount (5\$, 10\$,15\$):

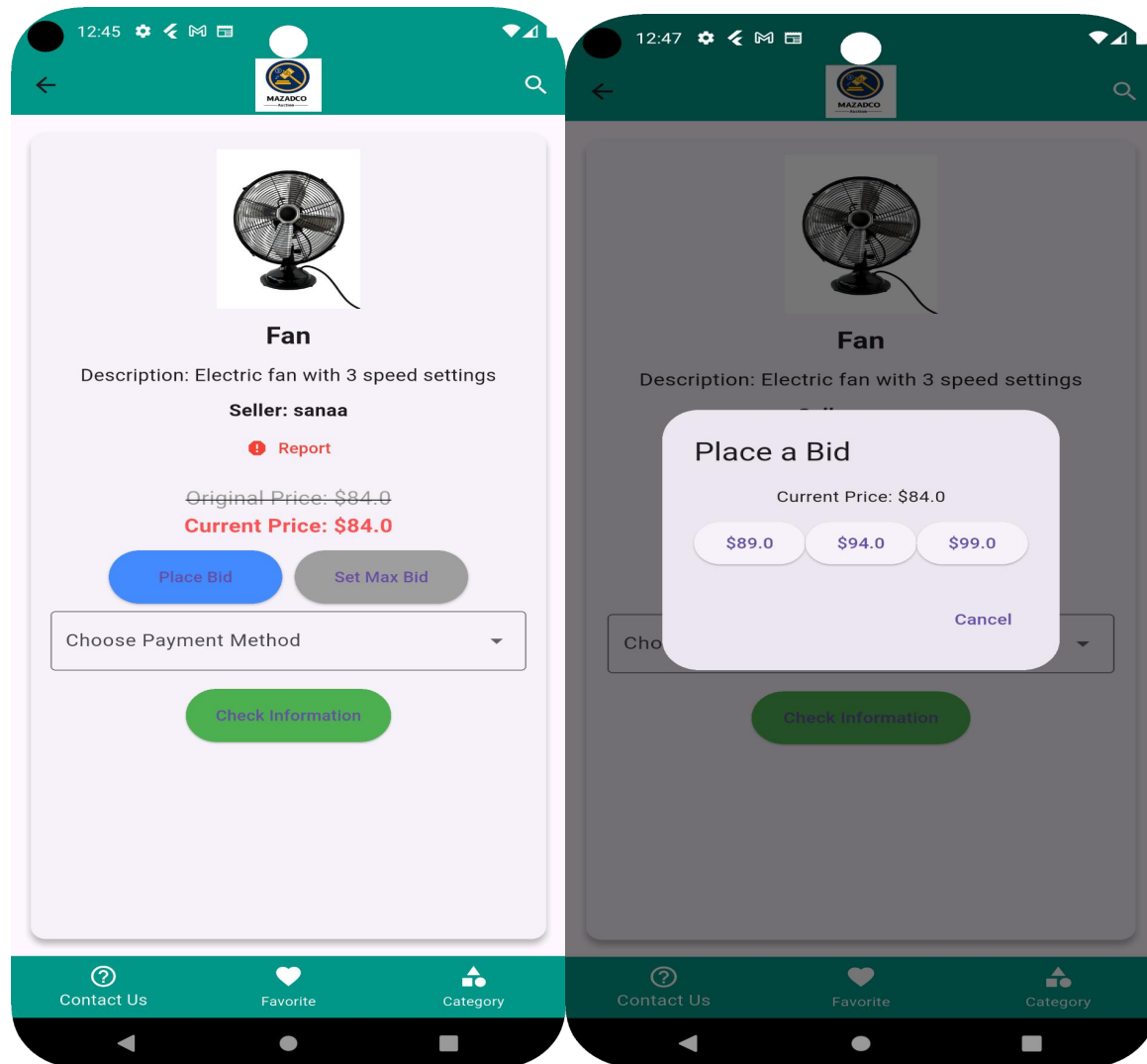


Figure 39 : range with price bellow 100 \$

Clicked in set Max Bid:

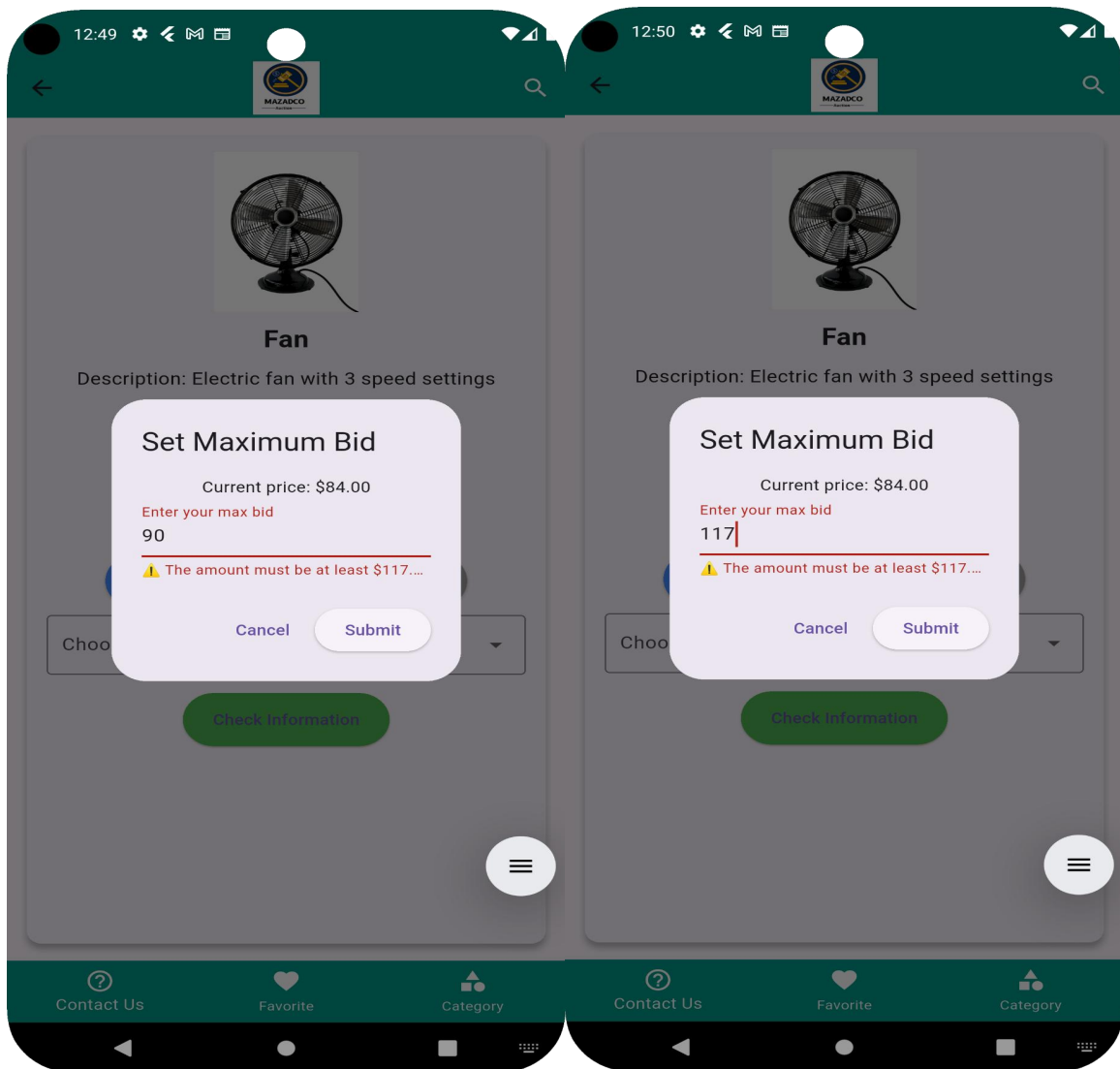


Figure 40 : set max bid

### Payment Method :

in our auction here we have a validity that the user can select the payment method depend on his current risk status

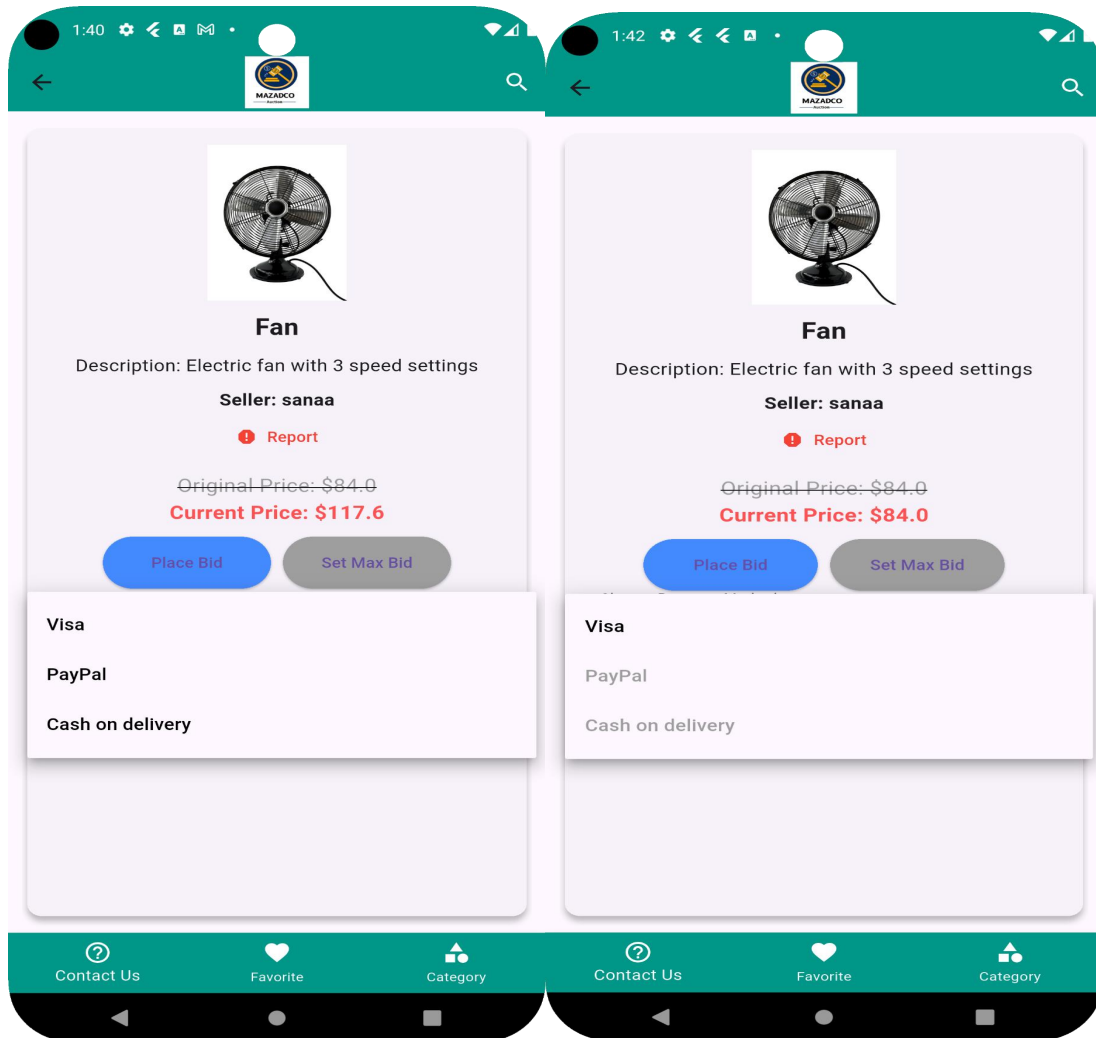


Figure 41: payment method option base on categorization

the user is low risk categorization

the user is high risk categorization

the user is medium risk categorization

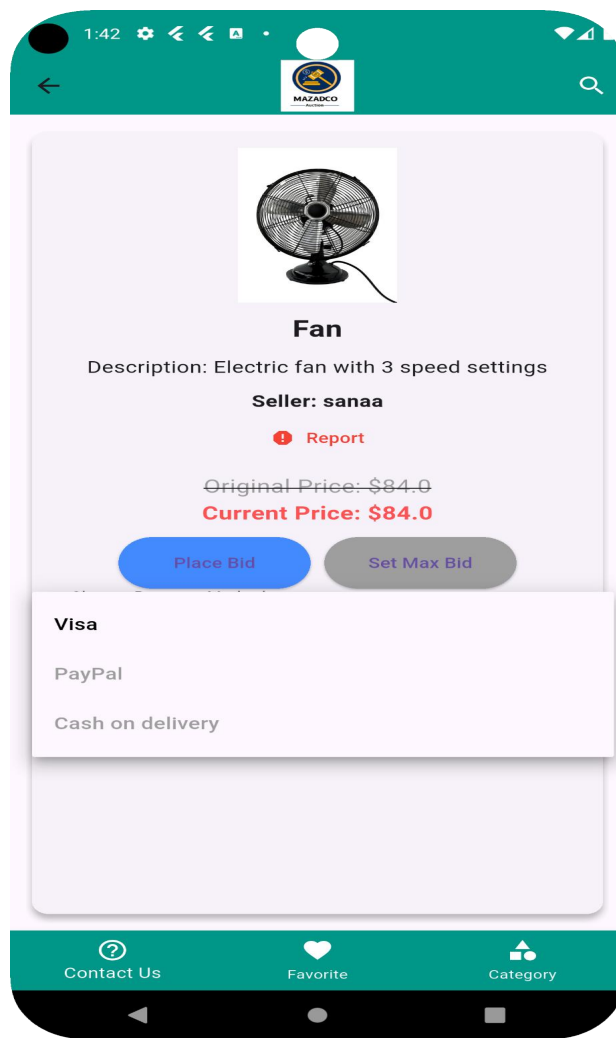


Figure 42 :payment method option base on categorization -medium

After placing a bid and selecting "Visa" as the payment method, the user is prompted to enter their Visa card details, including card number, expiration date, and CVV. Once the information is submitted, the system verifies the provided Visa credentials. If the verification is successful, the Visa payment details are securely stored, and the bid is confirmed. A confirmation message is displayed, showing the Visa information used and indicating that the bid has been placed successfully.

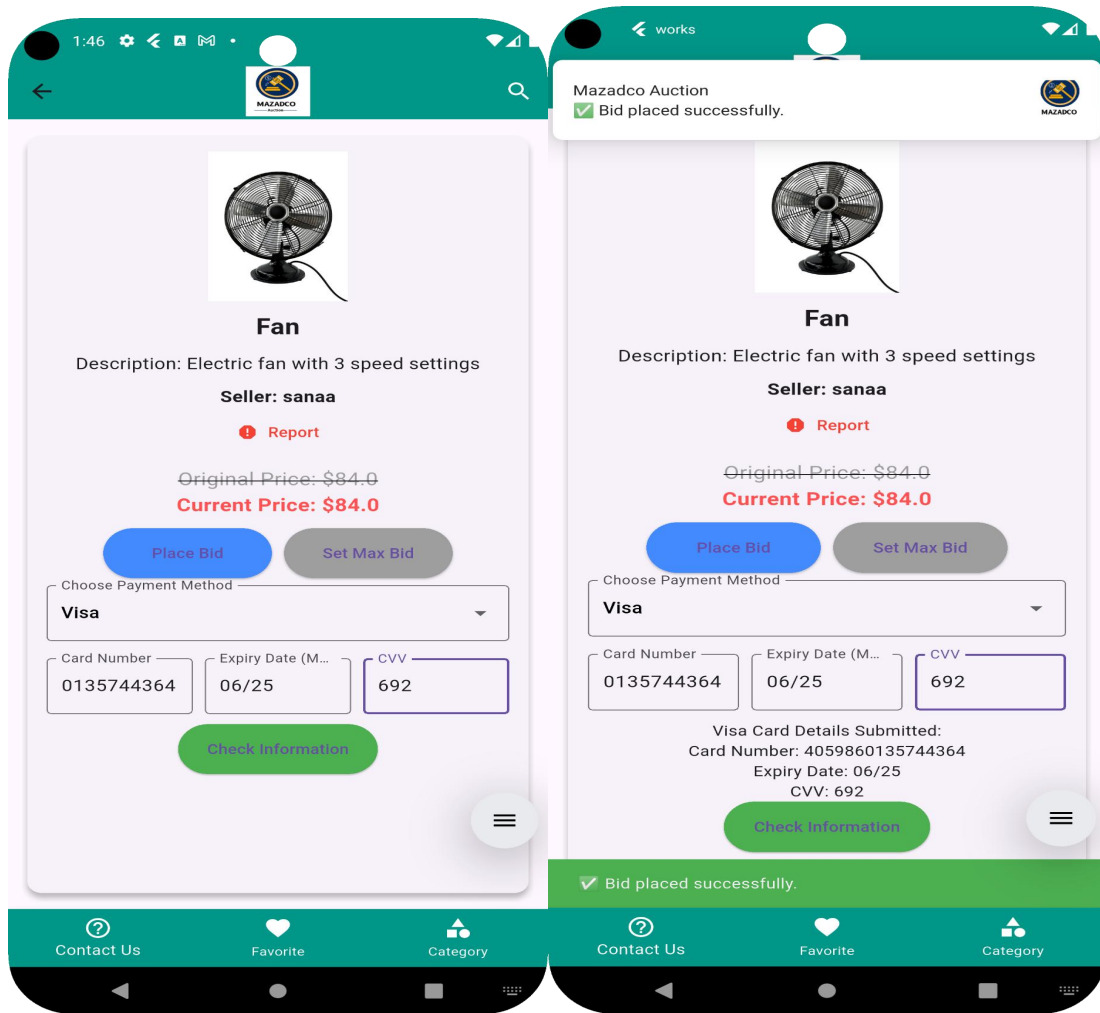


Figure 43 : check information button

However, if the user has reached their daily bidding limit, the system will block the bid attempt and display the message: " Daily bid limit reached based on risk level." This restriction is based on the user's assigned risk classification. High Risk accounts are normally limited to one bid per day, Medium Risk accounts may place up to three bids daily, while Low Risk accounts normally have no such restrictions.

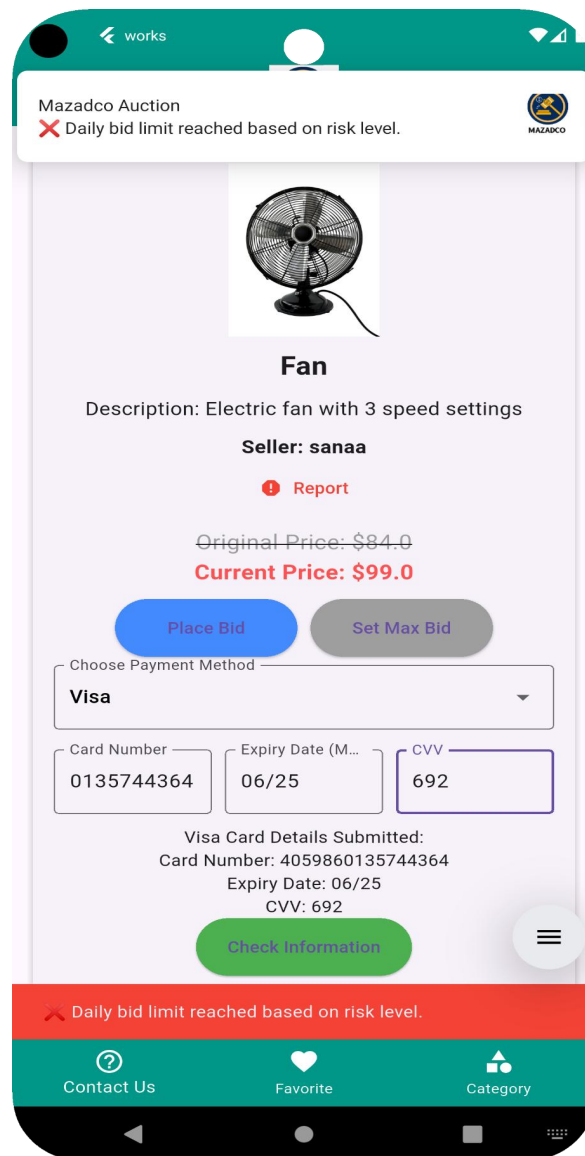


Figure 44 : limit bid based on categorization



### The seven interface (Auction details) :

This interface contains details of the items that the user (account owner) has bid on. If the auction is still, the user can **click the "Cancel" button** next to the bid to cancel their participation in the auction.

| Bid ID | Item ID | Bid Time            | User ID | Bid Amount | Title |
|--------|---------|---------------------|---------|------------|-------|
| 60     | 15      | 2025-06-18 17:15:54 | 1       | 3.00       | car   |
| 59     | 15      | 2025-06-18 17:15:45 | 1       | 2.00       | car   |
| 58     | 15      | 2025-06-18 17:15:32 | 1       | 1.00       | car   |
| 57     | 6       | 2025-05-28 00:48:42 | 1       | 3.00       | Table |
| 56     | 6       | 2025-05-28 00:48:35 | 1       | 2.00       | Table |
| 55     | 6       | 2025-05-28 00:48:15 | 1       | 1.00       | Table |
| 46     | 2       | 2025-05-23 05:40:04 | 1       | 9.00       | Phone |
| 45     | 2       | 2025-05-23 05:39:50 | 1       | 8.00       | Phone |
| 44     | 2       | 2025-05-23 05:39:39 | 1       | 7.00       | Phone |
| 43     | 2       | 2025-05-22 02:42:10 | 1       | 6.00       | Phone |
| 42     | 2       | 2025-05-22 02:42:00 | 1       | 5.00       | Phone |

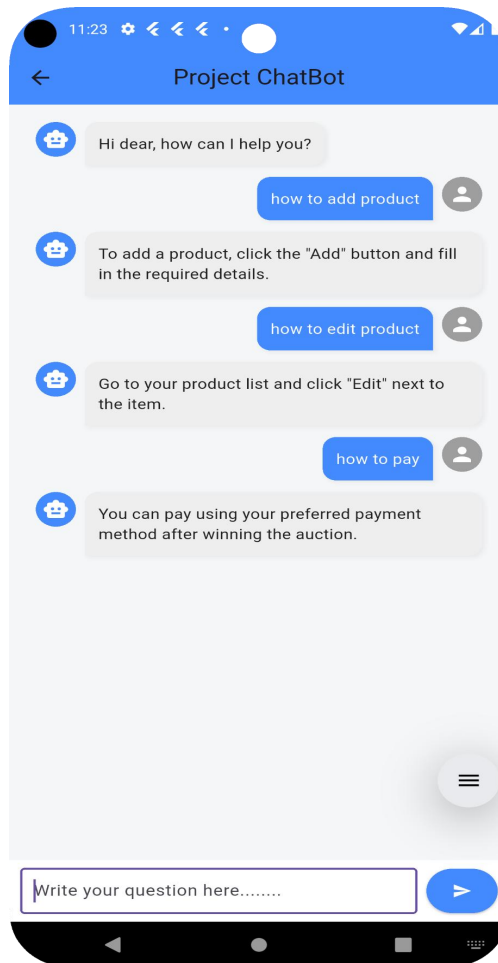
| Title | Description                                 | Starting |
|-------|---------------------------------------------|----------|
| car   | very fast                                   | \$350.   |
| car   | very fast                                   | \$350.   |
| car   | very fast                                   | \$350.   |
| Table | Wooden dining table for 6 people            | \$320.   |
| Table | Wooden dining table for 6 people            | \$320.   |
| Table | Wooden dining table for 6 people            | \$320.   |
| Phone | Latest model smartphone with great features | \$106.   |
| Phone | Latest model smartphone with great features | \$106.   |
| Phone | Latest model smartphone with great features | \$106.   |
| Phone | Latest model smartphone with great features | \$106.   |
| Phone | Latest model smartphone with great features | \$106.   |

| Auction Details |                                                                                     |        |                     |               |                     |                 |             |                                                                                       |
|-----------------|-------------------------------------------------------------------------------------|--------|---------------------|---------------|---------------------|-----------------|-------------|---------------------------------------------------------------------------------------|
| Starting Price  | Image                                                                               | Status | Start Time          | End           | End Time            | Location        | Seller Name | Cancel                                                                                |
| \$350.00        |    | open   | 2025-06-12 02:15:00 | 2025-07-10:00 | 2025-07-16 04:15:00 | Ramallah&#039;s | sanaa       |    |
| \$350.00        |    | open   | 2025-06-12 02:15:00 | 2025-07-10:00 | 2025-07-16 04:15:00 | Ramallah&#039;s | sanaa       |    |
| \$350.00        |    | open   | 2025-06-12 02:15:00 | 2025-07-10:00 | 2025-07-16 04:15:00 | Ramallah&#039;s | sanaa       |    |
| \$320.00        |    | open   | 2025-04-01 12:00:00 | 2025-07-10:00 | 2025-07-16 04:15:00 | aka             | user8       |    |
| \$320.00        |    | open   | 2025-04-01 12:00:00 | 2025-07-10:00 | 2025-07-16 04:15:00 | aka             | user8       |    |
| \$320.00        |    | open   | 2025-04-01 12:00:00 | 2025-07-10:00 | 2025-07-16 04:15:00 | aka             | user8       |    |
| \$106.00        |    | open   | 2025-04-01 10:00:00 | 2025-07-10:00 | 2025-07-16 04:15:00 | nablus          | user5       |    |
| \$106.00        |    | open   | 2025-04-01 10:00:00 | 2025-07-10:00 | 2025-07-16 04:15:00 | nablus          | user5       |    |
| \$106.00        |   | open   | 2025-04-01 10:00:00 | 2025-07-10:00 | 2025-07-16 04:15:00 | nablus          | user5       |   |
| \$106.00        |  | open   | 2025-04-01 10:00:00 | 2025-07-10:00 | 2025-07-16 04:15:00 | nablus          | user5       |  |
| \$106.00        |  | open   | 2025-04-01 10:00:00 | 2025-07-10:00 | 2025-07-16 04:15:00 | nablus          | user5       |  |

Figure 45 : auction details

## Chat Bot:

Chatbot is integrated to assist users with common questions, guide them through features, and improve the overall user experience in real-time .



*Figure 46 : chatbot*

## Search all item Page :

In this interface, we implemented the initial functionality to classify users based on their risk level by categorizing items using the percentile algorithm. Items are divided into three categories: low risk, medium risk, and high risk as previously described.

The displayed images represent different user accounts within the same risk category. A search and filter feature has been added to help users quickly find relevant items. Additionally, users can now click "Bid Now" to place a bid or select favorites to save preferred items.

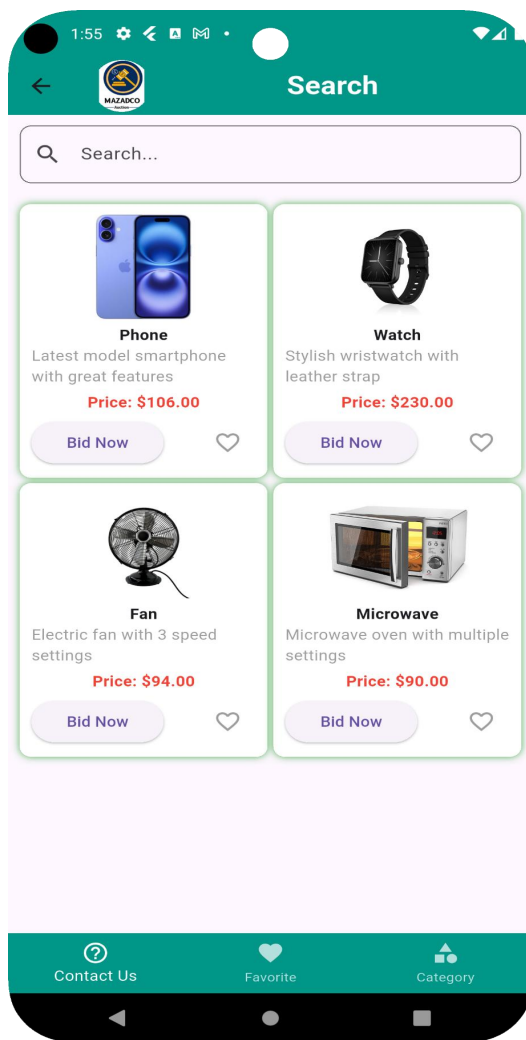


Figure 47 : High Risk (categorization)

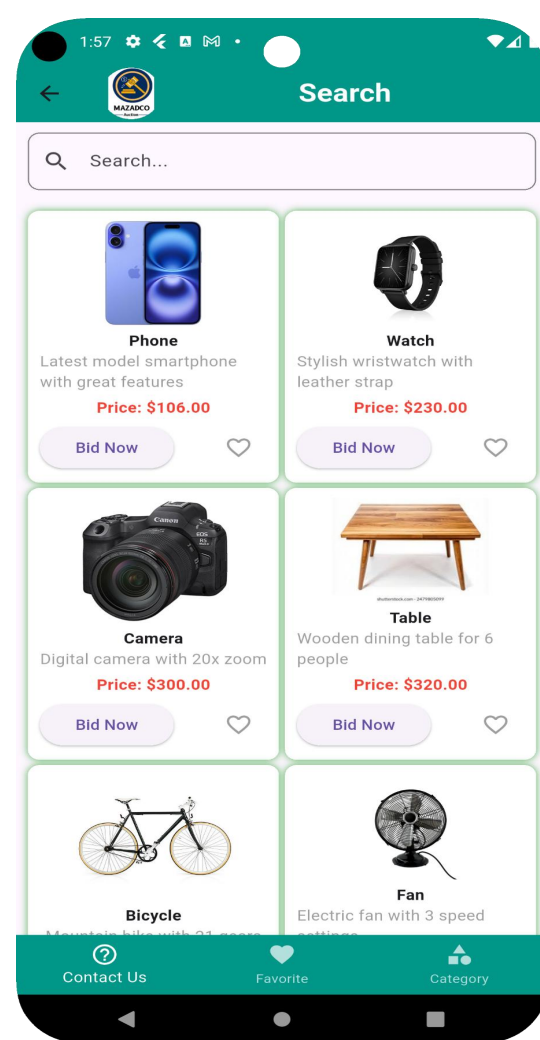


Figure 48: Medium Risk (categorization)

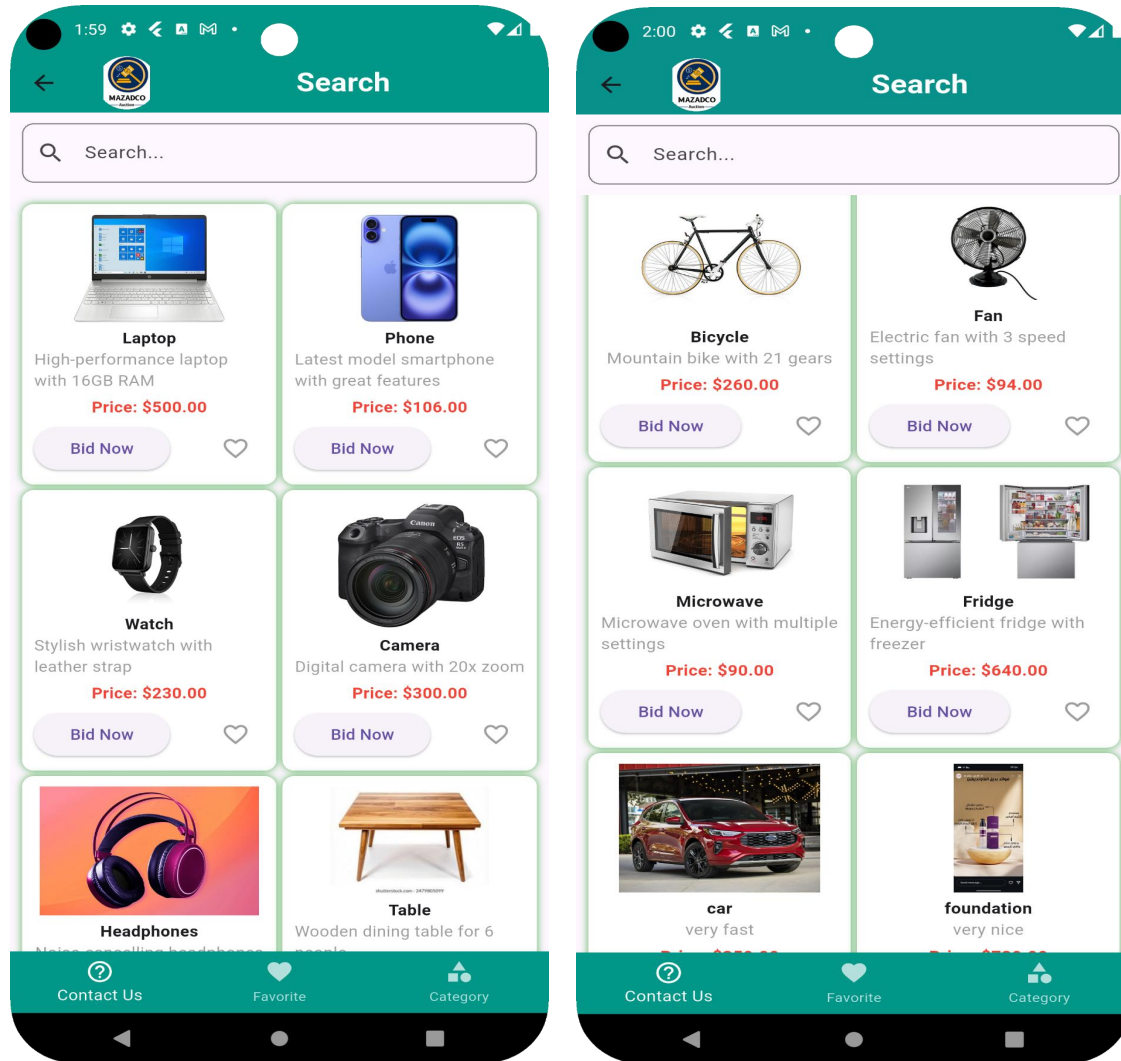


Figure 49 :Low Risk (categorization)

When an item is added to favorites, a notification appears confirming the action with the message: "Added to favorite!"

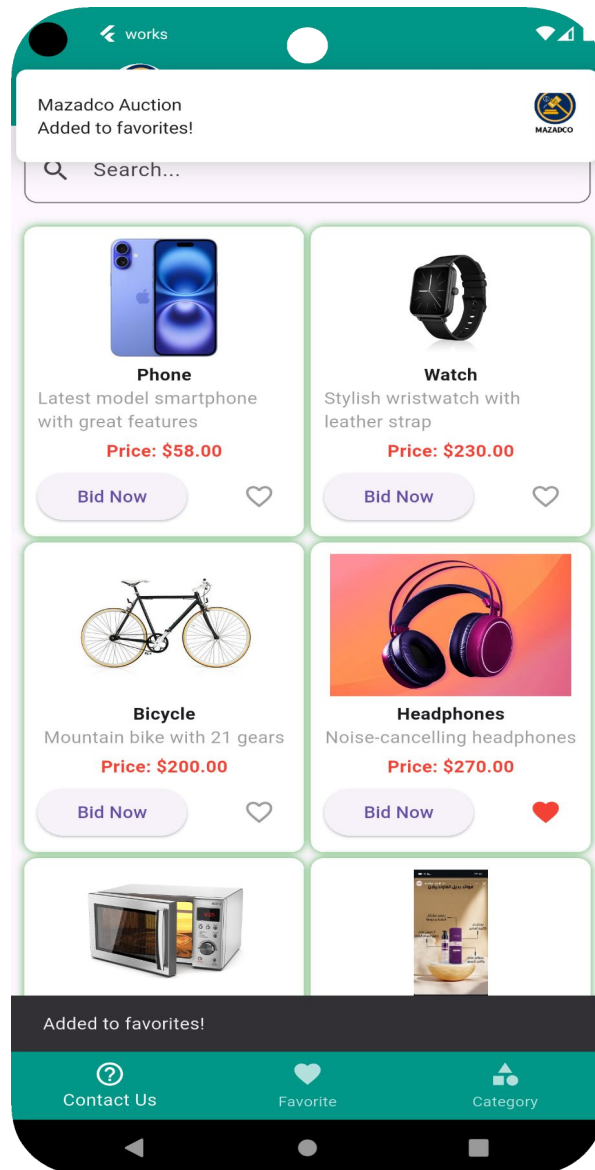


Figure 50 : Added to favorite

## Favorite Page:

When the favorite icon in the navigation bar is clicked, the application displays the list of items added to favorites.

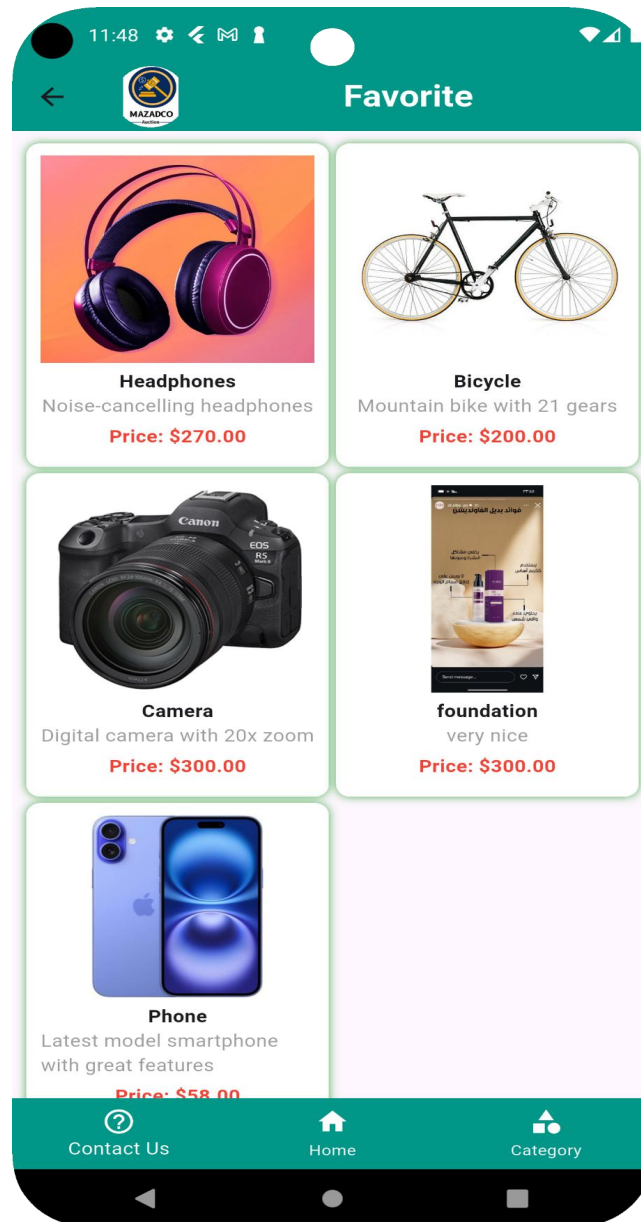


Figure 51: favorite page

## Category Page:

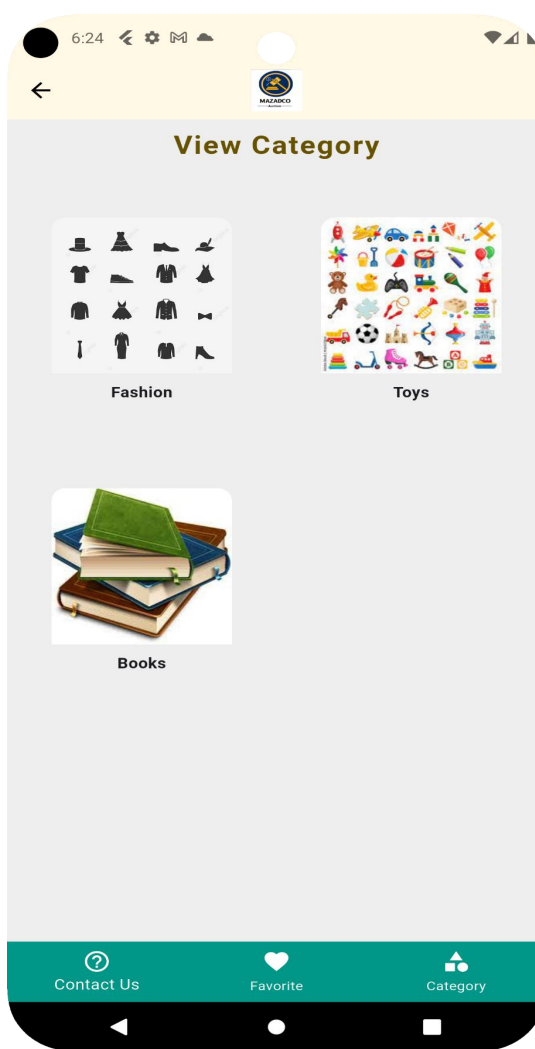


Figure 52 : High risk categorization

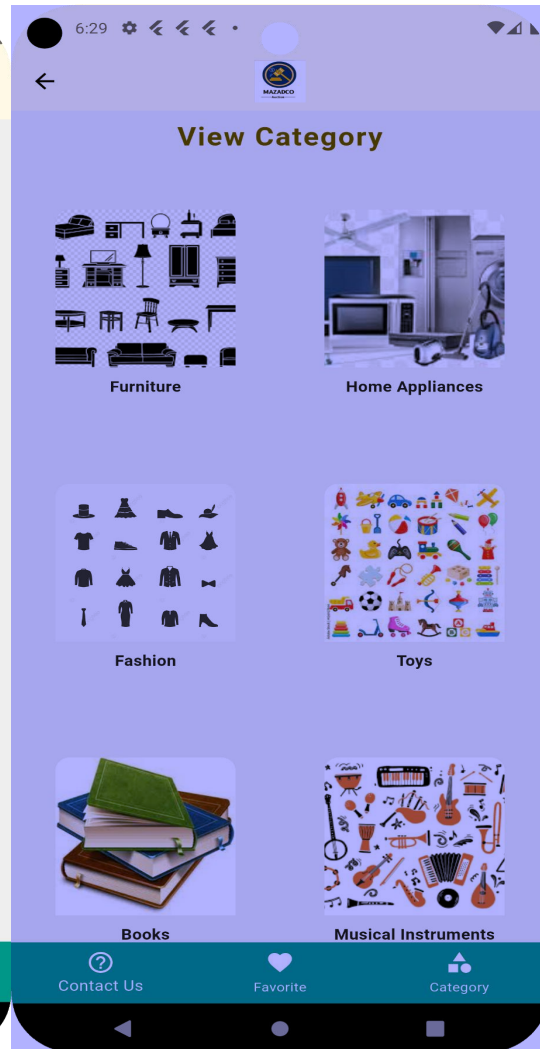
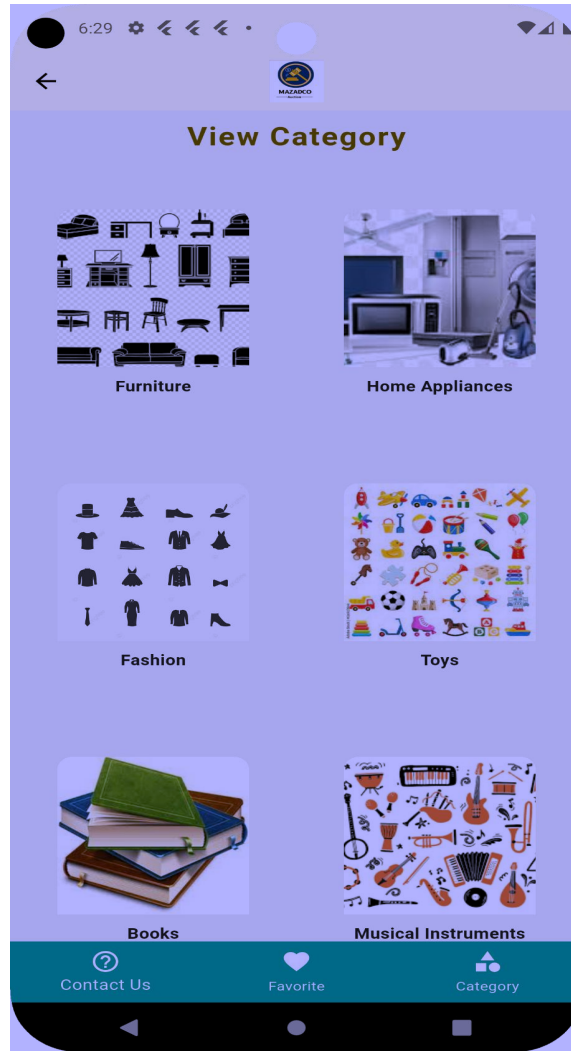


Figure 53 : low risk (all Category)





*Figure 54 : medium risk category*

The app's category classification interface is based on the user's classification level (Low, Medium, High Risk). When the user enters the app, only the categories they are allowed to access are displayed, according to their current classification.

High-risk users can only see categories rated as "high risk," such as books, games, and fashion. Medium-risk users are shown only categories rated as "high" and "medium" risk, such as musical instruments, home appliances, and furniture, in addition to the high-risk categories. Low-risk users are shown all categories without restrictions.

### When Click Electronics Category Show:

After selecting an electronics category, the application navigates to a search page that displays items based on the selected category. This page allows users to explore items depending on their risk level by categorizing items using the percentile algorithm, In addition, users can now click "Bid Now" to place a bid.

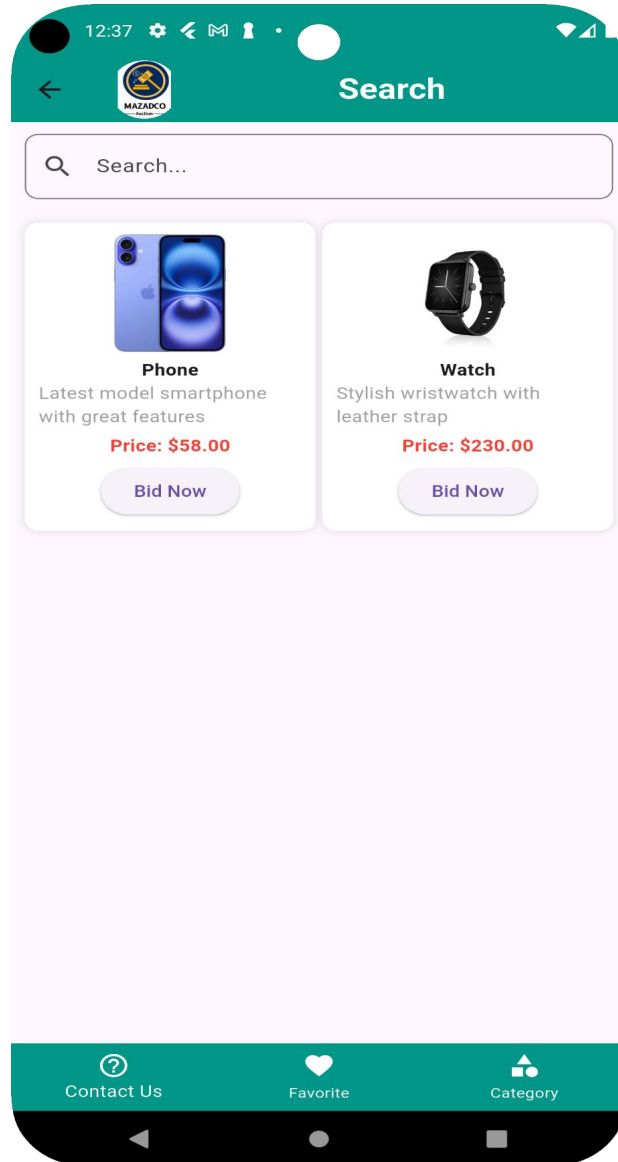
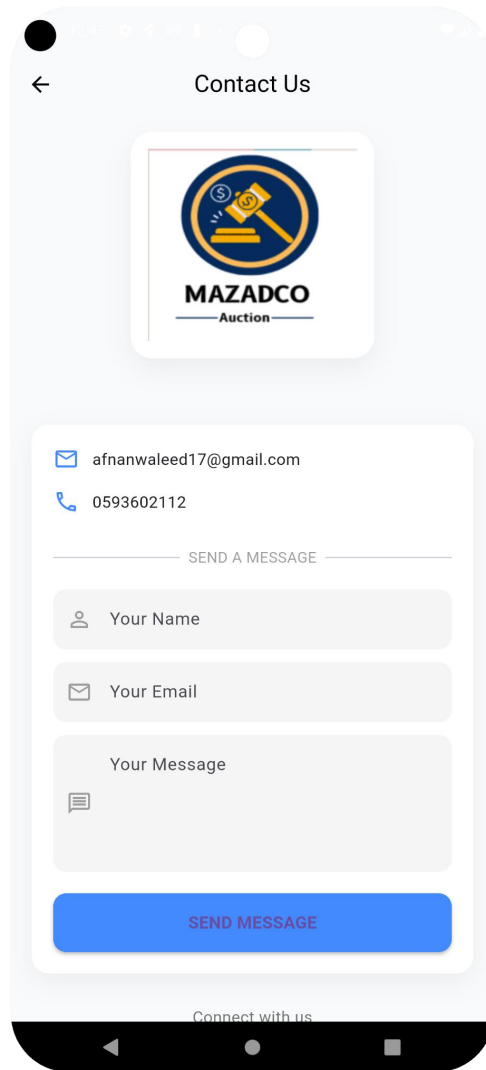


Figure 55 : search page

### When Click Contact Us:

interface contain simple form or info section that allows users to reach out to support or the app's administrators for help, feedback, or inquiries

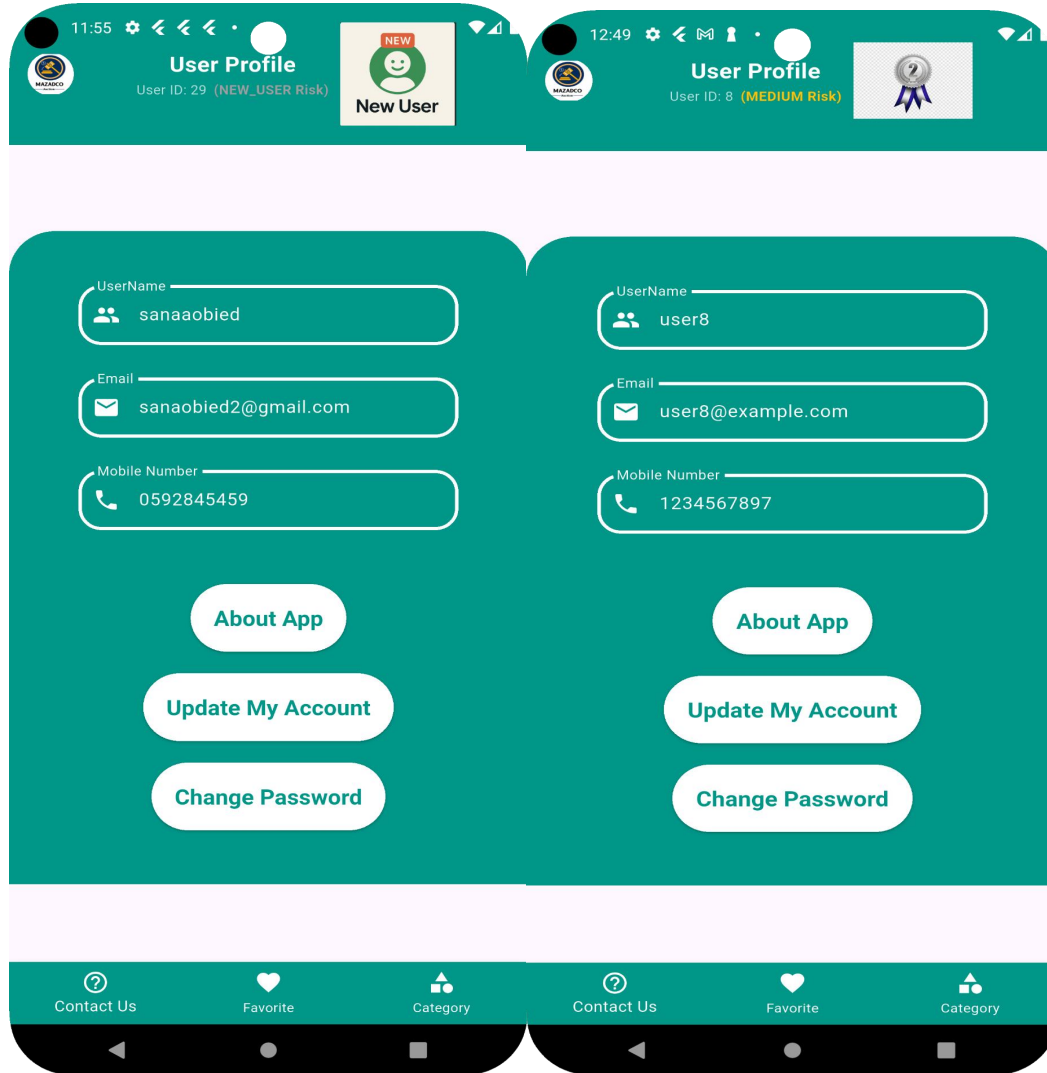


The image shows a mobile application interface for a contact page. At the top, there is a back arrow and the title "Contact Us". Below the title is a logo for "MAZADCO Auction" featuring a blue circle with a yellow auctioneer's gavel and a dollar sign. Under the logo, there is a white box containing contact information: an email icon followed by "afnanwaleed17@gmail.com" and a phone icon followed by "0593602112". Below this information is a horizontal line with the text "SEND A MESSAGE" in the center. Underneath the line are three input fields: "Your Name" with a person icon, "Your Email" with an envelope icon, and "Your Message" with a speech bubble icon. At the bottom of the form is a blue button with the text "SEND MESSAGE". Below the form is a section titled "Connect with us" with social media icons for Facebook, Twitter, and Instagram. The entire interface is displayed on a smartphone screen with a black home indicator bar at the bottom.

*Figure 56 : contact us page*

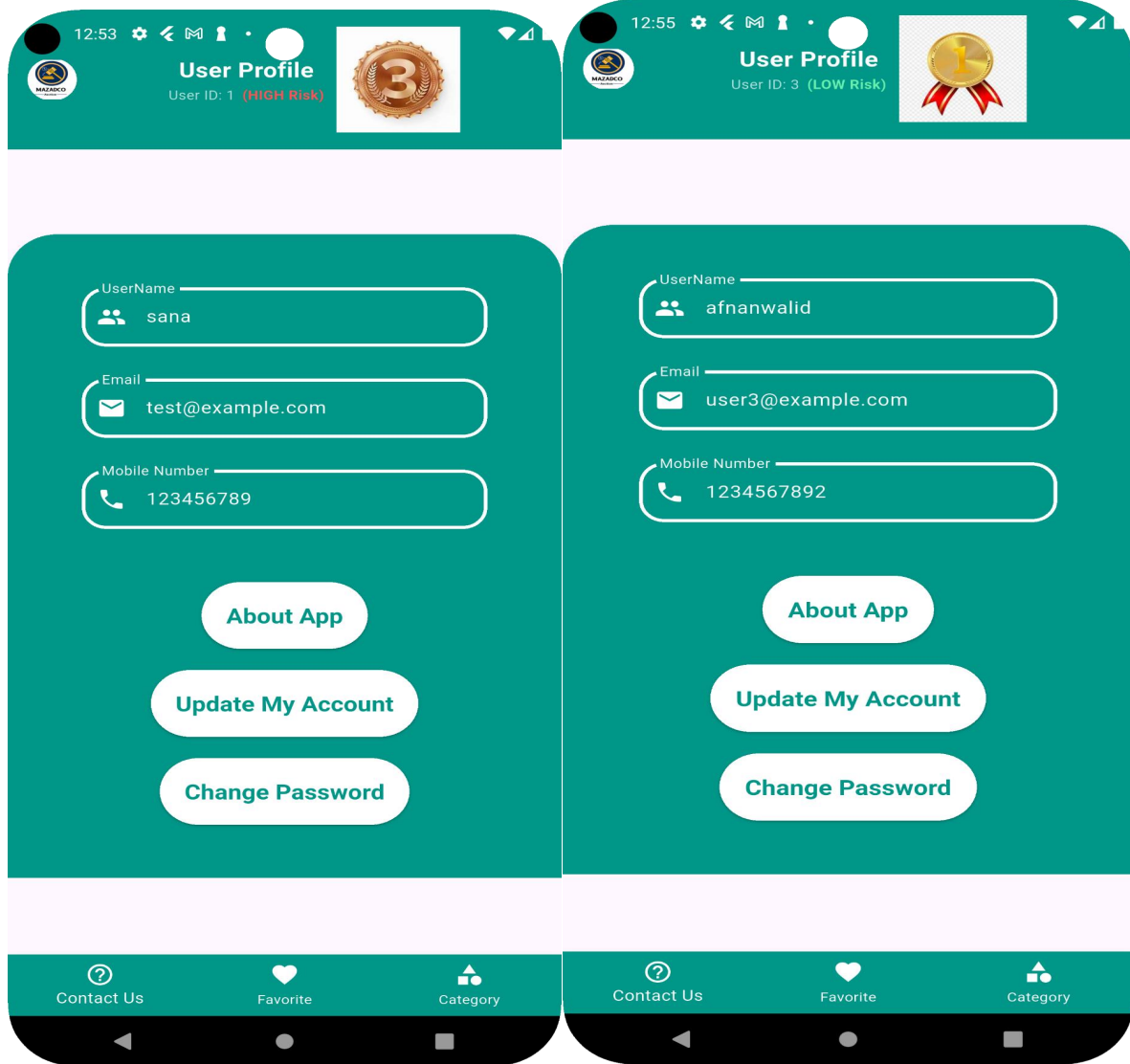
## User Profile page:

In the User Profile page, the user's name, email, mobile number and user's risk category: Low Risk, Medium Risk, or High Risk are displayed.



**New User risk**

**medium risk**



**High risk**

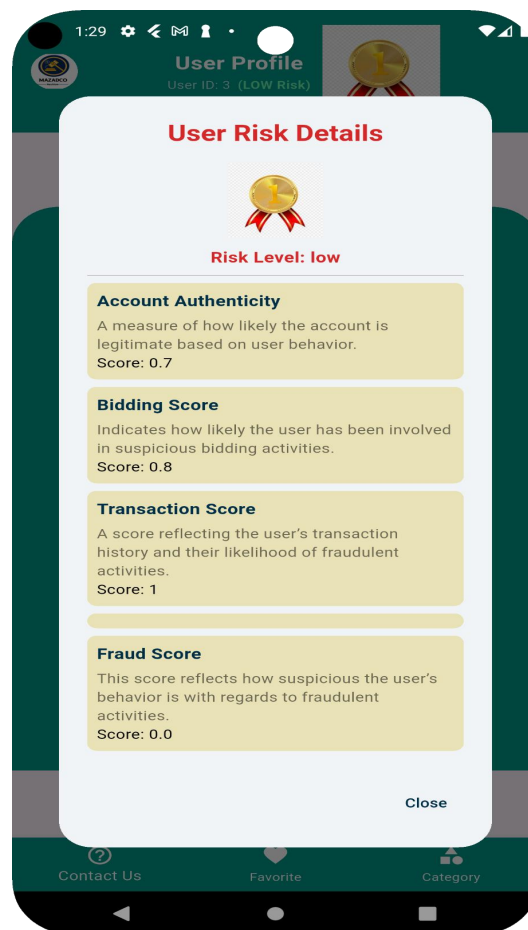
**Low risk**

*Figure 57: user profile page*

Additionally, when the user clicks the risk level icon, a dialog window appears titled "User Risk Details", which includes the following risk evaluation metrics:

- **Account Authenticity:** A measure of how likely the account is legitimate based on user behavior. The score is displayed.
- **Bidding Score:** Indicates how likely the user has been involved in suspicious bidding activities. The score is displayed.
- **Transaction Score:** Reflects the user's transaction history and their likelihood of fraudulent activities. The score is displayed.
- **Fraud Score:** Reflects how suspicious the user's behavior is regarding fraudulent activities. The score is displayed.

Each score is shown along with a corresponding visual indicator icon that reflects the user's risk category: Low Risk, Medium Risk, or High Risk, So when click in icon Low risk show :



*Figure 58 : user risk details*

When the user clicks "Update My Account", the profile fields become editable, allowing the user to update their personal information ,After update information in text field:

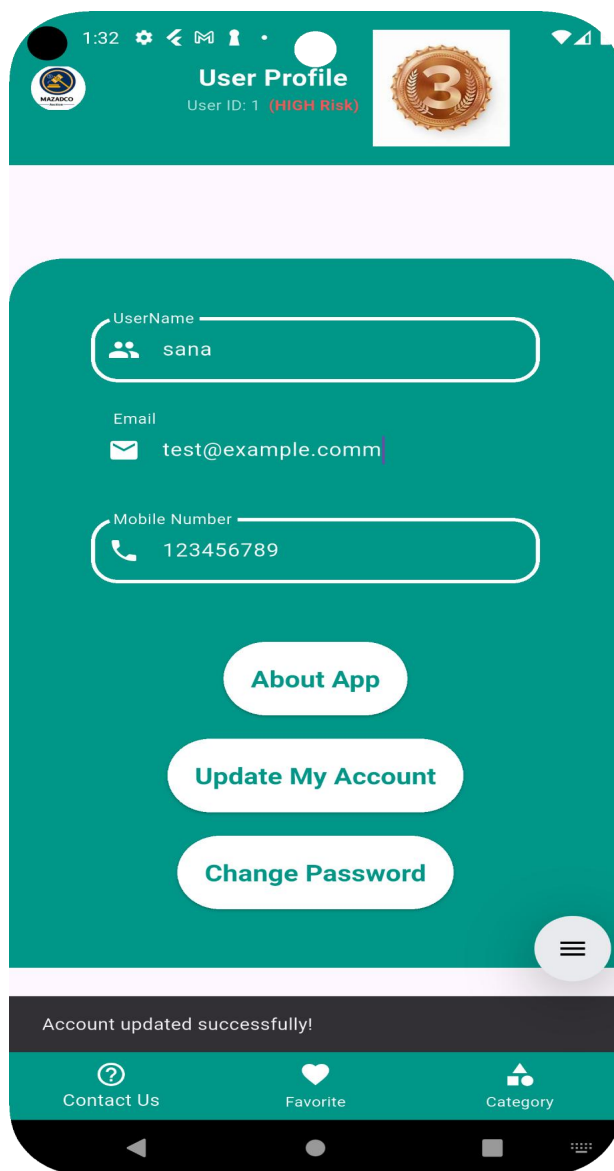
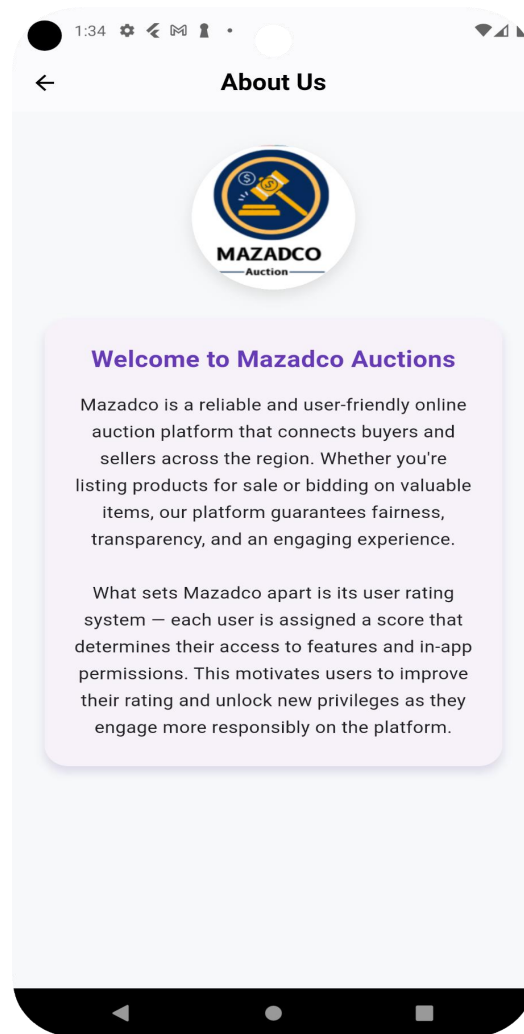


Figure 59 :updated successfully

Selecting "About App" opens a page that displays information about the application and its purpose.



*Figure 60 : about us page*



## Change Password Page:

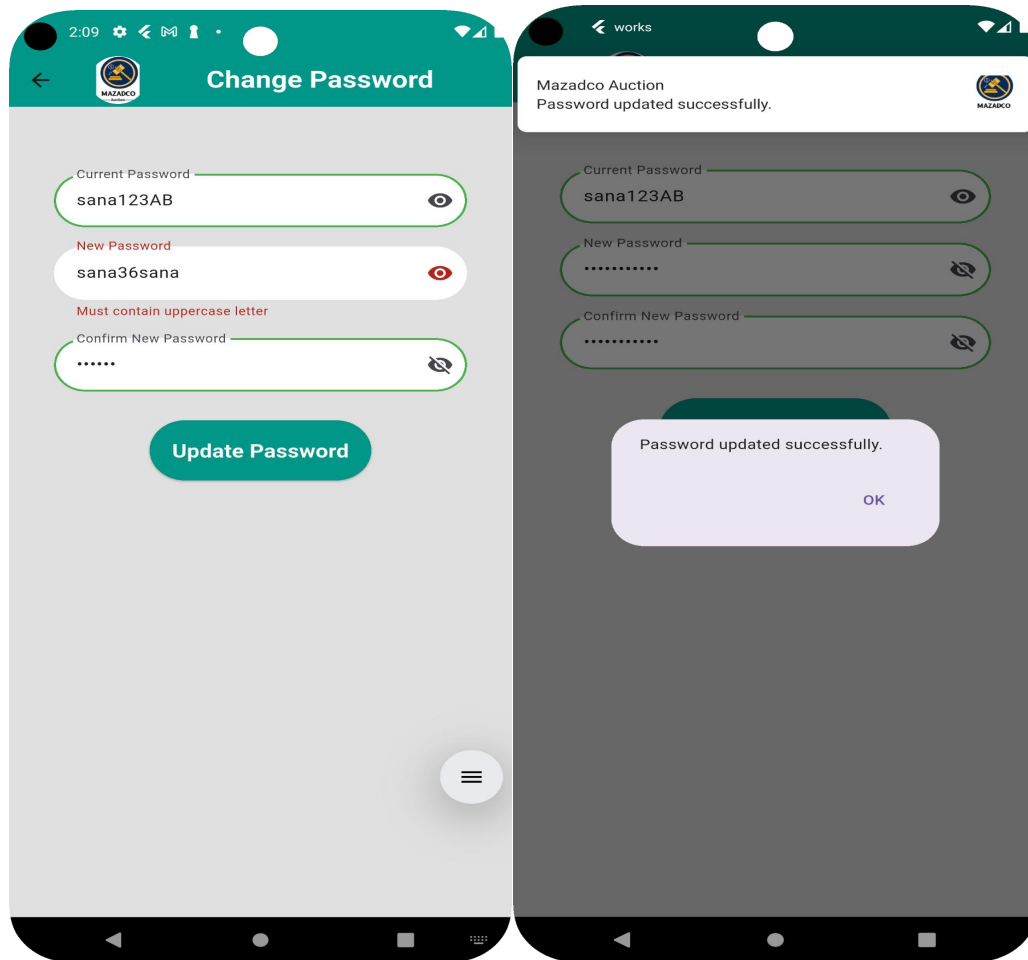
When the user clicks "Change Password", the form contains validation to ensure strong passwords: it must be at least 8 characters long and contain at least one lowercase and one uppercase letter. Users can toggle the visibility of the password fields. Once the information is entered correctly, clicking the "Update Password" updates the user's password. After a successful update, a notification is sent to the user confirming that their password has been changed.

The image displays two side-by-side screenshots of a mobile application's "Change Password" page. Both screens feature a green header with a back arrow, the MAZADCO logo, and the title "Change Password".

**Left Screenshot (Time: 1:54):** The form contains three input fields: "Current Password" (masked with dots), "New Password" (masked with dots), and "Confirm New Password" (masked with dots). Each field has a toggle icon for visibility. Below the fields is a green "Update Password" button.

**Right Screenshot (Time: 2:07):** The "Current Password" field now contains the text "sana123AB" and has a visibility toggle icon. The "New Password" field is empty and has a red error message below it: "Password must be at least 8 characters". The "Confirm New Password" field is also empty. A green "Update Password" button is at the bottom. A hamburger menu icon is visible in the bottom right corner.

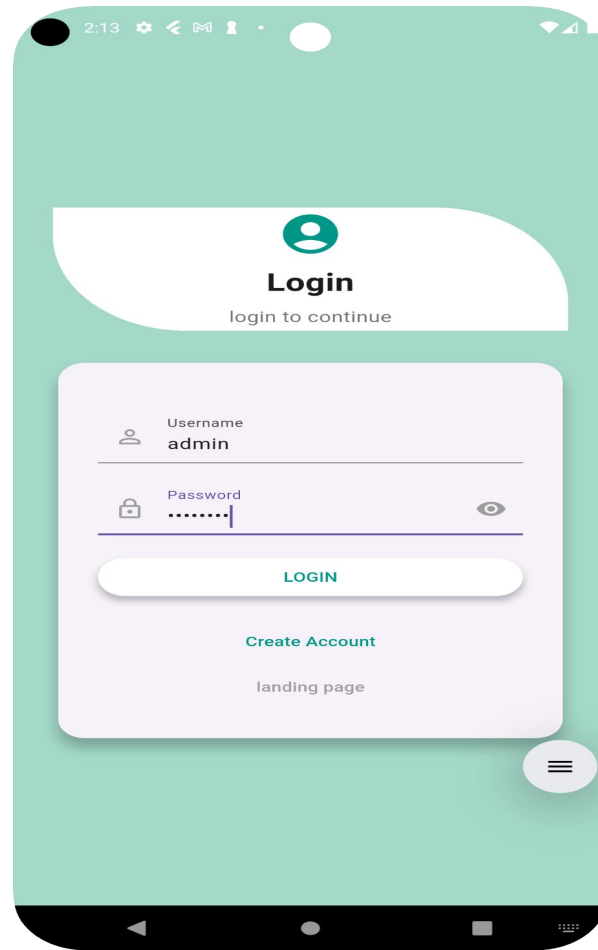
Figure 61: change password page



*Figure 62 : updated password successfully*

## Admin part:

The Admin panel offers an overview of platform activity, including recent auctions, registered users, and categories. Admins can manage content and take actions like editing or deleting or add auctions and shows all users with his risk level and send message to users



*Figure 63 : login for admin*

This interface contains two main sections that display the latest activity on the platform: Recent Auctions and Recent Users. The first section displays a summary of recently created auctions, such as the product title, starting price, start and end dates, and associated category and The second section displays information about newly registered users. To view the details, click Show.

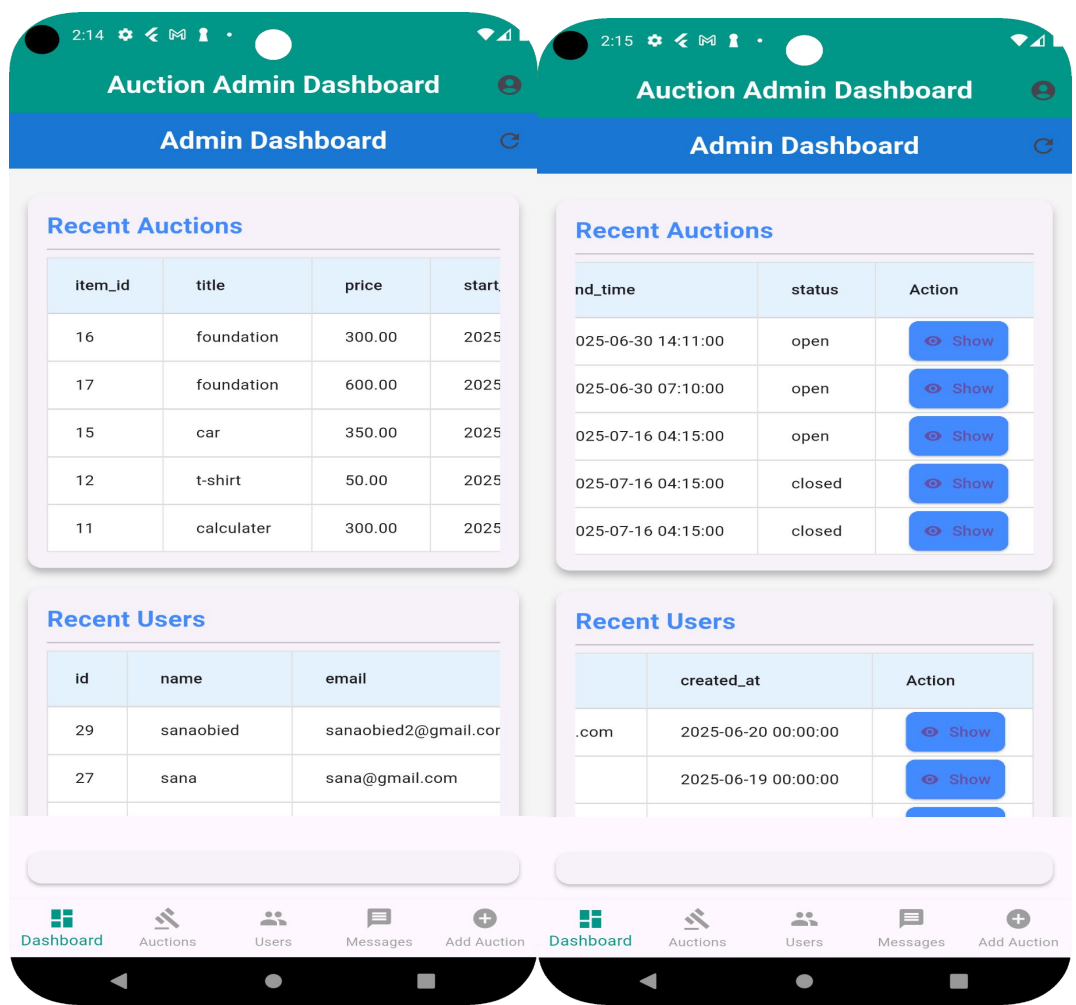
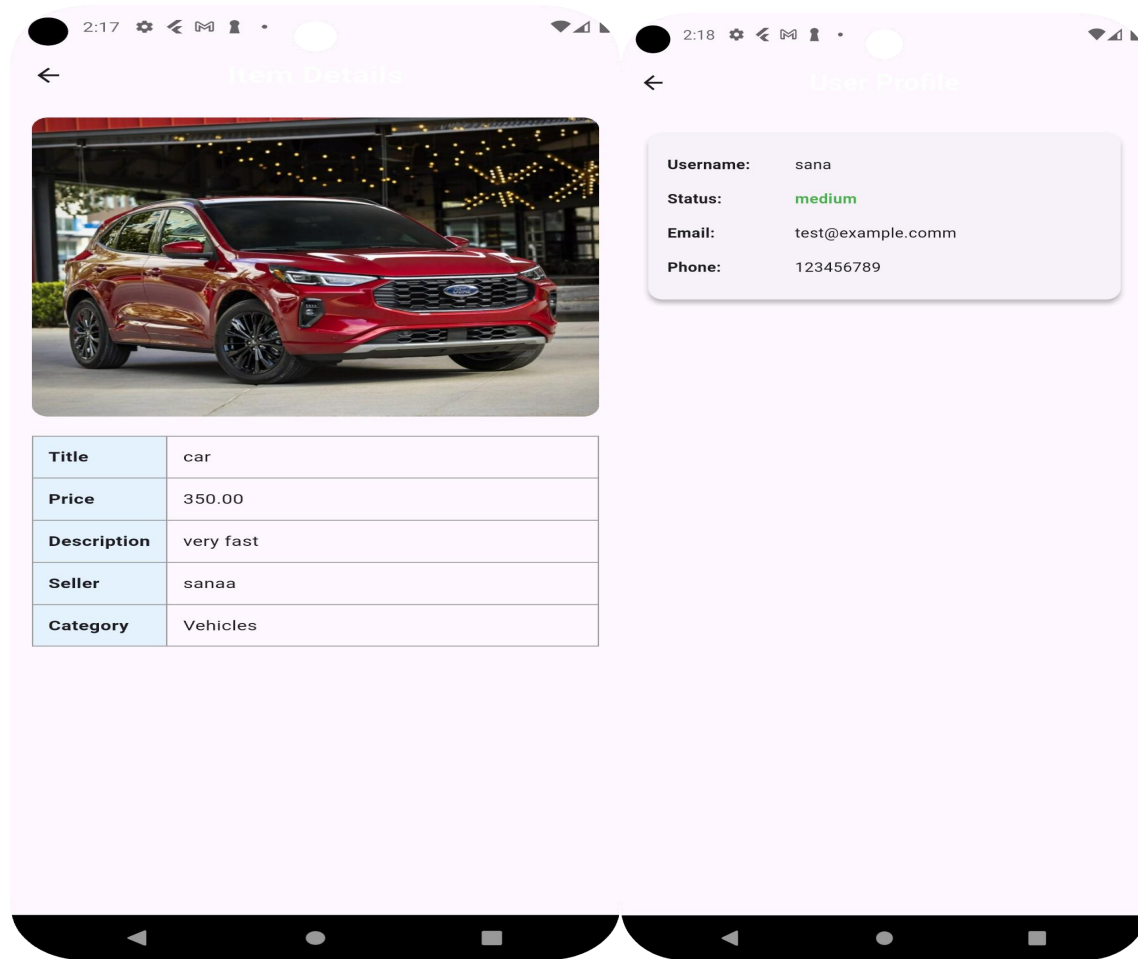


Figure 64: admin dashboard



Recent Auctions

Recent Users

Figure 65 : details page

This interface displays all categories available in the system, and each category contains a list of associated auctions. Through this interface, the admin can browse the categories and then view the auctions within each category in an organized manner. For each auction, information such as the product title, starting price, auction status (open/closed), and start and end dates is displayed. The interface also provides the administrator with direct controls, such as the "Edit" button to update auction data and the "Delete" button to remove it.

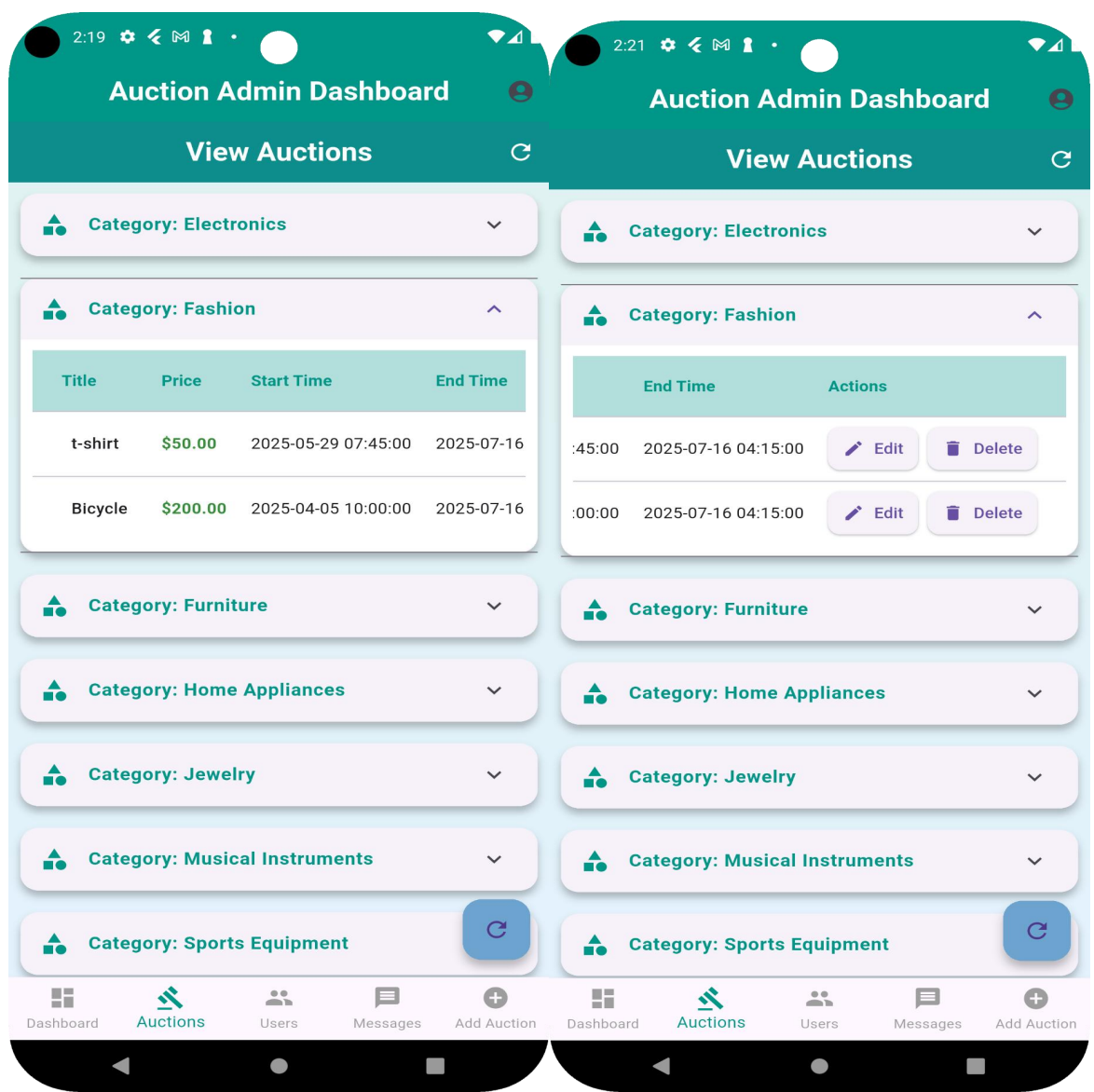


Figure 66 : auction and category details

The Edit Auction interface allows the administrator to modify the information of any previously created auction. Through this interface, basic auction data can be modified, such as the product title, starting price, category, duration (start and end dates), description, location, and product image. The interface is easy to use and has an organized design that allows the admin to make changes quickly and accurately.

The screenshot displays a mobile application interface for editing an auction. At the top, the status bar shows the time as 2:24 and various system icons. The app's header includes a back arrow, the title 'Edit Auction', and a refresh icon. The main content area is a light purple card titled 'Update Auction Info'. It contains several input fields: 'Title' with the value 't-shirt', 'Price' with '50.00', 'Start Time' with '2025-05-29 07:45:00', 'End Time' with '2025-07-16 04:15:00', 'Description' with 'very beautiful', 'Location' with 'salwad' (which is currently selected with a blue cursor and a small blue circle), and 'Seller Name' with 'sanaa'. At the bottom of the card is a button with a purple icon and the text 'Update'. A floating action button with a hamburger menu icon is located at the bottom right of the screen. The bottom of the image shows the standard Android navigation bar.

Figure 67: edit auction

## Customer Risk Level Report:

This page displays the User Risk Level Report, listing all users along with their risk levels. In the "Action" column, a delete icon is provided next to each user entry. When the admin clicks this delete icon, the selected user is removed from the system after a notification is displayed confirming the successful deletion.

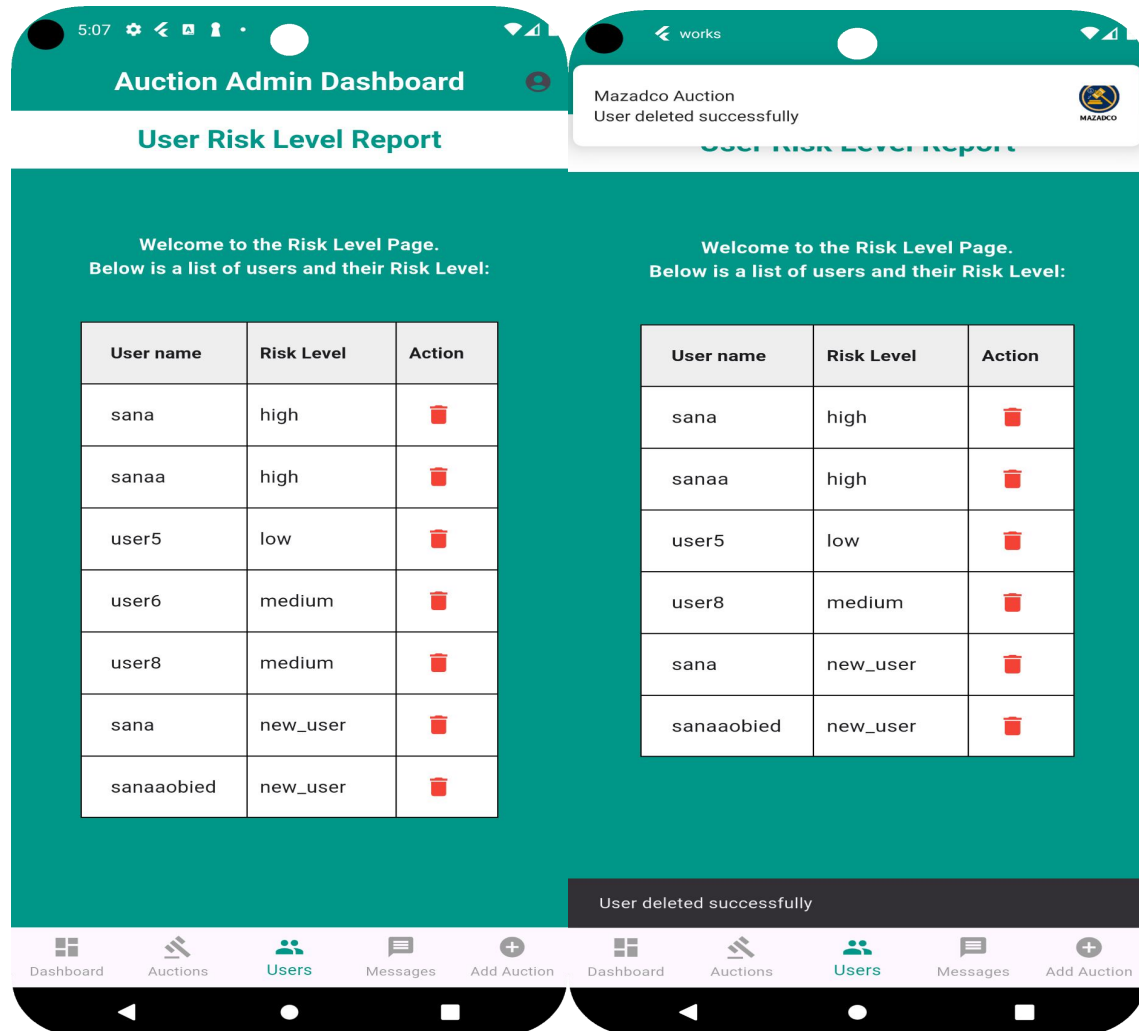


Figure 68 : user risk report



## Add Auctions:

This interface allows Admin and users type medium or Low to list a new product for auction by filling in details such as title, category, starting price, image, and auction time frame. It provides a clean, user-friendly form.

The image displays two side-by-side mobile app screenshots of the 'Add New Auction Product' form. Both screens have a green header with 'Auction Admin Dashboard' and a blue bar with 'Add New Auction Product'. The left screen shows the initial form with a 'Create New Auction' button and a placeholder for the product image. The right screen shows the form filled out with product details.

**Product Image**

MAZADCO Auction

**Create New Auction**

Fill in all required details to list your product

**Product Information**

Product Name  
foundation

Starting Price  
\$ 700

Description  
very nice

**Auction Details**

Start Date & Time  
2025-06-26 14:29

The bottom navigation bar includes icons for Dashboard, Auctions, Users, Messages, and Add Auction.

Figure 69 : add auction page

## Admin Profile Page:

In the User Profile page, the user's name, email and mobile number are displayed

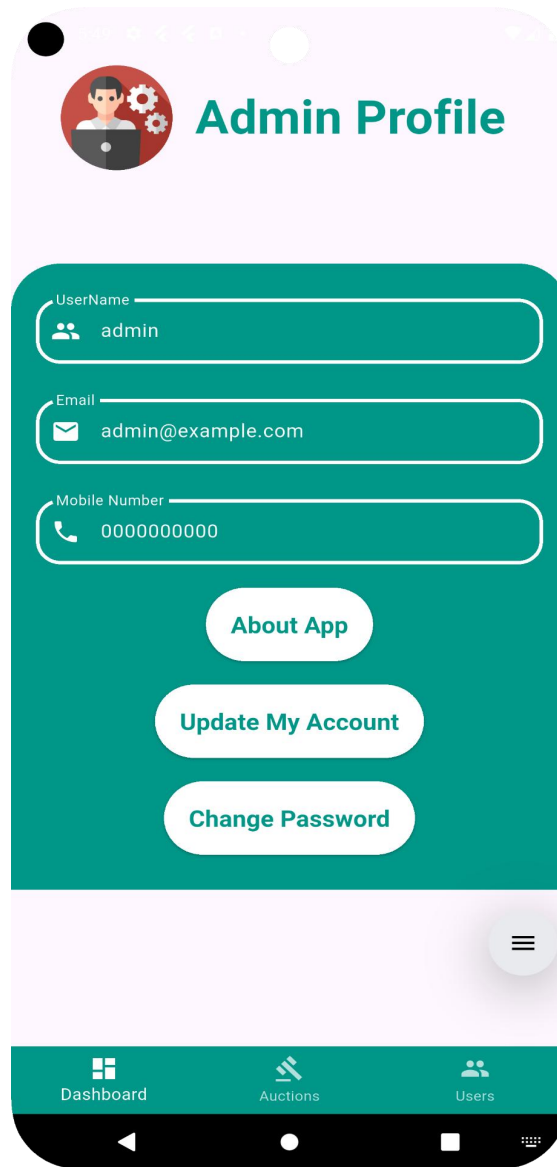
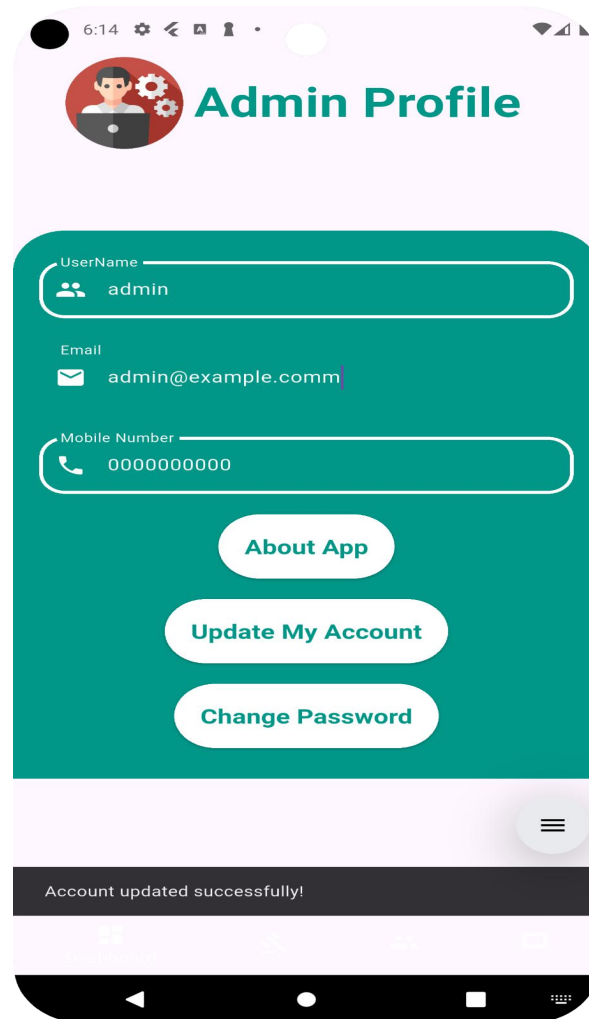


Figure 70 : admin profile page

## Update Account Page:

When the user clicks "Update My Account", the profile fields become editable, allowing the user to update their personal information, After update information in text field :



*Figure 71: Update admin Account Page*

## Change Password Page:

When the user clicks "Change Password", the form contains validation to ensure strong passwords: it must be at least 8 characters long and contain at least one lowercase and one uppercase letter. Users can toggle the visibility of the password fields . Once the information is entered correctly, clicking the "Update Password" updates the user's password

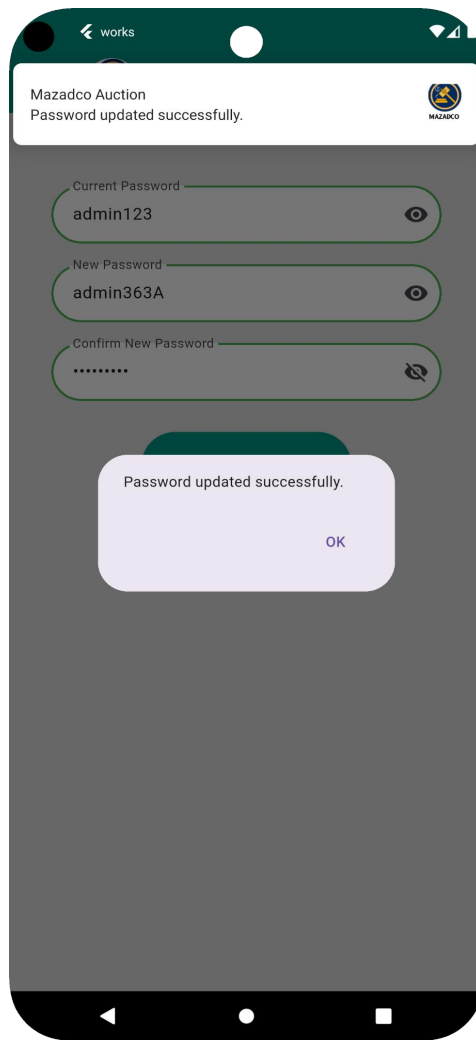
The image displays two side-by-side screenshots of a mobile application's "Change Password" page. Both screenshots feature a green header with a back arrow, a logo, and the title "Change Password". The status bar at the top shows the time as 6:17 and 6:18 respectively.

The left screenshot shows the "Current Password" field with the text "admin123" and an eye icon. The "New Password" field is empty, with a red error message "New Password" and a red eye icon. Below it, a message states "Password must be at least 8 characters". The "Confirm New Password" field is empty with a red eye icon. A green "Update Password" button is at the bottom.

The right screenshot shows the "Current Password" field with the text "admin123" and an eye icon. The "New Password" field contains the text "admin363" and a red eye icon. Below it, a message states "Must contain uppercase letter". The "Confirm New Password" field contains the text "\*\*\*\*\*" and a red eye icon. A green "Update Password" button is at the bottom.

Figure 72 : Change Password Page

After a successful update, a notification is sent to the user confirming that their password has been changed.



*Figure 73:notification password updated successfully*

## CHAPTER 6. TEST CASE EXECUTION

In this chapter, we present the testing process applied to ensure the usability, and security of the Mazadco auction platform. All test cases were planned, executed, and tracked using [Jira](#)

### Test Case #001

**TEST CASE TITLE :**Testing bidding with valid bid by low-risk user — positive case

**TEST DATE:**2025-06-24

**TEST SCENARIO:** Low-risk user places a bid within allowed limits

**TEST DESIGNED BY:**Sanaa Obied

**TEST DESCRIPTION:**User updates bid price, chooses payment method, enters payment info, and submits bid successfully.

#### PRE-CONDITIONS:

- User is logged in as low-risk
- Auction item is available with current price
- User can change bid price
- Allowed payment methods for low-risk user shown
- Payment info fields available based on selected method

*Table 7 : Testing bidding with valid bid by low-risk user — positive case*

| STEP ID | STEP DESCRIPTION                      | EXPECTED RESULTS                                                                             | ACTUAL RESULTS                                                       | PASS / FAIL |
|---------|---------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------|
| 1       | Log in as low-risk user               | User logged in; status shows 'low'                                                           | User logged in; status confirmed as 'low'                            | PASS        |
| 2       | Update bid price to valid amount      | Bid price updates to new amount                                                              | Bid price updated from \$81 to \$91                                  | PASS        |
| 3       | Select allowed payment method         | Dropdown shows payment options for low-risk user                                             | Dropdown shows Visa, PayPal ,Cash on delivery ; Credit Card selected | PASS        |
| 4       | Enter payment details                 | Payment info accepted                                                                        | Payment details entered and accepted                                 | PASS        |
| 5       | Click <b>Check Information</b> button | User sees success notification message confirming bid submission and display Payment details | User saw green notification:<br><b>Bid placed successfully!</b>      | PASS        |

## Test Case #002

**TEST CASE TITLE :**Testing bidding with valid bid by medium-risk user — positive case

**TEST DATE:**2025-06-24

**TEST SCENARIO:** Medium-risk user places a bid within daily limit

**TEST DESIGNED BY:**Sanaa Obied

**TEST DESCRIPTION:** User updates bid price, chooses payment method, enters payment info, and submits bid successfully.

### PRE-CONDITIONS:

- User is logged in as medium-risk
- User has placed fewer than 3 bids today
- Auction item is available
- Allowed payment methods for medium-risk user shown

*Table 8 :Testing bidding with valid bid by medium-risk user — positive case*

| STEP ID | STEP DESCRIPTION                      | EXPECTED RESULTS                                                                             | ACTUAL RESULTS                                                  | PASS / FAIL | ADDITIONAL NOTES                |
|---------|---------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------|---------------------------------|
| 1       | Log in as medium-risk user            | User logged in; status shows 'medium'                                                        | User logged in; status confirmed as 'medium'                    | PASS        |                                 |
| 2       | Update bid price to valid amount      | Bid price updates to new amount                                                              | Bid price updated from \$330 to \$360                           | PASS        |                                 |
| 3       | Select allowed payment method         | Dropdown shows payment options for medium-risk user                                          | Dropdown shows Credit Card only; Credit Card selected           | PASS        |                                 |
| 4       | Enter payment details                 | Payment info accepted                                                                        | Payment details entered and accepted                            | PASS        |                                 |
| 5       | Click <b>Check Information</b> button | User sees success notification message confirming bid submission and display Payment details | User saw green notification:<br><b>Bid placed successfully!</b> | PASS        | User has less than 3 bids today |

### Test Case #003

**TEST CASE TITLE :** Testing bid rejection when exceeding daily limit — negative case

**TEST DATE:**2025-06-24

**TEST SCENARIO:** High-risk user tries to place a bid exceeding daily limit

**TEST DESIGNED BY:**Sanaa Obied

**TEST DESCRIPTION:** User tries to bid but has already reached the daily bid limit based on risk.

#### **PRE-CONDITIONS:**

- User is logged in as high-risk
- User already placed 1 bid today (limit reached)
- Auction item is available

*Table 9 :Testing bid rejection when exceeding daily limit — negative case*

| STEP ID | STEP DESCRIPTION                      | EXPECTED RESULTS                                   | ACTUAL RESULTS                                                                                         | PASS / FAIL | ADDITIONAL NOTES                |
|---------|---------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------|---------------------------------|
| 1       | Log in as high-risk user              | User logged in; status shows 'high'                | User logged in; status confirmed as 'high'                                                             | PASS        |                                 |
| 2       | Click <b>Check Information</b> button | User sees error notification about daily bid limit | User saw red notification:<br><b>✗ You have reached your daily bid limit based on your risk level.</b> | PASS        | User already placed 1 bid today |



**Test Case Title:** Filtering Auction Items by User Risk Level Using Percentiles (positive test case )

**Test Date:** 2025-06-24

**Test Scenario:** Displaying Items Based on User Risk Level Using Percentiles

**Test Designer:** Safa Alawi

**Test Case Description:** The system displays a variety of auction items based on the user's risk level (low, medium, or high). The items are ranked by price, and percentages are then calculated to determine which items will be displayed.

**Pre -Condition :**

- User is logged in to the system
- Auction items are available and have specific prices
- User's risk level is determined
- Stable internet connection
- A sufficient number of items are available to calculate percentages

*Table 10: Filtering Auction Items by User Risk Level Using Percentiles (positive test case )*

| STEP ID | STEP DESCRIPTION                     | EXPECTED RESULTS                                      | ACTUAL RESULTS                                     | PASS / FAIL |
|---------|--------------------------------------|-------------------------------------------------------|----------------------------------------------------|-------------|
| 1       | Log in as low-risk user              | User sees all auction items regardless of price       | Displayed all 30 items (full list)                 | PASS        |
| 2       | Log in as medium-risk user           | User sees items from lowest to ~66th percentile       | Displayed 20 items with prices in lower two-thirds | PASS        |
| 3       | Log in as high-risk user             | User sees items in bottom third of price range only   | Displayed 10 cheapest items                        | PASS        |
| 4       | Edge Case: All items have same price | All users see same set (no difference in percentiles) | All saw same 30 items                              | PASS        |
| 5       | Edge Case: Less than 3 items         | All risk levels see same available items              | No filtering applied; all items shown              | PASS        |

**Negative Test Case Title:** Item filtering failed due to an error in the price data (presence of a non-numeric price)

**Test Date:** 2025-06-24

**Test Scenario:** The system cannot calculate percentages due to an item with a non-numeric price.

**Test Designer:** Safa Alawi

**Test Case Description:**

An auction item contains an invalid price value (such as "N/A," null, or text), causing the filtering algorithm to fail and the items to be loaded to the user based on risk.

**Pre condition :**

- The user is logged in.
- The auction data contains at least one item with a non-numeric price.
- The system applies price filtering and a percentile algorithm.
- The connection is stable.

*Table 11 : Item filtering failed due to an error in the price data (presence of a non-numeric price)*

| Step ID. | Step Description                     | Expected Result                                                 | Actual Result                                            | Pass / Fail | Notes                                   |
|----------|--------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------|-------------|-----------------------------------------|
| 1        | Log in as a user with any risk level | System identifies user and determines their risk level          | Login successful and risk level determined               | Pass        |                                         |
| 2        | Load auction items                   | System attempts to calculate percentiles                        | Error occurred during price sorting due to invalid value | Pass        |                                         |
| 3        | Display error message                | Message appears: "Data error occurred. Please try again later." | Expected message was displayed                           | Pass        | System crash was successfully prevented |

## Test Case #006

**TEST CASE TITLE :**Testing Add Auction — positive case

**TEST DATE:**2025-06-24

**TEST SCENARIO:**A user adds a new auction with complete and valid information

**TEST DESIGNED BY:**Afnan Kharosha

**TEST DESCRIPTION:**The user fills in all required fields correctly and submits the form successfully. The auction is created and displayed in the system.

### PRE-CONDITIONS:

- User is logged in
- User has permission to create an auction

*Table 12 : Testing Add Auction — positive case*

| Step ID | Description                                                          | Expected Result                                                                                                        | Actual Result                                                                                                       | Fail/Pass |
|---------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------|
| 1       | user Log in as admin or medium or low risk users and go to dashboard | login success and if user show product                                                                                 | login success and if user show product                                                                              | Pass      |
| 2       | user click to icon of Add Auction                                    | Open the form of add in new page                                                                                       | Open the form of add in new page                                                                                    | Pass      |
| 3       | Fill in all fields in form                                           | user add his all product information                                                                                   | user add his all product information                                                                                | Pass      |
| 4       | Click "Submit"                                                       | if all data fields fill correct data Confirmation message appears at the bottom of the page "Add your product success" | all data fields fill correct data Confirmation message appears at the bottom of the page "Add your product success" | Pass      |
| 5       | if Navigate the category you add your product you see it             | the Auction created in category and appears to users                                                                   | the Auction created in category and appears to users                                                                | Pass      |

**Test Case #007****TEST CASE TITLE :**Testing Add Auction — Negative case**TEST DATE:**2025-06-24**TEST SCENARIO:**A user tries to submit an auction without filling all required fields**TEST DESIGNED BY:**Afnan Kharosha**TEST DESCRIPTION:**The user forgets or omits required fields (e.g., title or image), and the system blocks the submission with validation errors**PRE-CONDITIONS:**

- User is logged in
- User has permission to create an auction

*Table 13 : Testing Add Auction — Negative case*

| Step ID | Description                                                                 | Expected Result                                                              | Actual Result                                                                | Fail/Pass |
|---------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------|
| 1       | user Log in as admin or medium or low risk users and go to dashboard        | login success and if user show product                                       | login success and if user show product                                       | Pass      |
| 2       | user click to icon of Add Auction                                           | Open the form of add in new page                                             | Open the form of add in new page                                             | Pass      |
| 3       | not Fill in all fields in form such as Leave the image or price field empty | the red sign appear contain Note (“this field required”)                     | the red sign appear contain Note (“this field required”)                     | Pass      |
| 4       | Click "Submit"                                                              | Form is not submitted and Validation error appears so Auction is not created | Form is not submitted and Validation error appears so Auction is not created | Pass      |

## Test Case #008

**TEST CASE TITLE :**Testing ChatBot-Positive Case

**TEST DATE:**2025-06-25

**TEST SCENARIO :**A user asks a question unrelated to auction features such as rockets

**TEST DESIGNED BY:**Afnan Kharosha

**TEST DESCRIPTION:**The user types a message that has nothing to do with the auction system. The bot should not give a misleading or incorrect answer, and must respond with a generic fallback response

### PRE-CONDITIONS:

- User is logged in

*Table 14 : ChatBot-Positive Case*

| Step ID | Description                                                         | Expected Result                                                                                            | Actual Result                          | Fail/Pass |
|---------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------|-----------|
| 1       | user Login and open the chatbot screen                              | ChatBot screen opens normally                                                                              | ChatBot screen opens normally          | Pass      |
| 2       | The user writes a message like “ what time are missiles from iran?” | Bot replies with an answer like: “To add a product, press the Add button and fill in the required fields.” | Bot replies with with expected message | Pass      |
| 3       | User sends more unrelated message                                   | Bot still returns the fallback message                                                                     | Bot still returns the fallback message | Pass      |
| 4       | Use random words or symbols only                                    | Bot still returns the fallback message                                                                     | Bot still returns the fallback message | pass      |

## Test Case #009

**TEST CASE TITLE :**Testing ChatBot-Positive Case

**TEST DATE:**2025-06-25

**TEST SCENARIO :**A user asks a question related to auction features

**TEST DESIGNED BY:**Afnan Kharosha

**TEST DESCRIPTION :** The user enters a clear and valid question like "How do I add a product?" and the chatbot responds with an accurate and relevant answer based on the knowledge base

### PRE-CONDITIONS:

- User is logged in

*Table 15 :ChatBot-Positive Case*

| Step ID | Description                                                          | Expected Result                                                                  | Actual Result                                                                    | Fail/Pass |
|---------|----------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------|
| 1       | user Login and open the chatbot screen                               | ChatBot screen opens normally                                                    | ChatBot screen opens normally                                                    | Pass      |
| 2       | The user writes a message like "How can I add a product?"            | Bot replies with fallback response: "Sorry, I did not understand your question " | Bot replies with fallback response: "Sorry, I did not understand your question " | Pass      |
| 3       | User sends more related message such as "Ask what happens if I win " | Bot return "you will be notified"                                                | Bot return "you will be notified"                                                | Pass      |

## Test Case #0010

### TEST CASE TITLE :

High Risk User Using Visa Payment-Positive Case

**TEST DATE:**2025-06-25

**TEST SCENARIO :**A user classified as "High Risk" selects Visa as a payment method and successfully submits

**TEST DESIGNED BY:** Safa Alawi

### TEST DESCRIPTION :

The user attempts to place a bid using a Visa card (which is allowed for high-risk users), and the system accepts the transaction successfully.

#### PRE-CONDITIONS:

- User is logged in
- User's risk level is High
- Auction is open

*Table 16 :High Risk User Using Visa Payment-Positive Case*

| Step ID | Description                                                         | Expected Result                                                 | Actual Result                                                   | Fail/Pass |
|---------|---------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|-----------|
| 1       | User logs in and navigates to auction item                          | Auction page loads successfully                                 | Auction page loads successfully                                 | Pass      |
| 2       | Click on place bid button and select to visa then enter visa number | the text field appears allowing to enter visa number            | the text field appears allowing to enter visa number            | Pass      |
| 3       | click to check to verify visa number                                | if number valid send confirmation notification and update price | if number valid send confirmation notification and update price | Pass      |

## Test Case #0011

**TEST CASE TITLE :** High Risk User Tries Apple Pay – Negative Case

**TEST DATE:**2025-06-25

**TEST SCENARIO :** A user classified as "High Risk" attempts to choose PayPal (which is not allowed for their level).

**TEST DESIGNED BY:** Safa Alawi

**TEST DESCRIPTION :**The user selects pay pal while placing a bid, but the system rejects **PRE-CONDITIONS:**

- User is logged in
- User's risk level is **High**
- Auction is open

*Table 17: High Risk User Tries Apple Pay – Negative Case*

| Step ID | Description                                                                                          | Expected Result                                           | Actual Result                                             | Fail/Pass |
|---------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|-----------|
| 1       | User logs in and navigates to auction item                                                           | Auction page loads successfully                           | Auction page loads successfully                           | Pass      |
| 2       | User selects bid amount and clicks on place bid and try to select PayPal option while it unclickable | System blocks                                             | System blocks                                             | Pass      |
| 3       | User cannot proceed to payment                                                                       | User is required to choose another valid method like Visa | User is required to choose another valid method like Visa | Pass      |



## Chapter 7: Conclusion and Future works

### 7.1 Review of the project and reiteration of important findings:

Our project will help people to sell their product and goods or buy and can use this application online without having to go to Auction and you can participate in an auction outside your country without the need to travel. Our application has high privacy.

We have added new features to our application based on the research and some apps to help users to supply products and to buy the best products with the best price.

Our project is an online auction application for Android mobile phones that allows the user to create or participate in auctions in any country (we have worldwide shipping) with benefits and financial benefits.

### 7.2 Next Semester Plan:

*Table 18 :Next Semester Plan*

| Semester            | Goals to be Achieved Expected                                                                                                                                                                                                                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First Semester      | 1. Collect Ideas and choose one<br>2. Read About the Main Topics in the Project and Write the Proposal.<br>3. Work in System Design and Requirements.<br>4. Finish Writing the Final Report & Prepare the Presentation.<br>from September until January                                                                          |
| Second Semester     | 1. Find Out About the Required Technology.<br>2. Design Interface.<br>3. Start Writing Code.<br>4. Write the Final Report.<br>5. Final Presentation.<br>from March until June                                                                                                                                                    |
| Next plan in Future | 1. support multiple languages<br>2. add internal shipping in application<br>3. add more payment method<br>4. add more criteria and features to Algorithm<br>5. The user can give feedback about the supplier and be public for others.<br>6. Increase the number of permissions based on classification<br>in (July and August ) |

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”وأخر دعوانهم أن الحمد لله رب العالمين”

